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Local anesthetics combined with vasoconstrictors in patients with cardiovascular disease undergoing dental procedures: Systematic review and meta-analysis

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Title: Local anesthetics combined with vasoconstrictors in patients with cardiovascular disease undergoing dental procedures: Systematic review and meta-analysis

Running title: Local anesthetics combined with vasoconstrictors in patients with cardiovascular disease undergoing dental procedures

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ABSTRACT

[Background] Local anesthetic (LA) with vasoconstrictor is widely used in dental procedures to block pain transmission and control bleeding. In this systematic review, the authors evaluated the evidence regarding the safety of its use in patients with cardiovascular diseases (CVD).

[Objectives] To systematically review the literature and update the evidence-based clinical practice for the safety of LA with vasoconstrictor in patients with CVD.

[Methods] We have searched in Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE (via Ovid), EMBASE (via Ovid), Healthstar (via Ovid), CINAHL, Web of Science, and ClinicalTrials.gov for randomized controlled trials (RCTs) up to December 2019. We have included RCTs involving adults with CVD who underwent dental treatments within two groups, intervention group with LA with vasoconstrictor and control group with LA without vasoconstrictor. The primary outcomes assessed were pain, bleeding and arrhythmias. The secondary outcomes were ST segment depression, anxiety, adverse effects and changes in hemodynamic parameters. The data were pooled using random effects meta-analyses and the confidence in the estimates was verified using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE).

[Results] Ten RCTs (n=478 participants) were included. Most of the them had a high risk of bias. There were more cases of pain and bleeding in the group without vasoconstrictor. The meta-analysis demonstrated no difference in the arrhythmias between the groups (RR = 2.16, 95% CI: 0.87 to 5.41). Overall, for the other outcomes assessed there was no statistical difference. The evidence was considered low according to the GRADE profile.

[Conclusions] The obtained results suggested that the use of LA with vasoconstrictor in adult patients with CVD improves control of pain, reduces bleeding and reduces systolic blood pressure after the procedure. However, the low quality of evidence demonstrates that literature needs further studies in order to confirm these results.

Register number: PROSPERO (CRD42016045421)

Keywords: Local Anesthetics, Dental Anesthesia, Dentistry, Cardiovascular Disease.

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Strengths and limitations of this study

- This review was conducted with methodological rigor, including risk assessment of bias and of quality of evidence; such information is not available in previous systematic reviews related to the subject.
- This study has explicit eligibility criteria, priority of outcomes (primary and secondary), comprehensive and extensive search in the databases, with independent and peer reviews for each study and use of GRADE to assess the quality of the evidence and the strength of recommendation for the outcomes included in the meta-analysis.
- The primary studies included were a limiting factor for the findings of this review due to methodological bias observed in the included RCTs, as well as the differences observed in the types of CVDs, types and dosages of anesthetic salts and vasoconstrictors.

Register number: PROSPERO (CRD42016045421)

Keywords: Local Anesthetics, Dental Anesthesia, Dentistry, Cardiovascular Disease.

INTRODUCTION

Cardiovascular diseases (CVD) are one of the causes of most premature deaths worldwide, representing 31% of all deaths. It is estimated that, in 2012, there were 17.5 million deaths caused by CVDs, being the leading cause of death in the world. In 2002, three-quarters of CVD deaths occurred in low- or middle-income countries (1).

CVD comprises essential hypertension, rheumatic heart diseases, ischemic heart diseases, cerebrovascular diseases, heart inflammatory diseases, and so forth (1).

In dental clinic, it is increasingly common to treat patients with complex diseases, such as CVD. These patients have access to better treatments and medications, which makes it possible to perform treatment in private dental clinics that previously was only executed by specialized hospitals (2, 3). However, careful evaluations should be conducted before the begin of dental treatment in patients with CVD, in order to control stress and mainly to block nerve

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transmission of pain, thereby minimizing transient episodes that may trigger negative cardiovascular outcomes (3).

The use of local anesthetics (LA) and analgesics can control pain, while sedatives control the anxiety during dental procedures (3, 4). The baseline of pain control for performing dental procedures is the LA (5).

Anesthetic salts are combined with vasoconstrictors to increase duration of anesthetic effect, reduce systemic toxicity, and optimize soft tissue hemostasis (6). However, despite the beneficial properties of vasoconstrictors when combined with anesthetic salts, dental professionals are concerned about the adverse effects associated with the vasoconstrictor, mainly with epinephrine, which is the most used vasoconstrictor in dentistry (2, 7).

These systemic adverse effects are linked to inadequate procedures such as intravascular injection, high-dose injections, drug interactions, or instant absorption of the anesthetic agent when it is injected into an extremely vascular area that could induce adverse cardiovascular effects, especially in patients with CVD (3, 8-10). Another factor is the concentration used in LAs. Currently, a greater dilution of the vasoconstrictor is recommended, mostly used at a concentration of 1: 100,000 for surgical procedures and, for other procedures, with a dilution of up to 1: 200,000 (2, 6). Nevertheless, for long and/or surgical procedures, LAs without vasoconstrictor do not offer a satisfactory anesthetic duration, which can cause pain resulting in systemic endogenous release of catecholamines, particularly noradrenaline (11), which can lead to negative cardiovascular outcomes (12, 13).

A previous study reported that the stress-induced release of catecholamines could be more than 10 times greater than the basal level (14). In stressful situations, such as pain and anxiety, the released endogenous catecholamines may reach concentrations higher than the low epinephrine concentrations used in dental LA (14, 15). Thus, the literature recommends (9, 13) using the lowest effective concentration of epinephrine as possible for CVD patients, both for dermatological and dental procedures. The maximum recommended dose for epinephrine is 40 µg per procedure (2, 6, 9).

Two systematic reviews analyzed the most frequent complications in patients with CVD after dental local anesthesia with vasoconstrictor, disclosed in electrocardiogram arrhythmias (12, 13). However, both reviews differed from the

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present study. Bader et al (2002), did not describe the inclusion criteria and risk of bias, included non-randomized clinical trials and a group with patients without CVD. Godzieba et al (2014) also included non-randomized clinical trials and patients without CVD.

LAs have a long history of safety and efficacy in dental patients if used in recommended doses and techniques (2, 3). However, professionals still have doubts about their use in patients with CVD, especially when combined with vasoconstrictor (13). This systematic review evaluated the safety in the use of LA with vasoconstrictor and determined the risk of cardiovascular events in patients with CVD.

METHODS AND ANALYSES

The systematic review was performed according to the recommendations specified in the Cochrane Handbook for Interventional Reviews and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses statement (PRISMA) (16).

Protocol and Registration

Our review protocol has been registered with the International Prospective Register of Systematic Reviews (PROSPERO-CRD42016045421).

Eligibility criteria

Inclusion criteria

Patients: adult patients with CVD: essential hypertension, rheumatic heart diseases, ischemic heart diseases, cerebrovascular diseases, and heart inflammatory diseases (1).

Interventions: one arm wherein patients received LAs with vasoconstrictors compared to another arm wherein patients received LAs without vasoconstrictors.

Procedures: patients who undergo tooth extraction, dental restorations, treatment and periodontal surgery, among others.

Type of study: considering that observational studies increase the risk of bias to answer this question, the study design that can generate high quality evidence, according to the evidence hierarchy, would be the randomized controlled studies

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(RCTs). We include two types of RCT designs. In the first type, patients are randomized to receive either LAs with vasoconstrictors during the first dental procedure or LAs without vasoconstrictors during the second dental procedure or vice versa. In the second type, patients are randomized to receive only one type of LA, with or without vasoconstrictors, during the dental procedure.

Language: any language, no restriction

Outcomes measures

Our primary and secondary outcomes were:

Primary outcomes:

- death;
- mortality by a specific cause;
- stroke;
- acute myocardial infarction;
- hospitalization;
- pain;
- bleeding;
- arrhythmias;

Secondary outcomes:

- ST segment depression;
- anxiety;
- adverse effects;
- changes in results obtained via oximetry;
- changes in blood pressure;
- changes in heart rate;

Exclusion criteria

We exclude studies involving patients with untreated or out-of-control essential hypertension or any type of CAD, who were pregnant or breastfeeding, who were allergic to the LAs used in the studies, with out-of-control diabetes

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mellitus, or who have had recent myocardial infarction, cancer, and malignant hypertension.

Search methods for primary studies

Electronic searches

The following electronic databases were searched: the Cochrane Central Register of Controlled Trials (CENTRAL) part of The Cochrane Library; MEDLINE (Ovid); EMBASE (Ovid); Healthstar (Ovid); CINAHL (Cumulative Index to Nursing and Allied Health Literature); and Web of Science, from their inception to January of 2020, without restrictions on the status of publication or date.

Searching other resources

We performed search in registration of clinical trials: https://clinicaltrials.gov, WHO clinical trials registry, http://www.ensaiosclinicos.gov.br; trials registry and bank of Brazil thesis (CAPES); conference proceedings of the Brazilian Congress of Cardiology, in the Brazilian Congress of Anesthesiology, and in the International Congress of Dentistry (CIOSP). Two reviewers analyzed the reference list or quotations found in secondary studies to verify and identify possible eligible studies.

Search strategy

The search strategy was individually conducted by type of dental intervention; type of anesthetic; and type of CVD. We have adapted the search strategy according to each database. The search strategy in Ovid Medline is in S1 Appendix.

Eligibility determination

Six reviewers (CCG vs CCB vs JOA and RHLM vs NKA vs JCR) working in trio independently evaluated whether summaries are in accordance with eligibility criteria. Discordances were resolved by consensus, and contentious issues were discussed with a third reviewer (LCL).

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To exclude duplicate articles, reviewers analyzed all eligible articles and identify those with one or more authors in common. In case of duplicate publications, we used the article with more complete data.

Data extraction

Six reviewers (CCG vs CCB vs JOA and RHLM vs NKA vs JCR), working in trio, independently extracted data and recorded information regarding patients, methods, interventions, outcomes, and missing outcome data using standardized and pretested data extraction forms with instructions. Before initiating data abstraction, we conducted calibration exercises to ensure consistency among the reviewers, both in screening and data extraction. We contacted the study authors as possible to resolve any uncertainties. Reviewer's resolved disagreements by discussion, and one arbitrator (LCL) judge unresolved disagreements.

Risk of bias in individual studies

Using a modified version of the Cochrane collaboration risk of bias tool (17, 18), the same pairs of reviewers independently rated the risk of bias for each RCT according to the following criteria: random sequence; allocation concealment; blinding of the patient, healthcare professionals, outcome assessors, data collectors, and data analysts; incomplete outcome data; selective outcome reporting; and major baseline imbalance. Reviewers gave response options of "definitely yes," "probably yes," "probably no," and "definitely no" for each of the domains, with the options "definitely yes" and "probably yes" ultimately being assigned a low risk of bias and "definitely no" and "probably no" as having a high risk of bias (19). Reviewer's resolved disagreements by discussion, and one arbitrator (LCL) judge unresolved disagreements. Risk of bias of RCTs included are in S2 Figure and S2 Appendix.

Explaining the heterogeneity of evidence

Possible explanations for heterogeneity included: a) age- the older the age, the higher the risk of cardiovascular transient episodes; b) gender- women outnumber men in deaths due to CVD; c) vasoconstrictor type- vasoconstrictors

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are linked to receptors α and β. However, some of these are more often linked to cardiac receptor β (except for felypressin, which links to the vasopressin receptor v1, present in the smooth muscles of blood vessel walls, raise cardiac frequency, and thus, higher risks of transient episodes are expected (20); d) vasoconstrictor concentration - which may vary from a 1:2,500 to a 1:200,000 greater risk is expected with higher vasoconstrictor concentration; e) dental procedure duration- the longer the duration to perform the procedure (surgical or periodontal procedures take longer than restorative procedures), the higher the concentration of anesthetic agent necessary, and the stronger the toxicity to the cardiovascular system, thereby increasing the risks of transient episodes in long-duration procedures; f) dental procedure type: usually surgical procedures (periodontal, extraction, and implantation) trigger great stress in the patient, thus increasing the risk of transient episodes. All to outcomes, subgroup analyses were not possible due to the low number of studies included.

We ranked heterogeneity associated with pooled effect estimates with the use of the χ^2 test and the I^2 statistic (21). The following heterogeneities were considered: 0–25% (low heterogeneity), 50% (moderate heterogeneity), and 75% (high heterogeneity) (18).

Data synthesis

Analyses were conducted for each vasoconstrictor and for each outcome of interest. Confidence in the findings was determined by estimates for each body of evidence. The studies which reported continuous outcomes presented the date in Standard Mean Difference (SMD) and 95% Confidence Interval (95% CI) and dichotomous outcomes were reported in the risk ratio (RR) was calculated combined with a 95% confidence interval (95% CI). Other details about the methods adopted for the synthesis of the data can be found in the publication of the protocol of this review (22). The random effect in meta-analysis was performed using the Review Manager 5 Software (version 5.3).

Summarizing evidence

The quality of evidence was independently evaluated (confidence in effect estimates) for each for each outcome reported using GRADE (23). Results are

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presented in evidence profiles, as recommended by the GRADE Working Group (24, 25).

In this approach, randomized trials begin with evidence of high quality may have their qualities diminished according to the judgment of one or more of the five categories of limitations presented below: risk of bias (assessed for each study as described previously), inconsistency, indirect evidence, imprecision and publication bias (23, 26). We used GRADE methodology to rate confidence in estimates of effect for each outcome as high, moderate, low or very low. Evidence profiles are provided brief presentations of evidence quality and effect magnitude. With the help of the software program GRADEpro (http://ims.cochrane.org/gradepro).

Patient and public involvement

No patient involved in this study.

RESULTS

Literature search results

From 4.048 publications retrieved from the databases, 1.924 articles remained for the titles and abstracts selection, after removing duplicates. From the 28 pre-selected papers, 18 were excluded (Appendix 3) and ten met eligibility criteria (27-36). Six studies (27, 29, 30, 32, 33, 36) were able to be included in the meta-analysis (S1 Figure).

Description of the studies (S1 Table)

Methods

Ten RCTs were included, encompassing a total of 478 patients with CVD, from which 226 patients had coronary artery disease, 226 patients had essential hypertension, 59 patients had valvular diseases, 33 patients had Chagas disease, and nine patients had unspecified cardiovascular disease. S1 Table describes the characteristics of the studies, where it was observed that the majority of the studies used lidocaine as a LA, and epinephrine as a vasoconstrictor. The studies were mainly published in Brazil, between 2007 and 2014. The reasons for exclusion of articles are found in S3 Appendix.

Intervention

Eight studies (28, 30-36) used the same anesthetic salt both in the control and intervention groups; two studies (27, 36) included more than one group in the intervention group; five studies (31-35) used anesthetic salt lidocaine both in the intervention and control groups; one study (28) used anesthetic salt prilocaine both in the intervention and control groups; two studies (30, 36) used anesthetic salt mepivacaine both in the intervention and control groups; and three studies (27, 29, 36) used different anesthetic salts between the intervention and control groups.

Eight studies (27, 30-36) used the vasoconstrictor epinephrine; followed by felypressin, present in three studies (27-29) and only one study (27) used both epinephrine and felypressin (S1 Table).

Participants

Five studies (27, 28, 31, 34, 36) included patients with essential hypertension. Two studies (30, 35) included patients with coronary artery disease. Two studies (31, 33) included patients with essential hypertension and coronary artery disease. One study (32) included patients with valvular diseases, and one study (29) included patients with Chagas disease and coronary artery disease.

Risk of bias in the included studies (S2 Appendix and S2 Figure)

Randomization and allocation

From the ten RCTs included, three (28, 31, 36) used the crossover design, what reduces the quality of the evidence, the studies performed wash-out periods between 10 minutes to 3 weeks. Randomization by computer was performed by only one study (36). Two studies (31, 32) used the random number table. Two studies (27, 30) did a draw of the ID cards. Five studies (28, 29, 33-35) did not mention the randomization method.

Blinding

Two study (32, 35) reported performing double-blinding, but only one study (32) partially described. One study (35) only mentioned that the blinding was

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performed, without an explanation of the method. Five studies (27-29, 33, 34) were not conducted blinded.

Only two studies (30, 32) used allocation concealment with delivered in sealed envelopes, and the cartridges were covered with a white strip.

The allocation concealment method was considered of low risk of bias in one study (36), when randomization was performed by computer.

Loss to follow-up

The studies reported no loss to follow-up.

Selective reporting of outcomes

Only three studies (27, 32, 35) had a registry of clinical trial reporting pre-specified outcomes. Two studies (32, 35) made the registry at ClinicalTrials.gov and one study (27) at the Institute Research Center.

Other potential source of bias

Four studies (27-29, 32) mentioned that there was no external funding. One study (35) received Brazilian funding through the São Paulo State Research Support Foundation (FAPESP), which is non-profit.

Six studies (27-30, 32, 33) declared no potential conflicts of interest. One study (32) mentioned both no external funding sources and no potential conflict of interest only in the registry of clinical trials

Reported primary outcomes

None of the studies reported death, mortality by specific cause, stroke, acute myocardial infarction or hospitalization. The only primary outcomes reported by the authors included in this review were pain, bleeding and arrhythmias (S1 Table). Due to the small number of studies and the lack of numerical data, it was not possible to perform the data meta-analysis of pain and bleeding. The reports were descriptive and qualitative.

Pain

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Two studies (30, 32) evaluated the pain outcome. Conrado et al. (2007) assessed the pain outcome at the extraction site in 54 patients with coronary disease and observed pain report in only one patient from the intervention group (mepivacaine 2% with epinephrine 1:100,000), and in 18 patients from the control group (mepivacaine 3% without vasoconstrictor). The authors did not describe how they have assessed pain and explained that the only case of pain in the intervention group was caused by a tooth that was ankylosed. Laragnoit, et al. (2009) examined 59 participants with several valvular diseases and described the results only in a qualitative way, reporting that there were more complaints of pain in the control group (lidocaine 2% without vasoconstrictor) compared to the intervention group (lidocaine 2% with epinephrine 1:100,000). The authors did not describe how they assessed pain.

Bleeding

Two studies (29, 31) evaluated the bleeding outcome. Caceres, et al. (2008) did not note bleeding reports during the procedures. In both groups, authors did not describe how they assessed this outcome. Davenport, et al. (1990) examined nine participants in the control and intervention groups and observed that all participants in the intervention group (lidocaine 2% with epinephrine 1:100,000) obtained adequate hemostasis versus seven participants with inadequate hemostasis in the control group (lidocaine 2% without vasoconstrictor). The authors did not describe how they assessed bleeding.

Arrhythmia

Three studies (29, 32, 33) assessed this outcome. Caceres et al. (2008) analyzed the presence of ventricular extrasystoles that remained stable in different measured moments, with a similar behavior in both groups.

Laragnoit, et al. (2009) reported eight patients with arrhythmia, being five patients from the intervention group (lidocaine 2% with epinephrine 1:100,000) and three patients in the control group (lidocaine 2%), with no observed statistical difference between both groups. Neves, et al. (2007) reported seven patients with arrhythmia, being three (11.1%) from the intervention group (lidocaine 2% with epinephrine 1:100,000) and four (13.8%) from the control group (lidocaine 2% without vasoconstrictor), with no observed significant statistical difference

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between the two groups. The meta-analysis demonstrated no difference in the arrhythmias between the groups (RR = 2.16, 95% CI: 0.87 to 5.41 low quality evidence) (S3 Figure, S2 Table).

Reported secondary outcomes

Adverse effects were not reported in the studies.

ST segment depression

Two studies (30, 33) evaluated the presence of ischemic episodes (measured as ST segment depression). Conrado, et al. (2007) observed the occurrence of the ST segment depression greater than 1mm in three patients in the intervention group (mepivacaine 2% with epinephrine 1:100,000) during anesthetic injection. No event was observed in the control group (mepivacaine 3% without vasoconstrictor) and no statistical difference was observed between the groups. Neves et al. (2007) observed ST segment depression greater than 1mm in four patients (14.8%) in the intervention group (lidocaine 2% with epinephrine 1:100,000) and in six patients (20.7%) in the control group (lidocaine 2% without vasoconstrictor), with no significant statistical difference between the groups. The meta-analysis demonstrated no difference in the ST segment depression between the groups (RR: 1.53, 95% CI: 0.17 to 13.72 low quality evidence) (S4 Figure, S2 Table).

Anxiety

Two studies (32, 35) evaluated the anxiety outcome. Laragnoit et al. (2009) assessed the anxiety of patients using the STAI scale (Spielberger State Anxiety Invetory) before the procedure and reported no difference in the mean values of anxiety levels between the intervention and control groups. Santos-Paul et al. (2015) examined patients using the Buchanan and Niven method, one hour before the procedure, five minutes after the procedure and one hour after the procedure. They reported no significant difference between the control and intervention groups at any of the evaluated time periods.

Mean blood pressure

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Four studies (29, 31, 32, 36) evaluated the mean blood pressure values. Two studies (29, 31) mention their results only in graphs and it was not possible to perform meta-analysis. Torres-Lagares et al. (2012) reported increase in the average of mean blood pressure in the intervention group (mepivacaine 2% with epinephrine 1:100,000) in comparison to the other three intervention groups (articaine 4% with epinephrine 1:200,000, articaine 4% with epinephrine 1:100,000, mepivacaine 3% without vasoconstrictor).

Meta-analysis was conducted with the Laragnoit et al. (2009) and Torres-Lagares et al. (2012) studies and showed no difference in the mean blood pressure between the groups during the procedure period (SMD = -0.05, 95% CI: -0.93 to 0.82, very low-quality evidence), and after the procedure period (SMD = -0.26, 95% CI: -0.71 to 0.18, low quality evidence). Time periods have been adapted to carry out the meta-analysis (S5 and S6 Figure, S2 Table)

Systolic blood pressure

Eight studies (27, 28, 30, 32-36) evaluated this outcome.

In five (28, 30, 32, 34, 35) there were no significant statistical difference between the assessed groups. In Abu-Mostafa et al. (2015) study there was a significant increase in the systolic blood pressure after dental extraction in the intervention group (lidocaine 2% with epinephrine 1:80,000) compared to the control group (mepivacaine 3% without vasoconstrictor), during after the procedure period.

Five studies (28, 30, 32, 34, 35) could not be included in the meta-analysis because their data were in graphs, even after attempts to contact the authors and to extract the data using the DigitizeIt software.

Meta-analysis showed no difference in systolic blood pressure while using anesthetic with vasoconstrictor compared to using only anesthetic without vasoconstrictor during the procedure period (SMD = -0.39, 95% CI: -0.80 to 0.03, very low-quality evidence) (S7 Figure, S2 Table). In after the procedure period, meta-analysis demonstrated a decrease in the systolic blood pressure with the use of anesthetic with vasoconstrictor (SMD = -0.95, 95% CI: -1.35 to -0.55, very low-quality evidence). The time periods have been adapted to carry out the meta-analysis (S8 Figure, S2 Table).

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Diastolic blood pressure

Eight studies (27, 28, 30, 32-36) evaluated the average diastolic blood pressure. In five studies (28, 30, 32, 34, 35) there were no significant statistical difference between the assessed groups and they could not be included in meta-analysis because their data were in graphs, even after attempts to contact the authors and to extract the data using the Digitizelt software.

In Bronzo et al. (2012), there was an increase in diastolic blood pressure during the dental procedures in both groups. However, in the intervention group (prilocaine 2% with felypressin 0,03 IU) diastolic blood pressure was greater than in the control group (prilocaine 4% without vasoconstrictor). Torres-Lagares et al. (2012) reported diastolic blood pressure values significantly higher in the intervention group (mepivacaine 2% with epinephrine 1:100.000) in comparison to the other two groups intervention groups (articaine 4% with epinephrine 1:200,000; articaine 4% with epinephrine 1:100,000) and control group (mepivacaine 3% without vasoconstrictor).

Meta-analysis showed no statistical difference between the diastolic blood pressure during the procedure (SMD = -0.51, 95% CI: -1.07 to 0.06) and after the procedure periods (SMD = -0.53, 95% CI: -1.06 to 0.01).

However, it was noted that there was a tendency, with no statistical significance, in favor of the use of LA with vasoconstrictor (S9 and S10 Figure). The quality of evidence was considered very low for both during and after the procedure periods. The time periods have been adapted to carry out the meta-analysis (S2 Table).

Heart rate

Nine studies (27, 29-36) assessed this outcome and in any of them significant difference between the groups was found. Six studies (29-32, 34, 35) could not be included in the meta-analysis, because their data were in graphs, even after attempts to contact the authors and to extract the data using the Digitizelt software. Only three (27, 33, 36) could be included in meta-analysis, from which numerical data was available. Meta-analysis showed no statistical difference for the periods during (SMD = -0.24, 95% CI: -0.61 to 0.13) after the procedure (SMD = -0.20, 95% CI: -0.57 to 0.17) for heart rate between the groups

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(S11 and S12 Figure), being the evidence considered of very low quality. The time periods have been adapted to carry out the meta-analysis (S2 Table).

DISCUSSION

Discussion of main findings with the literature

There was more pain and bleeding in the control group (without vasoconstrictor) when compared to the intervention group (with vasoconstrictor), but neither outcome could be included in the meta-analysis due to the lack of descriptive data. In the meta-analysis, there were only significant differences for the systolic blood pressure result in the period after the procedure, in favor of anesthetic with vasoconstrictor, where the systolic blood pressure means were lower in the intervention group compared to the control group.

In this study, no significant differences were observed comparing LA with and without the use of vasoconstrictor for the outcomes anxiety, arrhythmia, ST segment depression, oxygen saturation and hemodynamic outcomes.

Sensitivity analysis was not performed because important heterogeneity related to problems in the design of the included studies was not detected.

Considering previous systematic reviews found in literature related to this subject (12, 13), methodological differences and in the proposed objectives, as well as the absence of meta-analysis, did not allow the direct comparison of their results with the present study.

Bader et al. (2002) evaluated the cardiovascular adverse effects risk in the use of LA with epinephrine and in the use of epinephrine-impregnated retraction cord as well. The authors included six studies with a total of 325 patients, being 177 patients with essential hypertension. Four of their studies included healthy patients and the assessed outcomes were systolic and diastolic blood pressure, and heart rate. In two groups, the first one of hypertensive patients using LA with vasoconstrictor and the second one of healthy patients using LA with vasoconstrictor and LA without vasoconstrictor, despite presenting the results in a table, it has not been reported whether there was any significant statistical difference between the groups. The studies with the epinephrine-impregnated retraction cord were not included because they have not found studies with this inclusion criterion. As a conclusion, the authors reported that the adverse effects

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in patients with essential hypertension are uncommon and that the only risk are the small hemodynamic changes. They have enhanced that the methods employed in the studies are not efficient to assess the risks of adverse cardiovascular outcomes associated with the use of epinephrine.

Godzieba et al. (2014) assessed the limited use for LA with vasoconstrictor in patients with CVD. The authors included 11 studies, being six RCTs and five clinical trials, with 567 patients. Five studies were the same as the ones we included in this systematic review; two included studies had no comparator group; one study included healthy patients, a non-randomized trial; one study comparing two different types of LA with vasoconstrictors; and one study comparing two groups with the same LA with vasoconstrictor, although in the first group without the use of midazolam and in the second one with the use of midazolam. There was no description of the assessed outcomes, they just reported that, among 560 patients, there were 87 complications after the anesthetic administration, being arrhythmia the most frequent one, followed by asymptomatic ischemic changes in three patients, and one case of transient local neuropathy; no adverse clinical effects were reported. The authors concluded that the use of up to four tubes of lidocaine with epinephrine 1:100,000 appears to be relatively safe with CVD.

Strengths and limitations of this study

This study was conducted with methodological rigor, including risk assessment of bias and of quality of evidence; such information is not available in reviews related to the subject (12, 13). Furthermore, the study has explicit eligibility criteria, priority of outcomes (primary and secondary), comprehensive and extensive search in the databases, with independent and peer reviews for each study and use of GRADE to assess the quality of the evidence and the strength of recommendation for the outcomes included in the meta-analysis.

The primary studies included were a limiting factor for the findings of this review due to methodological bias observed in the included RCTs, as well as the differences observed in the types of CVDs, types and dosages of anesthetic salts and vasoconstrictors. Thus, it was noted a need for more primary studies about the subject, with better methodological accuracy and prioritization of outcomes in order to increase confidence in findings.

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Implications for clinical practice and research

The LA associated with vasoconstrictor is essential in dental treatment because vasoconstrictor agents counteract the vasodilating action of anesthetic salts, increasing anesthetic time and reducing bleeding (6, 8). The findings of this systematic review showed that there was no statistical difference between the use of LA with vasoconstrictor and without vasoconstrictor in patients with CVDs for the assessed outcomes. However, the low quality of evidence demonstrates that literature needs further studies in order to confirm these results.

Thus, new RCTs with higher methodological rigor are needed to obtain more reliable results, in order to verify the safety of LAs with vasoconstrictor in patients with CVDs. It is also suggested that the professionals use the recommended doses as well as perform the procedure respecting the anesthetic techniques (2, 6, 9). Moreover, communication with the patient's cardiologist can help clarify the patient's clinical condition, to evaluate the care and limitations of the safest dental treatment for each patient (9).

Conclusion

Based on the studies included in this review, the use of LA with vasoconstrictor in adult patients with CVD improves control of pain, reduces bleeding and reduces systolic blood pressure after procedures when compared to the use of LA without vasoconstrictor. Furthermore, the use of the vasoconstrictor does not alter the risk of arrhythmia, ST segment depression, mean blood pressure, systolic and diastolic blood pressure and heart rate, demonstrating safety in its use for these outcomes. However, the low quality of evidence demonstrates that literature needs further studies in order to confirm these results, caution is recommended in these findings.

Contributors

CCG is the principal investigator and led the writing of the manuscript. LCL and RHLM are the project managers and co-investigators and contributed to the writing and revision of the manuscript. CCG, CCB, RHLM, JCR, JOA, NKA are co-investigators who contributed to screening the eligible studies and extracted the data of the studies, also participated the revision of the manuscript. MTS

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performed the statistical analysis, also participated the revision of the manuscript. All authors read and approved the final manuscript.

Competing interests

The authors declare that they have no competing interests.

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Provenance and peer review

Not commissioned, externally peer reviewed.

Abbreviations

Local anesthetic (LA), cardiovascular disease (CVD), Cochrane Central Register of Controlled Trials (CENTRAL), Medical literature Analysis and Retrieval System Online (MEDLINE), Excerpta Medica dataBASE (EMBASE), Cumulative Index to Nursing and Allied Health Literature (CINAHL), Grading of Recommendations Assessment, Development, and Evaluation (GRADE), randomized clinical trial (RCT), bank of Brazil thesis (CAPES), Congresso Internacional de Odontologia de São Paulo (CIOSP), confidence interval (CI), weighted mean difference (WMD), minimally important difference (MID), and standardized mean difference (SMD), standard deviation (SD).

References

1. World Health Organization.WHO. Cardiovascular diseases (CVDs). WHO [Internet]. 2017 2017-05-1714:41:03. Available from: http://www.who.int/mediacentre/factsheets/fs317/en/.

2. Becker DE, Reed KL. Local anesthetics: review of pharmacological considerations. Anesth Prog. 2012;59(2):90-101; quiz 2-3. doi: 10.2344/0003-3006-59.2.90.

3. Jowett NI, Cabot LB. Patients with cardiac disease: considerations for the dental practitioner. Br Dent J. 2000;189(6):297-302.

For peer review only

4. American Dental Association A. Guideline for teaching pain control and sedation to dentist and dental students. 2012:1-18.

5. St George G, Morgan A, Meechan J, Moles DR, Needleman I, Ng YL, et al. Injectable local anaesthetic agents for dental anaesthesia. Cochrane Database Syst Rev. 2018;7(7):Cd006487.

6. Haas DA. An update on local anesthetics in dentistry. J Can Dent Assoc. 2002;68(9):546-51.

7. Jage J. Circulatory effects of vasoconstrictors combined with local anesthetics. Anesth Pain Control Dent. 1993;2(2):81-6.

8. Fencl JL. Guideline implementation: local anesthesia. AORN J. 2015;101(6):682-9; quiz 90-2.

9. Kouba DJ, LoPiccolo MC, Alam M, Bordeaux JS, Cohen B, Hanke CW, et al. Guidelines for the use of local anesthesia in office-based dermatologic surgery. J Am Acad Dermatol. 2016;74(6):1201-19.

10. Wadlund DL. Local Anesthetic Systemic Toxicity. Aorn j. 2017;106(5):367-77.

11. Brown RS. Local anesthetics. Dent Clin North Am. 1994;38(4):619-32.

12. Bader JD, Bonito AJ, Shugars DA. A systematic review of cardiovascular effects of epinephrine on hypertensive dental patients. Oral Surg Oral Med Oral Pathol Oral Radiol Endod. 2002;93(6):647-53.

13. Godzieba A, Smektala T, Jedrzejewski M, Sporniak-Tutak K. Clinical assessment of the safe use local anaesthesia with vasoconstrictor agents in cardiovascular compromised patients: a systematic review. Med Sci Monit. 2014;20:393-8.

14. Brand HS, Gortzak RA, Palmer-Bouva CC, Abraham RE, Abraham-Inpijn L. Cardiovascular and neuroendocrine responses during acute stress induced by different types of dental treatment. Int Dent J. 1995;45(1):45-8.

15. Meyer FU. Haemodynamic changes under emotional stress following a minor surgical procedure under local anaesthesia. Int J Oral Maxillofac Surg. 1987;16(6):688-94.

16. Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. PLoS Med. 2009;6(7):e1000097.

For peer review only

17. Altman R, Alarcon G, Appelrouth D, Bloch D, Borenstein D, Brandt K, et al. The American College of Rheumatology criteria for the classification and reporting of osteoarthritis of the hand. Arthritis Rheum. 1990;33(11):1601-10.

18. Higgins JP, Green S. Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0 (updated March 2011). The Cochrane Collaboration 2011.

19. Akl EA, Sun X, Busse JW, Johnston BC, Briel M, Mulla S, et al. Specific instructions for estimating unclearly reported blinding status in randomized trials were reliable and valid. J Clin Epidemiol. 2012;65(3):262-7.

20. Manning M, Stoev S, Chini B, Durroux T, Mouillac B, Guillon G. Peptide and non-peptide agonists and antagonists for the vasopressin and oxytocin V1a, V1b, V2 and OT receptors: research tools and potential therapeutic agents. Prog Brain Res [Internet]. 2008; 170:[473-512 pp.].

21. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. Stat Med. 2002;21(11):1539-58.

22. Guimaraes CC, Lopes Motta RH, Bergamaschi CC, Araujo JO, Andrade NK, Figueiro MF, et al. Local anaesthetics combined with vasoconstrictors in patients with cardiovascular disease undergoing dental procedures: systematic review and meta-analysis protocol. BMJ Open. 2017;7(11):e014611.

23. Guyatt GH, Oxman AD, Kunz R, Woodcock J, Brozek J, Helfand M, et al. GRADE guidelines: 7. Rating the quality of evidence--inconsistency. J Clin Epidemiol. 2011;64(12):1294-302.

24. Guyatt GH, Oxman AD, Santesso N, Helfand M, Vist G, Kunz R, et al. GRADE guidelines: 12. Preparing summary of findings tables-binary outcomes. J Clin Epidemiol. 2013;66(2):158-72.

25. Guyatt GH, Thorlund K, Oxman AD, Walter SD, Patrick D, Furukawa TA, et al. GRADE guidelines: 13. Preparing summary of findings tables and evidence profiles-continuous outcomes. J Clin Epidemiol. 2013;66(2):173-83.

26. Guyatt GH, Oxman AD, Montori V, Vist G, Kunz R, Brozek J, et al. GRADE guidelines: 5. Rating the quality of evidence--publication bias. J Clin Epidemiol. 2011;64(12):1277-82.

27. Abu-Mostafa N, Aldawssary A, Assari A, Alnujaidy S, Almutlaq A. A prospective randomized clinical trial compared the effect of various types of local

For peer review only

anesthetics cartridges on hypertensive patients during dental extraction. Journal of clinical and experimental dentistry. 2015;7(1):e84-8.

28. Bronzo AL, Cardoso CG, Jr., Ortega KC, Mion D, Jr. Felypressin increases blood pressure during dental procedures in hypertensive patients. Arquivos brasileiros de cardiologia. 2012;99(2):724-31.

29. Caceres MT, Ludovice AC, Brito FS, Darrieux FC, Neves RS, Scanavacca MI, et al. Effect of local anesthetics with and without vasoconstrictor agent in patients with ventricular arrhythmias. Arquivos brasileiros de cardiologia. 2008;91(3):128-33, 42-7.

30. Conrado VC, de Andrade J, de Angelis GA, de Andrade AC, Timerman L, Andrade MM, et al. Cardiovascular effects of local anesthesia with vasoconstrictor during dental extraction in coronary patients. Arquivos brasileiros de cardiologia. 2007;88(5):507-13.

31. Davenport RE, Porcelli RJ, Iacono VJ, Bonura CF, Mallis GI, Baer PN. Effects of anesthetics containing epinephrine on catecholamine levels during periodontal surgery. Journal of periodontology. 1990;61(9):553-8.

32. Laragnoit AB, Neves RS, Neves IL, Vieira JE. Locoregional anesthesia for dental treatment in cardiac patients: a comparative study of 2% plain lidocaine and 2% lidocaine with epinephrine (1:100,000). Clinics (Sao Paulo, Brazil). 2009;64(3):177-82.

33. Neves RS, Neves IL, Giorgi DM, Grupi CJ, Cesar LA, Hueb W, et al. Effects of epinephrine in local dental anesthesia in patients with coronary artery disease. Arquivos brasileiros de cardiologia. 2007;88(5):545-51.

34. Ogunlewe MO, James O, Ajuluchukwu JN, Ladeinde AL, Adeyemo WL, Gbotolorun OM. Evaluation of haemodynamic changes in hypertensive patients during tooth extraction under local anaesthesia. The West Indian medical journal. 2011;60(1):91-5.

35. Santos-Paul MA, Neves IL, Neves RS, Ramires JA. Local anesthesia with epinephrine is safe and effective for oral surgery in patients with type 2 diabetes mellitus and coronary disease: a prospective randomized study. Clinics (Sao Paulo, Brazil). 2015;70(3):185-9.

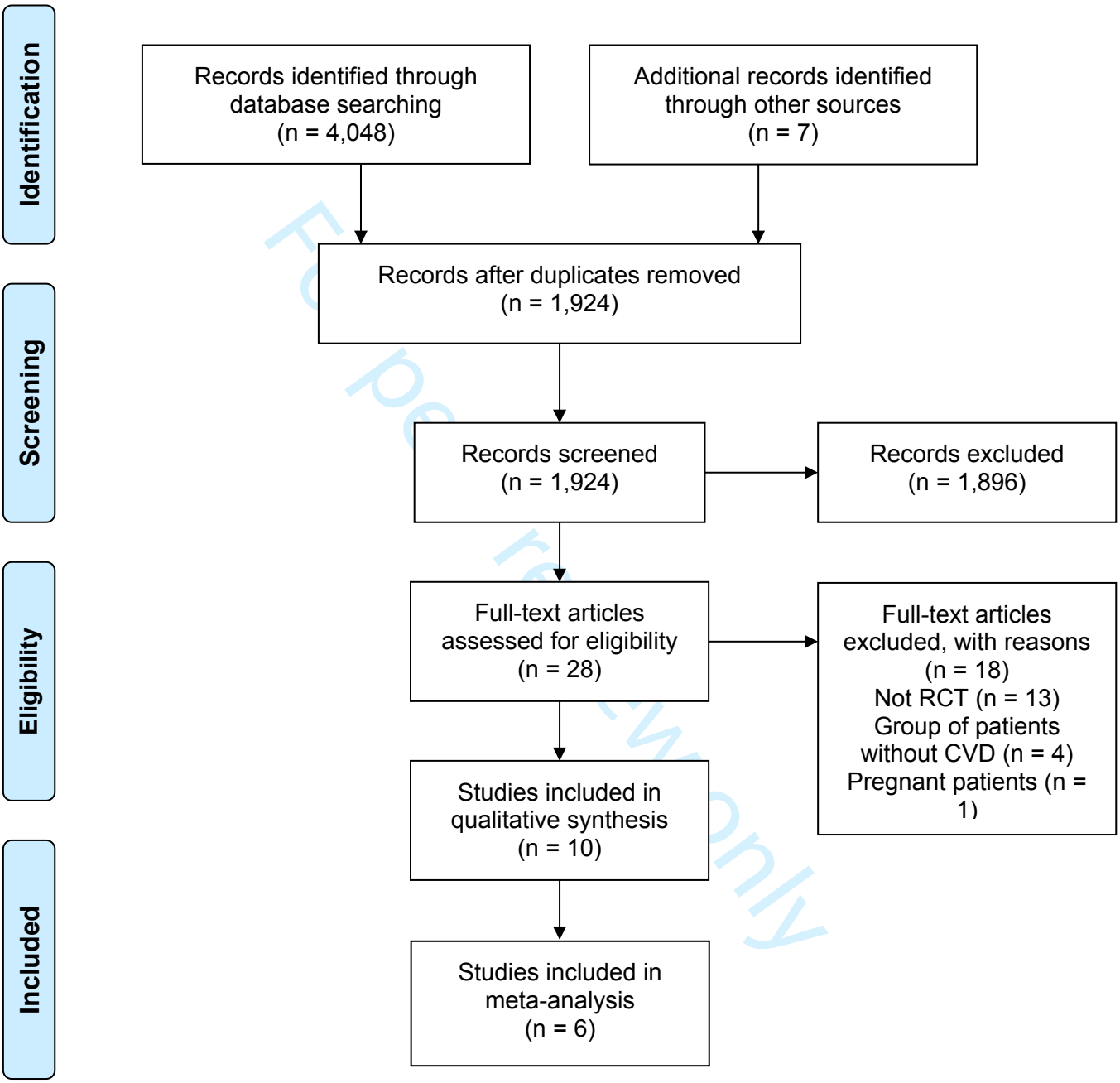
36. Torres-Lagares D, Serrera-Figallo MA, Machuca-Portillo G, Corcuera-Flores JR, Machuca-Portillo C, Castillo-Oyague R, et al. Cardiovascular effect of dental anesthesia with articaine (40 mg with epinefrine 0,5 mg % and 40 mg with

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epinefrine 1 mg%) versus mepivacaine (30 mg and 20 mg with epinefrine 1 mg%) in medically compromised cardiac patients: a cross-over, randomized, single blinded study. Medicina oral, patologia oral y cirugia bucal. 2012;17(4):e655-60.

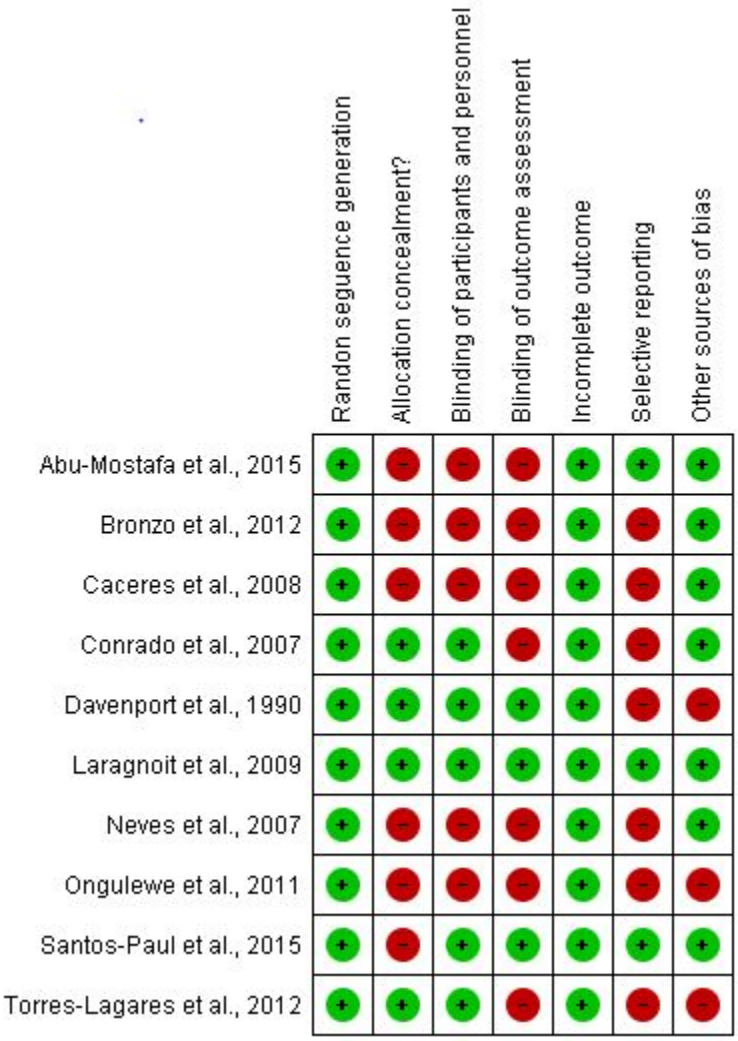
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S1 Figure. Flowchart for literature search and study selection



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S2 Figure. Risk of bias summary: review authors' judgments



S1 Table. Characteristic of included studies

STUDY	Population	Sample size (N)	Country	Female (n)	Mean age (y)	Local anesthetic + vasoconstrictor	Anesthetic without vasoconstrictor	Procedures	Outcomes measured
Abu-mostafa et al., 2015	Essential hypertension	45	Saudia Arabia	14	55.66±NR	Lidocaine 2% (Epinephrine$) Prilocaine 3% (Felypressin&)	Mepivacaine 3%	Dental extraction	Systolic and diastolic blood pressure, and heart rate
Bronzo et al., 2012	Essential hypertension	71	Brazil	44	48±12	Prilocaine 3% (Felypressin&)	Prilocaine 4%	Periodontal treatment	Anxiety, systolic, and diastolic blood pressure
Caceres et al., 2008	Chagas'disease, coronary artery disease	65	Brazil	21	54.73± 7.94	Prilocaine 3% (Felypressin&)	Lidocaine 2%	Dental procedures	Bleeding, arrhythmia, mean blood pressure, and heart rate
Conrado et al., 2007	Coronary artery disease	54	Brazil	20	56.65±8.27	Mepivacaine 2% (Epinephrine$)	Mepivacaine 3%	Dental extraction	Pain, ST segment depression, systolic and diastolic blood pressure and heart rate
Davenport et al., 1990	Essential hypertension, coronary artery disease	9	US	0	62±NR	Lidocaine 2% (Epinephrine$)	Lidocaine 2%	Periodontal surgery	Bleeding, mean blood pressure, heart rate and plasma epinephrine, and norepinephrine levels
Laragnoit et al., 2009	Valvular disease	59	Brazil	NR	41.25±10.6	Lidocaine 2% (Epinephrine$)	Lidocaine 2%	Dental extraction or restorative procedure	Pain, arrhythmia, anxiety, mean blood pressure, and heart rate
Neves et al., 2007	Coronary artery disease and essential hypertension	62	Brazil	11	58.7±8.8	Lidocaine 2% (Epinephrine$)	Lidocaine 2%	Restorative procedure	Arrhythmia, ST segment depression, systolic and diastolic blood pressure and heart rate
Ongulewe et al., 2011	Essential hypertension	33	Nigeria	20	50±11.7	Lidocaine 2% (Epinephrine@)	Lidocaine 2%	Dental extraction	Systolic and diastolic blood pressure and heart rate
Santos-paul et al., 2015	Coronary disease	70	Brazil	20	63.4±8.3	Lidocaine 2% (Epinephrine$)	Lidocaine 2%	Dental extraction	Anxiety, systolic and diastolic blood pressure, heart rate, and blood glucose levels
Torres-lagares et al., 2012	Essential hypertension	10	Spain	NR	63±12,5	Articaine 4% (Epinephrine$, Epinephrine#) Mepivacaine 2% (Epinephrine$)	Mepivacaine 3%	Periodontal treatment	Mean blood pressure, systolic and diastolic blood pressure, heart rate, and oximetry

NR: not reported

Dose of Epinephrine: $ 1:100.000, # 1:200.000, @ 1:80.000. Dosage of Felypressin: & 0.03 IU/ml

S2 Table. - GRADE Assessment of quality of evidence

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Local Anesthesic with vasoconstrictor	Local Anesthesic without vasoconstrictor	Relative (95% CI)	Absolute (95% CI)		
Mean blood pressure during the procedure (assessed with: mmHg)												
1	randomised trials	serious [a]	not serious	serious [b]	serious [c]	none	10	10	-	SMD **0.05 SD lower** (0.93 lower to 0.82 higher)	⨁◯◯◯ VERY LOW	IMPORTANT
Systolic Blood Pressure during the procedure (assessed with: mmHg)												
3	randomised trials	very serious [d]	not serious	serious [b]	not serious	none	55	57	-	SMD **0.39 SD lower** (0.8 lower to 0.03 higher)	⨁◯◯◯ VERY LOW	NOT IMPORTANT
Diastolic Blood Pressure during the procedure (assessed with: mmHg)												
3	randomised trials	very serious [d]	serious [e]	serious [b]	not serious	none	55	57	-	SMD **0.51 SD lower** (1.07 lower to 0.06 higher)	⨁◯◯◯ VERY LOW	IMPORTANT
Heart Rate during the procedure (assessed with: bpm)												
3	randomised trials	very serious [d]	not serious	serious [f]	not serious	none	55	57	-	SMD **0.24 SD lower** (0.61 lower to 0.13 higher)	⨁◯◯◯ VERY LOW	NOT IMPORTANT
Mean blood Pressure after the procedure (assessed with: mmHg)												
2	randomised trials	serious [g]	not serious	serious [b]	not serious	none	38	41	-	SMD **0.26 SD lower** (0.71 lower to 0.18 higher)	⨁⨁◯◯ LOW	IMPORTANT
Systolic Blood Pressure after the procedure (assessed with: mmHg)												
3	randomised trials	very serious [h]	not serious	serious [b]	not serious	none	53	56	-	SMD **0.95 SD lower** (1.35 lower to 0.55 lower)	⨁◯◯◯ VERY LOW	NOT IMPORTANT
Diastolic Blood Pressure after the procedure (assessed with: mmHg)												
3	randomised trials	very serious [h]	serious [e]	serious [b]	not serious	none	53	56	-	SMD **0.53 SD lower** (1.06 lower to 0.01 higher)	⨁◯◯◯ VERY LOW	IMPORTANT
Heart Rate after the procedure (assessed with: HR)												

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Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Local Anesthesic with vasoconstrictor	Local Anesthesic without vasoconstrictor	Relative (95% CI)	Absolute (95% CI)		
3	randomised trials	very serious [i]	not serious	serious [f]	not serious	none	55	57	-	SMD **0.2 SD lower** (0.57 lower to 0.17 higher)	⨁◯◯◯ VERY LOW	NOT IMPORTANT
Arrhythmia												
3	randomised trials	serious [i]	not serious	serious [j]	serious [c]	none	12/94 (12.8%)	6/92 (6.5%)	**RR 2.16** (0.87 to 5.41)	**76 more per 1.000** (from 8 fewer to 288 more)	⨁◯◯◯ VERY LOW	NÃO IMPORTANTE
ST segment depression												
2	randomised trials	serious [k]	serious [e]	serious [l]	serious [m]	none	7/57 (12.3%)	6/59 (10.2%)	**RR 1.53** (0.17 to 13.72)	**54 more per 1.000** (from 84 fewer to 1.000 more)	⨁◯◯◯ VERY LOW	NÃO IMPORTANTE

CI: Confidence interval; **SMD:** Standardised mean difference; **RR:** Risk ratio

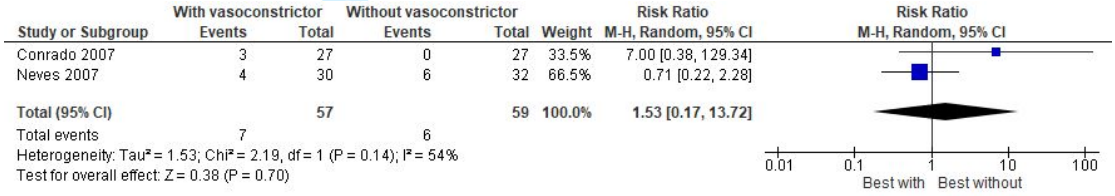
Explanations
a. The study was not blinding for outcome assessment. The protocol of the study was not registered previously.
b. Blood Pressor is a surrogate outcome.
c. Wide confidence interval considering the size of the generated effect
d. Three studies not blinded also alocation not concealment without register of protocols.
e. Moderate heterogeneity without explanation of differences between studies
f. Heart Rate is a surrogate outcome
g. One study was not blinded for outcome assessment.
h. Two studies from three of them were not blinded.
i. Three studies without blinding outcome assessment. Two of them not clear to allocation concealment and two without any previous register of protocol.
j. arrhythmias is a surrogate outcome
k. One study was not blinded for outcome assessment, and two without previous register of protocol.
l. ST segment depression is a surrogate outcome
m. one study with wide confidence interval consideringthe size of the generated effect

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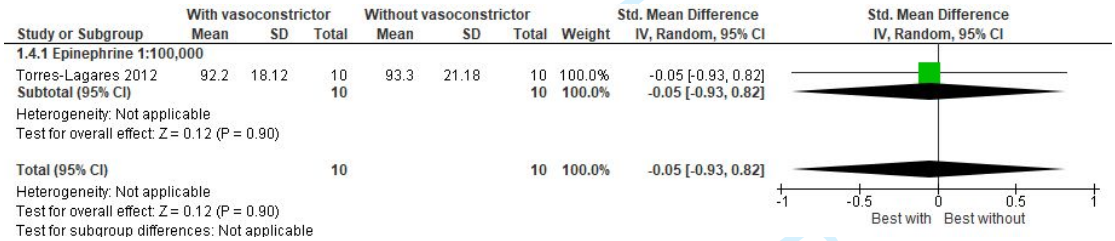
S3 Figure. Presence of arrhythmia associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor.



S4 Figure. Presence of ST segment depression associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor.



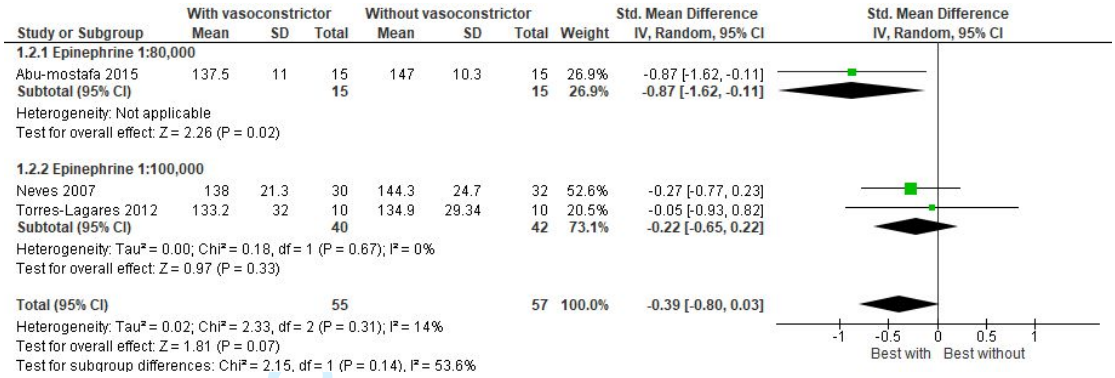
S5 Figure. Mean blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, during the procedure period.



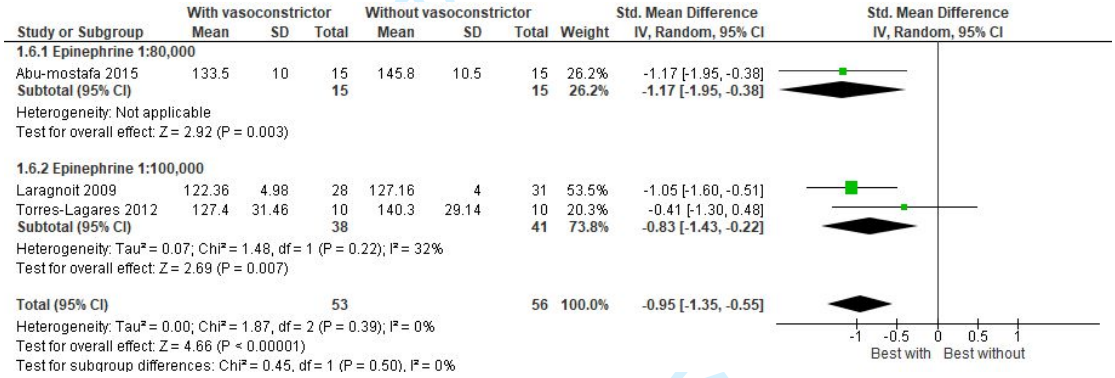
S6 Figure. Mean blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, after the procedure period.



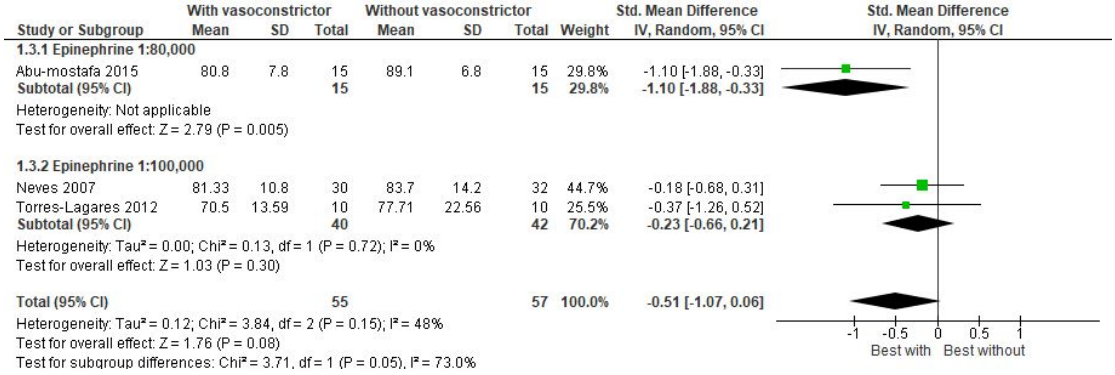
S7 Figure. Systolic blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, during the procedure period.



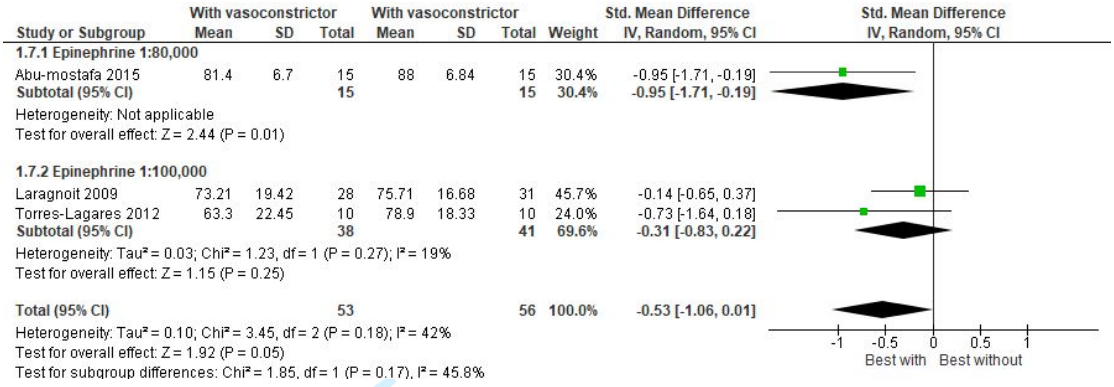
S8 Figure. Systolic blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, after procedure period.



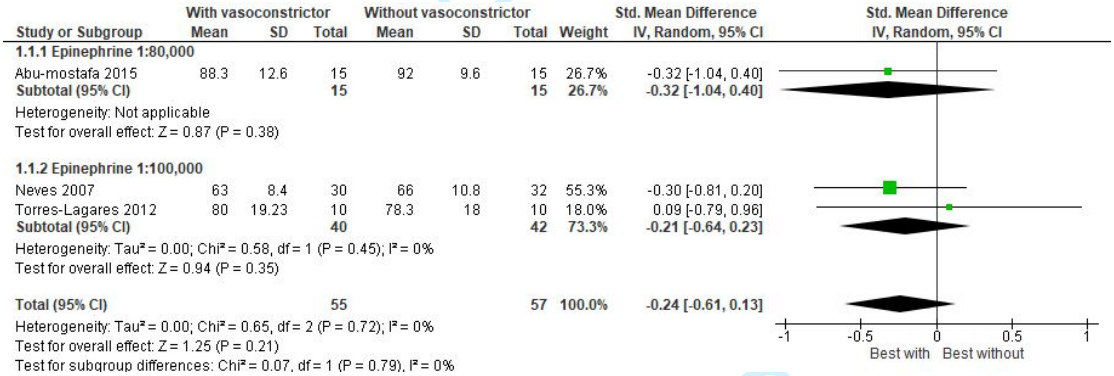
S9 Figure. Diastolic blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, during the procedure period.



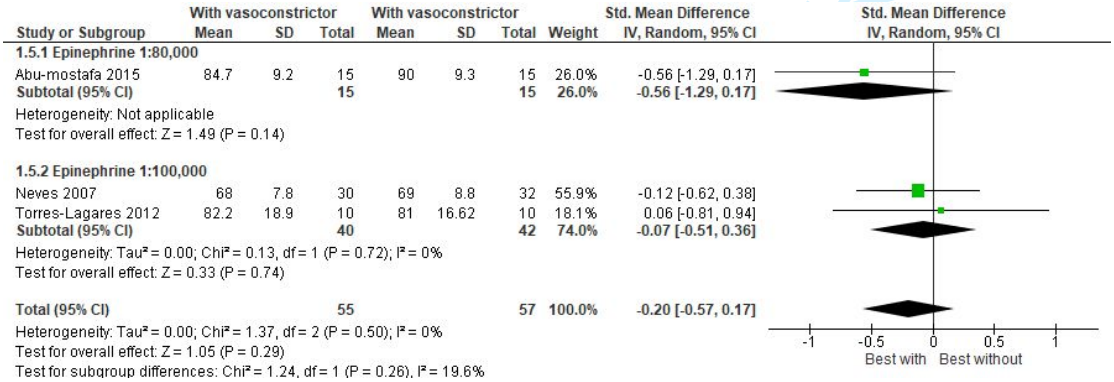
S10 Figure. Diastolic blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, after the procedure period.



S11 Figure. Heart rate associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, during the procedure period.



S12 Figure. Heart rate associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, after the procedure period.



S1 Appendix– Search strategy (via OVID, MEDLINE and EMBASE)

1 exp Dentistry/ (357819)
2 exp Dentistry, Operative/ (32163)
3 exp Dental Care/ (29438)
4 Dental Restoration, Permanent/ (18759)
5 Dental Restoration Repair/ (102)
6 Periodontal Debridement/ (191)
7 Subgingival Curettage/ (977)
8 Dental Scaling/ (3316)
9 Chronic Periodontitis/ (1971)
10 Periodontal Diseases/ (24137)
11 Periodontal Surgery.mp. (1302)
12 Periodontal treatment.mp. (2728)
13 Oral Surgical Procedures/ (5363)
14 exp Surgery, Oral/ (7419)
15 Tooth Extraction/ (17008)
16 Dental Prosthesis/ (3384)
17 “Root Canal Therapy”/ (11735)
18 exp Dental Implants/ (17311)
19 Dental Implants, Single Tooth/ (1901)
20 Dental Implantation/ (3773)
21 1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 (372164)
22 exp Anesthetics, Local/ (96504)
23 exp Anesthesia, Local/ (15673)
24 exp Anesthesia, Dental/ (10417)
25 Lidocaine/ (22512)
26 Prilocaine/ (2018)
27 Bupivacaine/ (10713)
28 Procaine/ (11313)
29 Mepivacaine/ (1899)
30 Carticaine/ (451)
31 Etidocaine/ (288)
32 22 OR 23 OR 24 OR 25 OR 26 OR 27 OR 28 OR 29 OR 30 OR 31 (114729)
33 exp Cardiovascular Diseases/ (2067079)
34 Cardiac.mp. (631932)
35 Coronary Disease/ (128393)
36 Coronary Artery Disease/ (47503)
37 Coronary arteriosclerosis.mp. (733)
38 Coronariopathy.mp. (29)
39 Arrhythmias, cardiac/ (56112)
40 Heart Valve Diseases/ (21116)
41 Heart Diseases/ (63528)
42 Heart Failure/ (63528)
43 Rheumatic Heart Disease/ (12263)
44 Myocardial Ischemia/ (34898)
45 Myocardial Infarction/ (151398)
46 Hypertension (210548)
47 Hypertensive Patients.mp. (25167)
48 33 OR 34 OR 35 OR 36 OR 37 OR 38 OR 39 OR 40 OR 41 OR 42 OR 43 OR 44 OR 45 OR 46 OR 47 (2314784)
49 21 AND 32 AND 48 (752)

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S2 Appendix– Risk of bias of RCts included (modified version of the Cochrane collaboration)

Author	Year	Random sequence generation	Allocation concealment	Blinding of participants and personnel	Blinding of outcome assessment	Incomplete outcome data	Selective reporting	Other sources of bias
Abu-Mostafa, et al. 2015)	2015	Definitely yes	Probably no	Probably no	Probably no	Probably yes	Probably yes	Probably yes
Bronzo, et al. 2012	2012	Probably yes	Probably no	Definitely no	Definitely no	Probably yes	Probably no	Probably yes
Caceres, et al. 2008	2008	Probably yes	Definitely no	Definitely no	Definitely no	Probably yes	Probably no	Probably yes
Conrado, et al. 2007	2007	Probably yes	Probably yes	Probably yes	Definitely no	Probably yes	Probably no	Probably yes
Davenport, et al. 1990	1990	Definitely yes	Probably yes	Probably yes	Probably yes	Probably yes	Probably no	Probably no
Laragnoit, et al. 2009	2009	Definitely yes	Probably yes	Definitely yes	Definitely yes	Probably yes	Probably yes	Probably yes
Neves, et al. 2007	2007	Probably yes	Probably no	Definitely no	Definitely no	Probably yes	Probably no	Probably yes
Ogunlewe, et al. 2011	2011	Probably yes	Probably no	Definitely no	Definitely no	Probably yes	Probably no	Probably no
Santos-Paul, et al. 2015)	2015	Probably yes	Probably no	Probably yes	Probably yes	Probably yes	Probably yes	Probably yes
Torres-Lagares, et al. 2012	2012	Definitely yes	Probably yes	Probably yes	Definitely no	Probably yes	Definitely no	Probably no

S3 Appendix- Characteristics of excluded studies

Excluded studies	**Exclusion reasons**
Abu-Mostafa et al. 2015	Group of patients without CVD
Agani et al. 2015	Not RCT
Bazhanov et al.1983	Not RCT
Brkovic et al. 2008	Group of patients without CVD
Caviedes et al. 1983	Not RCT
Cecava, Mocek, Sel. 1965	Not RCT
Harang. 1949	Not RCT
Montebugnoli, Pelliccioni, Borghi 1990	Not RCT
Nava, Torres-Martinez 1967	Not RCT
Neves et al. 2009	Pregnant patients
Niwa 1999	Group of patients without CVD
Ohgami, Suda 1979	Not RCT
Rollason et al. 1977	Not RCT
Salman, Schwartz 1955	Not RCT
Tucci, Gabrielli, Mendes 1977	Not RCT
Vieillefosse 1973	Not RCT
Yoshimura et al. 1987	Not RCT
Zivotic-Vanovic, Marjanovic 2006	Group of patients without CVD

CAD- Cardiovascular disease. RCT- Randomised controlled trial

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	S1 appendix
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	9
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	–
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	S1 Figure
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	S1 Table
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	S2 Figure
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	12
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	13
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	S2 Table
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	–
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	18
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	18
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	19
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	19

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Local anesthetics combined with vasoconstrictors in patients with cardiovascular disease undergoing dental procedures: Systematic review and meta-analysis

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Title: Local anesthetics combined with vasoconstrictors in patients with cardiovascular disease undergoing dental procedures: Systematic review and meta-analysis

Running title: Local anesthetics combined with vasoconstrictors in patients with cardiovascular disease undergoing dental procedures

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ABSTRACT

Objectives: There is a lack of evidence about the use of local anesthetics (LA) in patients with cardiovascular diseases (CVD) in dental procedures. Thus, this study evaluated the safety of using LA with vasoconstrictor to determine the risk of cardiovascular events in patients with CVD.

Design: Systematic Review and meta-analysis.

Methods: We have searched in Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE (via Ovid), EMBASE (via Ovid), Healthstar (via Ovid), CINAHL, Web of Science, and ClinicalTrials.gov for randomized controlled trials (RCTs) up to January of 2020. We have included RCTs involving adults with CVD within two groups: intervention group with LA with vasoconstrictor and control group with LA without vasoconstrictor. The primary outcomes assessed were death, mortality by a specific cause, stroke, acute myocardial infarction, hospitalization, pain, bleeding and arrhythmias. The secondary outcomes were ST segment depression, anxiety, adverse effects and changes in hemodynamic parameters. The data were pooled using random effects meta-analyses and the confidence in the estimates was verified using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE).

Results: Ten RCTs (n=478 participants) were included. Most of them had a high risk of bias. There were more cases of pain and bleeding in groups without vasoconstrictor. Meta-analysis demonstrated a decrease in the systolic blood pressure with the use of local anesthetic with vasoconstrictor (SMD: -0.95, 95% CI: -1.35 to -0.55) after procedure. Overall, for the other outcomes assessed there was no statistical difference. The quality of evidence was considered low according to the GRADE profile.

Conclusions: The results suggest that the use of LA with vasoconstrictors (epinephrine in low doses) is safe in patients with some types of CVD. However, the low quality of evidence demonstrated that literature needs further studies in order to confirm these results.

Protocol registration: PROSPERO (CRD42016045421)

Keywords: Local Anesthetics, Dental Anesthesia, Dentistry, Cardiovascular Disease.

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Strengths and limitations of this study

- This review was conducted with methodological rigor, including assessment of the risk of bias and the quality of evidence; such information is not available in previous systematic reviews related to the subject.
- This study has explicit eligibility criteria, priority of primary and secondary outcomes, comprehensive and extensive search in the databases, with independent and peer reviews for each study and use of GRADE to assess the quality of the evidence and the strength of recommendation for the outcomes included in the meta-analysis.
- The primary studies included were a limiting factor for the findings of this review due to methodological bias observed in the included RCTs, as well as the differences observed in the types of CVDs, types and dosages of local anesthetics and vasoconstrictors.

INTRODUCTION

Cardiovascular diseases (CVD) are one of the causes of most premature deaths worldwide, representing 31% of all deaths. It is estimated that in 2012 CVD were the main cause of death in the world with approximately 17.5 million deaths. In 2002, three-quarters of CVD deaths occurred in low- or middle-income countries. CVD comprises essential hypertension, rheumatic heart diseases, ischemic heart diseases, cerebrovascular diseases, heart inflammatory diseases, and so forth (1).

In the dental office, the presence of patients with complex diseases such as CVD is increasingly common. These patients have access to better treatments and medications, which makes it possible to perform treatment in private dental clinics that previously was only executed by specialized hospitals (2, 3). However, careful evaluations should be conducted before the begin of dental treatment in patients with CVD in order to control stress and mainly to block nerve

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transmission of pain, thereby minimizing transient episodes that may trigger negative cardiovascular outcomes (3).

The use of local anesthetics (LA) and analgesics can control pain, while sedatives control the anxiety during dental procedures (3, 4). However, the use of LA has been the baseline of pain control for dental procedures for several decades (5,6).

Local anesthetics are combined with vasoconstrictors to increase duration of anesthetic effect, reduce systemic toxicity, and optimize soft tissue hemostasis (6). However, despite the beneficial properties of vasoconstrictors when combined with local anesthetics, dental professionals are concerned about the adverse effects associated with the vasoconstrictor, mainly with epinephrine, which is the most used vasoconstrictor in Dentistry (2, 7).

These systemic adverse effects are linked to inadequate procedures such as intravascular injection, high-dose injections, drug interactions, or instant absorption of the anesthetic agent when it is injected into an extremely vascular area that could induce adverse cardiovascular effects, especially in patients with CVD (3, 8-10).

Another factor for adverse effects is the concentration used in LAs. Currently, a greater dilution of the vasoconstrictor is recommended, mostly used at a concentration of 1: 100,000 for surgical procedures and, for other procedures, with a dilution of up to 1: 200,000 (2, 6). Nevertheless, for long and/or surgical procedures, LAs without vasoconstrictor do not offer a satisfactory anesthetic duration, which can cause pain resulting in systemic endogenous release of catecholamines (particularly noradrenaline) and causing to negative cardiovascular outcomes (11, 12, 13).

A previous study reported that the stress-induced release of catecholamines could be more than 10 times greater than the basal level (14). In stressful situations, such as pain and anxiety, the released endogenous catecholamines may also reach concentrations higher than the low epinephrine concentrations used in dental LA (14, 15). Thus, the literature recommends using the lowest effective concentration of epinephrine as possible for CVD patients, both for dermatological and dental procedures (9, 13). A few decades ago, the maximum dose of epinephrine in patients with cardiovascular disease was set at up to 200 µg (16). However, over the years several studies and guidelines on this

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subject have been published and suggested that 36 to 54 µg of epinephrine appear to be tolerated in most patients with hypertension or other cardiovascular diseases (2,17).

Previously two systematic reviews analyzed the most frequent complications in patients with CVD after dental local anesthesia with vasoconstrictor, disclosed in electrocardiogram arrhythmias (12, 13). However, both reviews differed from the present study. Bader et al (2002) did not describe the inclusion criteria and risk of bias (12). In addition, the authors included non-randomized clinical trials and a group of patients without CVD. Godzieba et al (2014) also included non-randomized clinical trials and patients without CVD (13).

LA have a long history of safety and efficacy in dental patients if used in recommended doses and techniques (3,6). However, dental professionals still have doubts about their use in patients with CVD, especially when combined with vasoconstrictor (13). Thus, this systematic review evaluated the safety in the use of LA with vasoconstrictor and determined the risk of cardiovascular events in patients with CVD.

METHODS AND ANALYSES

The systematic review was performed according to the recommendations specified in the Cochrane Handbook for Interventional Reviews and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses statement (PRISMA) (18).

Protocol and Registration

The protocol of this systematic review was registered on the PROSPERO – *International Prospective Register of Systematic Reviews* (registration number - CRD42016045421).

Eligibility criteria

Inclusion criteria

Patients: adult patients with CVD (essential hypertension, rheumatic heart diseases, ischemic heart diseases, cerebrovascular diseases, and heart inflammatory diseases) (1).

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Interventions: one arm wherein patients received LA with vasoconstrictors compared to another arm wherein patients received LAs without vasoconstrictors.

Procedures: patients who undergo tooth extraction, dental restorations, treatment and periodontal surgery, among others.

Type of study: considering that observational studies increase the risk of bias to answer this question, the study design that can generate high quality evidence, according to the evidence hierarchy, would be the randomized controlled studies (RCTs). We include two types of RCT designs. In the first type, patients are randomized to receive either LAs with vasoconstrictors during the first dental procedure or LAs without vasoconstrictors during the second dental procedure or vice versa. In the second type, patients are randomized to receive only one type of LA, with or without vasoconstrictors, during the dental procedure.

Language: any language without restriction.

Outcomes measures

The safety of the use of local anesthetics with vasoconstrictors in patients with CVD was based on the number of reports considering the following primary and secondary outcomes:

Primary outcomes:

- death;
- mortality by a specific cause;
- stroke;
- acute myocardial infarction;
- hospitalization;
- pain;
- bleeding;
- arrhythmias;

Secondary outcomes:

- ST segment depression;
- anxiety;

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- adverse effects;
- changes in hemodynamic parameters.

Exclusion criteria

We exclude studies involving patients with untreated or out-of-control essential hypertension or any type of CAD, who were pregnant or breastfeeding, who were allergic to the LAs used in the studies, with out-of-control diabetes mellitus, or who have had recent myocardial infarction, cancer, and malignant hypertension.

Search methods for primary studies

Electronic searches

The following electronic databases were searched: the Cochrane Central Register of Controlled Trials (CENTRAL) part of The Cochrane Library; MEDLINE (Ovid); EMBASE (Ovid); Healthstar (Ovid); CINAHL (Cumulative Index to Nursing and Allied Health Literature); and Web of Science, from their inception to January of 2020, without restrictions on the language and date of publication.

Searching other resources

We performed search in registration of clinical trials: https://clinicaltrials.gov, WHO clinical trials registry, http://www.ensaiosclinicos.gov.br; trials registry and bank of Brazil thesis (CAPES); conference proceedings of the Brazilian Congress of Cardiology, in the Brazilian Congress of Anesthesiology, and in the International Congress of Dentistry (CIOSP). Two reviewers analyzed the reference list or quotations found in secondary studies to verify and identify possible eligible studies.

Search strategy

The search strategy was individually conducted by type of dental intervention; type of anesthetic; and type of CVD. We have adapted the search strategy according to each database. The search strategy in Medline (via Ovid) is in S1 Appendix.

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Eligibility determination

Six reviewers (CCG vs CCB vs JOA and RHLM vs NKA vs JCR) working in trio independently evaluated whether summaries are in accordance with eligibility criteria. Discordances were resolved by consensus, and contentious issues were discussed with a third reviewer (LCL).

To exclude duplicate articles, reviewers analyzed all eligible articles and identify those with one or more authors in common. In case of duplicate publications, we used the article with more complete data.

Data extraction

Six reviewers (CCG vs CCB, JOA vs RHLM and NKA vs JCR), working in pairs and independently extracted data and recorded information regarding patients, methods, interventions, outcomes, and missing outcome data using standardized and pretested data extraction forms with instructions. Before initiating data extraction, we conducted calibration exercises to ensure consistency among the reviewers, both in screening and data extraction. We contacted the study authors as necessary to resolve any uncertainties. Reviewer's resolved disagreements by discussion, and one arbitrator (LCL) judged unresolved disagreements.

Risk of bias in individual studies

Using a modified version of the Cochrane collaboration risk of bias tool (19, 20), The same pairs of reviewers independently rated the risk of bias for each RCT according to the following criteria: random sequence; allocation concealment; blinding of the patient, healthcare professionals, outcome assessors, data collectors, and data analysts; incomplete outcome data; selective outcome reporting; and major baseline imbalance.

Reviewers gave response options of "definitely yes," "probably yes," "probably no," and "definitely no" for each of the domains, with the options "definitely yes" and "probably yes" ultimately being assigned a low risk of bias and "definitely no" and "probably no" as having a high risk of bias (21). Reviewer's resolved disagreements by discussion, and one arbitrator (LCL) judge unresolved disagreements. Risk of bias of RCTs included are in S2 Figure and S2 Appendix.

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Explaining the heterogeneity of evidence

Possible explanations for heterogeneity included: a) age- the older the age, the higher the risk of cardiovascular transient episodes; b) gender- women outnumber men in deaths due to CVD; c) vasoconstrictor type- vasoconstrictors are linked to receptors α and β. However, some of these are more often linked to cardiac receptor β (except for felypressin, which links to the vasopressin receptor v1, present in the smooth muscles of blood vessel walls, raise cardiac frequency, and thus, higher risks of transient episodes are expected (22); d) vasoconstrictor concentration - which may vary from a 1:2,500 to a 1:200,000 greater risk is expected with higher vasoconstrictor concentration; e) dental procedure duration- the longer the duration to perform the procedure (surgical or periodontal procedures take longer than restorative procedures), the higher the concentration of anesthetic agent necessary, and the stronger the toxicity to the cardiovascular system, thereby increasing the risks of transient episodes in long-duration procedures; f) dental procedure type: usually surgical procedures (periodontal, extraction, and implantation) trigger great stress in the patient, thus increasing the risk of transient episodes. For the all outcomes, subgroup analyses were not possible due to the low number of studies included.

We ranked heterogeneity associated with pooled effect estimates with the use of the χ^2 test and the I^2 statistic (23). The following heterogeneities were considered: 0–25% (low heterogeneity), 50% (moderate heterogeneity), and 75% (high heterogeneity) (20).

Data synthesis

Analyses were conducted for each vasoconstrictor and for each outcome of interest. Confidence in the findings was determined by estimates for each body of evidence. The studies which reported continuous outcomes presented the date in Standard Mean Difference (SMD) and 95% Confidence Interval (95% CI) and dichotomous outcomes were reported as risk ratio (RR) combined with the 95% CI.

Other details about the methods adopted for the synthesis of the data can be found in the publication of the protocol of this review (24). The random effect in meta-analysis was performed using the Review Manager 5 Software (version 5.3).

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Summarizing evidence

The quality of evidence was independently evaluated (confidence in effect estimates) for each for each outcome reported using GRADE (25). Results are presented in evidence profiles, as recommended by the GRADE Working Group (26, 27).

In this approach, randomized trials begin with evidence of high quality may have their qualities diminished according to the judgment of one or more of the five categories of limitations presented below: risk of bias (assessed for each study as described previously), inconsistency, indirect evidence, imprecision and publication bias (25, 28). We used GRADE methodology to rate confidence in estimates of effect for each outcome as high, moderate, low or very low. Evidence profiles are provided brief presentations of evidence quality and effect magnitude. With the help of the software program GRADEpro (http://ims.cochrane.org/gradepro).

Patient and public involvement

No patient was involved in this study.

RESULTS

Literature search results

From 4,048 publications retrieved from the databases, 1,924 articles remained for the titles and abstracts selection after removing duplicates. From the 28 pre-selected papers, 18 were excluded (S3 Appendix) and 10 met eligibility criteria (29-38). Six studies (29, 31, 32, 34, 35, 38) were able to be included in the meta-analysis (S1 Figure).

Description of the studies (S1 Table)

Methods

Ten RCTs were included, encompassing a total of 478 patients with CVD, from which 226 patients had coronary artery disease, 226 patients had essential

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hypertension, 59 patients had valvular diseases, 33 patients had Chagas disease, and 9 patients had unspecified cardiovascular disease. S1 Table describes the characteristics of the studies in which it was observed that the majority of the studies used lidocaine as LA and epinephrine as a vasoconstrictor. The studies were mainly published in Brazil between 2007 and 2014. The different reasons for exclusion of articles are found in S3 Appendix.

Intervention

Eight studies (30, 32-38) used the same local anesthetic both in the control and intervention groups; two studies (29, 38) included more than one group in the intervention group; five studies (33-37) used lidocaine both in the intervention and control groups; one study (30) used prilocaine both in the intervention and control groups; two studies (32, 38) used mepivacaine both in the intervention and control groups; and three studies (29, 31, 38) used different local anesthetic solutions between the intervention and control groups.

Eight studies (29, 32-38) used the vasoconstrictor epinephrine; followed by felypressin, which was present in three studies (29-31) and only one study (29) used both epinephrine and felypressin (S1 Table).

Participants

Five studies (29, 30, 33, 36, 38) included patients with essential hypertension. Two studies (32, 37) included patients with coronary artery disease. Two studies (33, 35) included patients with essential hypertension and coronary artery disease. One study (34) included patients with valvular diseases, and one study (31) included patients with Chagas disease and coronary artery disease.

Risk of bias in the included studies (S2 Appendix and S2 Figure)

Randomization and allocation

From the 10 RCTs included, three studies (30, 33, 38) used the crossover design, what reduces the quality of the evidence, and performed wash-out periods between 10 minutes to 3 weeks. Randomization by computer was performed by only one study (38). Two studies (33, 34) used the random number

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table. Two studies (29, 32) did a draw of the ID cards. Five studies (30, 31, 35-37) did not mention the randomization method.

The allocation concealment method was considered of low risk of bias in one study (38), when randomization was performed by computer.

Blinding

Two study (34, 37) reported performing double-blinding, but only one study (34) partially described. One study (37) only mentioned that the blinding was performed, without an explanation of the method. Five studies (29-31, 35, 36) were not conducted blinded.

Only two studies (32, 34) did not mention the blinding but used allocation concealment with delivered in sealed envelopes, and the cartridges were covered with a white strip.

Loss to follow-up

The studies reported no loss to follow-up.

Selective reporting of outcomes

Three studies (29, 34, 37) had the registry of clinical trial reporting pre-specified outcomes. Two studies (34, 37) made the registry at ClinicalTrials.gov and one study (29) at the Institute Research Center.

Other potential source of bias

Four studies (29-31, 34) mentioned that there was no external funding. One study (37) received Brazilian funding through the São Paulo State Research Support Foundation (FAPESP), which is non-profit.

Six studies (29-32, 34, 35) declared no potential conflicts of interest. One study (34) mentioned both no external funding sources and no potential conflict of interest only in the registry of clinical trials

Reported primary outcomes

None of the studies reported death, mortality by specific cause, stroke, acute myocardial infarction or hospitalization. The only primary outcomes

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reported by the authors included in this review were pain, bleeding and arrhythmias (S1 Table). Due to the small number of studies and the lack of numerical data, it was not possible to perform the data meta-analysis of pain and bleeding. The reports of these outcomes were descriptive and qualitative.

Pain

Two studies (32, 34) evaluated the pain outcome. Conrado et al. (2007) assessed the pain outcome at the extraction site in 54 patients with coronary disease and observed pain report in only one patient from the intervention group (mepivacaine 2% with epinephrine 1:100,000), and in 9 patients from the control group (mepivacaine 3% without vasoconstrictor). The authors did not describe how they have assessed pain and explained that the only one case of pain in the intervention group was caused by a tooth that was ankylosed (32).

Laragnoit, et al. (2009) examined 59 participants with several valvular diseases and described the results only in a qualitative way. It was observed that there were more complaints of pain in the control group (lidocaine 2% without vasoconstrictor) compared to the intervention group (lidocaine 2% with epinephrine 1:100,000) (34). The authors did not describe how they assessed pain.

Bleeding

Two studies (31, 33) evaluated the bleeding outcome. Caceres et al. (2008) did not reported bleeding during the procedures. In both groups, authors did not describe how they assessed this outcome (31). Davenport, et al. (1990) examined 9 participants in the control and intervention groups and observed that all participants in the intervention group (lidocaine 2% with epinephrine 1:100,000) obtained adequate hemostasis versus 7 participants with inadequate hemostasis in the control group (lidocaine 2% without vasoconstrictor). The authors did not describe how they assessed bleeding (33).

Arrhythmia

Three studies (31, 34, 35) assessed this outcome. Caceres et al. (2008) analyzed the presence of ventricular extrasystoles that remained stable in different measured moments, with a similar behavior in both groups (31).

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Laragnoit, et al. (2009) reported eight patients with arrhythmia, being five patients from the intervention group (lidocaine 2% with epinephrine 1:100,000) and three patients in the control group (lidocaine 2%), with no observed statistical difference between both groups (34).

Neves et al. (2007) reported seven patients with arrhythmia, being three (11.1%) from the intervention group (lidocaine 2% with epinephrine 1:100,000) and four (13.8%) from the control group (lidocaine 2% without vasoconstrictor), with no observed significant statistical difference between the two groups (35).

The meta-analysis demonstrated no difference in the arrhythmias between the groups (RR: 1.24, 95% CI: 0.47 to 3.28 low quality evidence) (S3 Figure, S2 Table).

Reported secondary outcomes

Adverse effects were not reported in the studies.

ST segment depression

Two studies (32, 35) evaluated the presence of ischemic episodes (measured as ST segment depression).

Conrado et al. (2007) observed the occurrence of the ST segment depression greater than 1mm in three patients in the intervention group (mepivacaine 2% with epinephrine 1:100,000) during anesthetic injection. No event was observed in the control group (mepivacaine 3% without vasoconstrictor) and no statistical difference was observed between the groups (32).

Neves et al. (2007) observed ST segment depression greater than 1mm in four patients (14.8%) in the intervention group (lidocaine 2% with epinephrine 1:100,000) and in six patients (20.7%) in the control group (lidocaine 2% without vasoconstrictor), with no significant statistical difference between the groups (35).

The meta-analysis demonstrated no difference in the ST segment depression between the groups (RR: 1.53, 95% CI: 0.17 to 13.72 low quality evidence) (S4 Figure, S2 Table).

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Anxiety

Two studies (34, 37) evaluated the anxiety outcome. Laragnoit et al. (2009) assessed the anxiety of patients using the STAI scale (Spielberger State Anxiety Inventory) before the procedure and reported no difference in the mean values of anxiety levels between the intervention and control groups (34).

Santos-Paul et al. (2015) assessed the anxiety of patients using Buchanan and Niven facial scale in different periods (one hour before the procedure, five minutes after the procedure and one hour after the procedure). They reported no significant difference between the control and intervention groups at any of the evaluated time periods (37).

Mean blood pressure

Five studies (31, 33, 34, 35, 38) evaluated the mean blood pressure values. In four studies (31, 33, 34, 35) there were no significant statistical difference between the assessed groups. Torres-Lagares et al. (2012) reported increase in the average of mean blood pressure in the intervention group (mepivacaine 2% with epinephrine 1:100,000) in comparison to the other three intervention groups (articaine 4% with epinephrine 1:200,000, articaine 4% with epinephrine 1:100,000, mepivacaine 3% without vasoconstrictor) (38). Three studies (31, 33, 35) mention their results only in graphs and it was not possible to perform meta-analysis.

Meta-analysis was conducted with the Laragnoit et al. (2009) and Torres-Lagares et al. (2012) studies (34, 38). No difference was observed in the mean blood pressure between the groups during the procedure period (SMD: -0.05, 95% CI: -0.93 to 0.82, very low-quality evidence), and after the procedure period (SMD: -0.26, 95% CI: -0.71 to 0.18, low quality evidence). Time periods have been adapted to carry out the meta-analysis (S5 and S6 Figure, S2 Table).

Systolic blood pressure

Eight studies (29, 30, 32, 34-38) evaluated this outcome. In seven studies (30, 32, 34-38) there were no significant statistical difference between the assessed groups.

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Abu-Mostafa et al. (2015) observed a significant increase in the systolic blood pressure after dental extraction in the control group (mepivacaine 3% without vasoconstrictor) compared to the intervention group (lidocaine 2% with epinephrine 1:80,000) after the procedure (29).

Four studies (30, 32, 36, 37) could not be included in the meta-analysis because their data were presented in graphs. Even after several attempts to contact the authors, they did not respond. Additionally, unsuccessful attempts were made to extract data using the Digitizelt software.

Meta-analysis showed no difference in systolic blood pressure while using local anesthetic with vasoconstrictor compared to using only local anesthetic without vasoconstrictor during the procedure period (SMD: -0.39, 95% CI: -0.80 to 0.03, very low-quality evidence) (S7 Figure, S2 Table). After the procedure period, meta-analysis demonstrated a decrease in the systolic blood pressure with the use of local anesthetic with vasoconstrictor (SMD: -0.95, 95% CI: -1.35 to -0.55.) (S8 Figure, S2 Table).

Diastolic blood pressure

Eight studies (29, 30, 32, 34-38) evaluated the average diastolic blood pressure. In six studies (29, 32, 34-37) there was no significant statistical difference between the assessed groups.

In Bronzo et al. (2012), there was an increase in diastolic blood pressure during the dental procedures in both groups. However, in the intervention group (prilocaine 2% with felypressin 0,03 IU) diastolic blood pressure was greater than in the control group (prilocaine 4% without vasoconstrictor) (30). Torres-Lagares et al. (2012) reported diastolic blood pressure values significantly higher in the intervention group (mepivacaine 2% with epinephrine 1:100.000) in comparison to the other two groups intervention groups (articaine 4% with epinephrine 1:200,000; articaine 4% with epinephrine 1:100,000) and control group (mepivacaine 3% without vasoconstrictor) (38).

Four studies (30, 32, 36, 37) could not be included in the meta-analysis because their data were presented in graphs. Even after several attempts to contact the authors, they did not respond. Additionally, unsuccessful attempts were made to extract data using the Digitizelt software.

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Meta-analysis showed no statistical difference between the diastolic blood pressure during the procedure (SMD: -0.51, 95% CI: -1.07 to 0.06) and after the procedure periods (SMD: -0.53, 95% CI: -1.06 to 0.01). However, it was noted that there was a tendency, with no statistical significance, in favor of the use of LA with vasoconstrictor (S9 and S10 Figure). The quality of evidence was considered very low for both during and after the procedure periods (S2 Table).

Heart rate

Nine studies (29, 31-38) assessed this outcome and in any of them significant difference between the groups was found. Six studies (31-34, 36, 37) could not be included in the meta-analysis because their data were presented in graphs. Even after several attempts to contact the authors, they did not respond. Additionally, unsuccessful attempts were made to extract data using the DigitizeIt software.

Only three studies (29, 35, 38) could be included in meta-analysis from which numerical data was available. Meta-analysis showed no statistical difference for the periods during (SMD: -0.24, 95% CI: -0.61 to 0.13) after the procedure (SMD: -0.20, 95% CI: -0.57 to 0.17) for heart rate between the groups (S11 and S12 Figure), being the evidence considered of very low quality (S2 Table).

DISCUSSION

Discussion of main findings with the literature

The use of local anesthetic with vasoconstrictor decreased pain and bleeding, but neither outcome could be included in the statistical analysis due to the lack of data. The findings of meta-analysis showed decreasing on the systolic blood pressure with the use of local anesthetic with vasoconstrictor compared to control group after the procedure. No significant difference was observed comparing LA with and without the use of vasoconstrictor for the outcomes anxiety, arrhythmia, ST segment depression, mean blood pressure, diastolic blood pressure and rate heart during and after the dental procedure.

In general, the quality of the evidence was considered low for the outcomes evaluated mainly due to the risk of bias, such as lack of blinding,

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concealment of the allocation not clearly described and problems due to the high heterogeneity between clinical trials and the inaccuracy of the findings.

Considering previous systematic reviews found in literature related to this subject (12, 13), methodological differences and in the proposed objectives, as well as the absence of meta-analysis did not allow the direct comparison of their results with the present study.

Bader et al. (2002) evaluated the cardiovascular adverse effects risk on the use of LA with epinephrine and the use of epinephrine-impregnated retraction cord. The authors included healthy patients and patients with CVD using LA with or without vasoconstrictor, and they also included non-randomized clinical trials, different from the objectives and methods of our review. The authors reported that the adverse effects in patients with essential hypertension are uncommon and that there is a small risk for hemodynamic changes, such as blood pressure and heart rate.

Another systematic review (13) evaluated the use of LA with vasoconstrictor in patients with CVD. However, studies with healthy patients and non-randomized studies were included and outcomes were not prioritized. The review verified 87 complications that occurred after the administration of the local anesthetic. Although cardiac arrhythmia was the most frequent outcome, it was not clinically relevant. The authors concluded that use with the maximum limit of four dental cartridges of lidocaine with epinephrine 1: 100,000 appears to be relatively safe in patients with CVD.

Strengths and limitations of this study

This study was conducted with methodological rigor, including risk assessment of bias and of quality of evidence; such information is not available in reviews related to the subject (12, 13). Furthermore, the study has explicit eligibility criteria, priority of outcomes (primary and secondary), comprehensive and extensive search in the databases, with independent and peer reviews for each study and use of GRADE to assess the quality of the evidence and the strength of recommendation for the outcomes included in the meta-analysis.

Although the search strategy was carried out in different databases (Cochrane Central Register of Controlled Trials, MEDLINE, EMBASE, Healthstar,

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CINAHL, Web of Science, and ClinicalTrials.gov), coincidentally the included studies were mainly published in Brazil between 2007 and 2014.

Considering the possible outcomes, conducting clinical studies with CVD patients is highly complex. In addition, the difficulty of including patients with specific systemic profiles may explain the limited number of types of CVD included in this study.

The primary studies included were a limiting factor for the findings of this review due to methodological bias observed in the included RCTs, as well as the differences observed in the types of CVDs, types and dosages of LA and vasoconstrictors. Thus, it was noted a need for more primary studies about the subject, with better methodological accuracy and prioritization of outcomes in order to increase confidence in findings.

Implications for clinical practice and research

The LA associated with vasoconstrictor is essential in dental treatment because vasoconstrictor agents counteract the vasodilating action of local anesthetics, increasing anesthetic time and reducing bleeding (6, 8). The findings of this systematic review are limited to some types of CVD and the use of epinephrine and felypressin as vasoconstrictors. Additionally, the low quality of evidence demonstrates that literature needs further studies in order to confirm these results.

Thus, new RCTs with higher methodological rigor are needed to obtain more reliable results, in order to verify the safety of LAs with vasoconstrictor in patients with CVDs. It is also suggested that the professionals use the recommended doses as well as perform the procedure respecting the anesthetic techniques (2, 6, 9). Moreover, communication with the patient's cardiologist can help clarify the patient's clinical condition, to evaluate the care and limitations of the safest dental treatment for each patient (3,9).

Conclusion

None of the included studies in this review reported death, mortality by specific cause, stroke or acute myocardial infarction. The results suggest that the use of LA with epinephrine in low doses in adult patients with CVD reduced systolic blood pressure after dental procedures when compared to the use of LA

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without vasoconstrictor. Furthermore, the use of epinephrine (in low doses) does not alter the risk of arrhythmia, ST-segment depression, mean arterial pressure, systolic and diastolic blood pressure and heart rate, suggesting safety in its use for these outcomes in adult patients with some types of CVD. However, the low quality of evidence demonstrates that literature needs further studies in order to confirm these results.

Contributors

CCG is the principal investigator and led the writing of the manuscript. LCL and RHLM are the project managers and co-investigators and contributed to the writing and revision of the manuscript. CCG, CCB, RHLM, JCR, JOA, NKA are co-investigators who contributed to screening the eligible studies and extracted the data of the studies, also participated the revision of the manuscript. MTS performed the statistical analysis, also participated the revision of the manuscript. All authors read and approved the final manuscript.

Competing interests

None declared.

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Data sharing statement

All data relevant to the study are included in the article or uploaded as supplementary information. However, other information is available upon request.

Research Ethics Approval: Human Participants

Not applicable.

Provenance and peer review

Not commissioned, externally peer reviewed.

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Abbreviations

Local anesthetic (LA), cardiovascular disease (CVD), Cochrane Central Register of Controlled Trials (CENTRAL), Medical literature Analysis and Retrieval System Online (MEDLINE), Excerpta Medica dataBASE (EMBASE), Cumulative Index to Nursing and Allied Health Literature (CINAHL), Grading of Recommendations Assessment, Development, and Evaluation (GRADE), randomized clinical trial (RCT), bank of Brazil thesis (CAPES), Congresso Internacional de Odontologia de São Paulo (CIOSP), confidence interval (CI), weighted mean difference (WMD), minimally important difference (MID), and standardized mean difference (SMD), standard deviation (SD).

References

1. World Health Organization.WHO. Cardiovascular diseases (CVDs). WHO [Internet]. 2017 2017-05-1714:41:03. Available from: http://www.who.int/mediacentre/factsheets/fs317/en/.

2. Becker DE, Reed KL. Local anesthetics: review of pharmacological considerations. Anesth Prog. 2012;59(2):90-101; quiz 2-3. doi: 10.2344/0003-3006-59.2.90.

3. Jowett NI, Cabot LB. Patients with cardiac disease: considerations for the dental practitioner. Br Dent J. 2000;189(6):297-302.

4. American Dental Association A. Guideline for teaching pain control and sedation to dentist and dental students. 2012:1-18.

5. St George G, Morgan A, Meechan J, Moles DR, Needleman I, Ng YL, et al. Injectable local anaesthetic agents for dental anaesthesia. Cochrane Database Syst Rev. 2018;7(7):Cd006487.

6. Haas DA. An update on local anesthetics in dentistry. J Can Dent Assoc. 2002;68(9):546-51.

7. Jage J. Circulatory effects of vasoconstrictors combined with local anesthetics. Anesth Pain Control Dent. 1993;2(2):81-6.

8. Fencl JL. Guideline implementation: local anesthesia. AORN J. 2015;101(6):682-9; quiz 90-2.

9. Kouba DJ, LoPiccolo MC, Alam M, Bordeaux JS, Cohen B, Hanke CW, et al. Guidelines for the use of local anesthesia in office-based dermatologic surgery. J Am Acad Dermatol. 2016;74(6):1201-19.

For peer review only

10. Wadlund DL. Local Anesthetic Systemic Toxicity. Aorn j. 2017;106(5):367-77.

11. Brown RS. Local anesthetics. Dent Clin North Am. 1994;38(4):619-32.

12. Bader JD, Bonito AJ, Shugars DA. A systematic review of cardiovascular effects of epinephrine on hypertensive dental patients. Oral Surg Oral Med Oral Pathol Oral Radiol Endod. 2002;93(6):647-53.

13. Godzieba A, Smektala T, Jedrzejewski M, Sporniak-Tutak K. Clinical assessment of the safe use local anaesthesia with vasoconstrictor agents in cardiovascular compromised patients: a systematic review. Med Sci Monit. 2014;20:393-8.

14. Brand HS, Gortzak RA, Palmer-Bouva CC, Abraham RE, Abraham-Inpijn L. Cardiovascular and neuroendocrine responses during acute stress induced by different types of dental treatment. Int Dent J. 1995;45(1):45-8.

15. Meyer FU. Haemodynamic changes under emotional stress following a minor surgical procedure under local anaesthesia. Int J Oral Maxillofac Surg. 1987;16(6):688-94.

16. New York Heart Association. Use of epinephrine in connection with procaine in dental procedures. J Am Dent Assoc 1955;50:108.

17. Brown RS, Rhodus NL. Epinephrine and local anesthesia revisited. Oral Surg Oral Med Oral Pathol Oral Radiol Endod. 2005 Oct;100(4):401-8.

18. Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. PLoS Med. 2009;6(7):e1000097.

19. Altman R, Alarcon G, Appelrouth D, Bloch D, Borenstein D, Brandt K, et al. The American College of Rheumatology criteria for the classification and reporting of osteoarthritis of the hand. Arthritis Rheum. 1990;33(11):1601-10.

20. Higgins JP, Green S. Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0 (updated March 2011). The Cochrane Collaboration 2011.

21. Akl EA, Sun X, Busse JW, Johnston BC, Briel M, Mulla S, et al. Specific instructions for estimating unclearly reported blinding status in randomized trials were reliable and valid. J Clin Epidemiol. 2012;65(3):262-7.

22. Manning M, Stoev S, Chini B, Durroux T, Mouillac B, Guillon G. Peptide and non-peptide agonists and antagonists for the vasopressin and oxytocin V1a,

For peer review only

V1b, V2 and OT receptors: research tools and potential therapeutic agents. Prog Brain Res [Internet]. 2008; 170:[473-512 pp.].

23. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. Stat Med. 2002;21(11):1539-58.

24. Guimaraes CC, Lopes Motta RH, Bergamaschi CC, Araujo JO, Andrade NK, Figueiro MF, et al. Local anaesthetics combined with vasoconstrictors in patients with cardiovascular disease undergoing dental procedures: systematic review and meta-analysis protocol. BMJ Open. 2017;7(11):e014611.

25. Guyatt GH, Oxman AD, Kunz R, Woodcock J, Brozek J, Helfand M, et al. GRADE guidelines: 7. Rating the quality of evidence--inconsistency. J Clin Epidemiol. 2011;64(12):1294-302.

26. Guyatt GH, Oxman AD, Santesso N, Helfand M, Vist G, Kunz R, et al. GRADE guidelines: 12. Preparing summary of findings tables-binary outcomes. J Clin Epidemiol. 2013;66(2):158-72.

27. Guyatt GH, Thorlund K, Oxman AD, Walter SD, Patrick D, Furukawa TA, et al. GRADE guidelines: 13. Preparing summary of findings tables and evidence profiles-continuous outcomes. J Clin Epidemiol. 2013;66(2):173-83.

28. Guyatt GH, Oxman AD, Montori V, Vist G, Kunz R, Brozek J, et al. GRADE guidelines: 5. Rating the quality of evidence--publication bias. J Clin Epidemiol. 2011;64(12):1277-82.

29. Abu-Mostafa N, Aldawssary A, Assari A, Alnujaidy S, Almutlaq A. A prospective randomized clinical trial compared the effect of various types of local anesthetics cartridges on hypertensive patients during dental extraction. Journal of clinical and experimental dentistry. 2015;7(1):e84-8.

30. Bronzo AL, Cardoso CG, Jr., Ortega KC, Mion D, Jr. Felypressin increases blood pressure during dental procedures in hypertensive patients. Arquivos brasileiros de cardiologia. 2012;99(2):724-31.

31. Caceres MT, Ludovice AC, Brito FS, Darrieux FC, Neves RS, Scanavacca MI, et al. Effect of local anesthetics with and without vasoconstrictor agent in patients with ventricular arrhythmias. Arquivos brasileiros de cardiologia. 2008;91(3):128-33, 42-7.

32. Conrado VC, de Andrade J, de Angelis GA, de Andrade AC, Timerman L, Andrade MM, et al. Cardiovascular effects of local anesthesia with

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vasoconstrictor during dental extraction in coronary patients. Arquivos brasileiros de cardiologia. 2007;88(5):507-13.

33. Davenport RE, Porcelli RJ, Iacono VJ, Bonura CF, Mallis GI, Baer PN. Effects of anesthetics containing epinephrine on catecholamine levels during periodontal surgery. Journal of periodontology. 1990;61(9):553-8.

34. Laragnoit AB, Neves RS, Neves IL, Vieira JE. Locoregional anesthesia for dental treatment in cardiac patients: a comparative study of 2% plain lidocaine and 2% lidocaine with epinephrine (1:100,000). Clinics (Sao Paulo, Brazil). 2009;64(3):177-82.

35. Neves RS, Neves IL, Giorgi DM, Grupi CJ, Cesar LA, Hueb W, et al. Effects of epinephrine in local dental anesthesia in patients with coronary artery disease. Arquivos brasileiros de cardiologia. 2007;88(5):545-51.

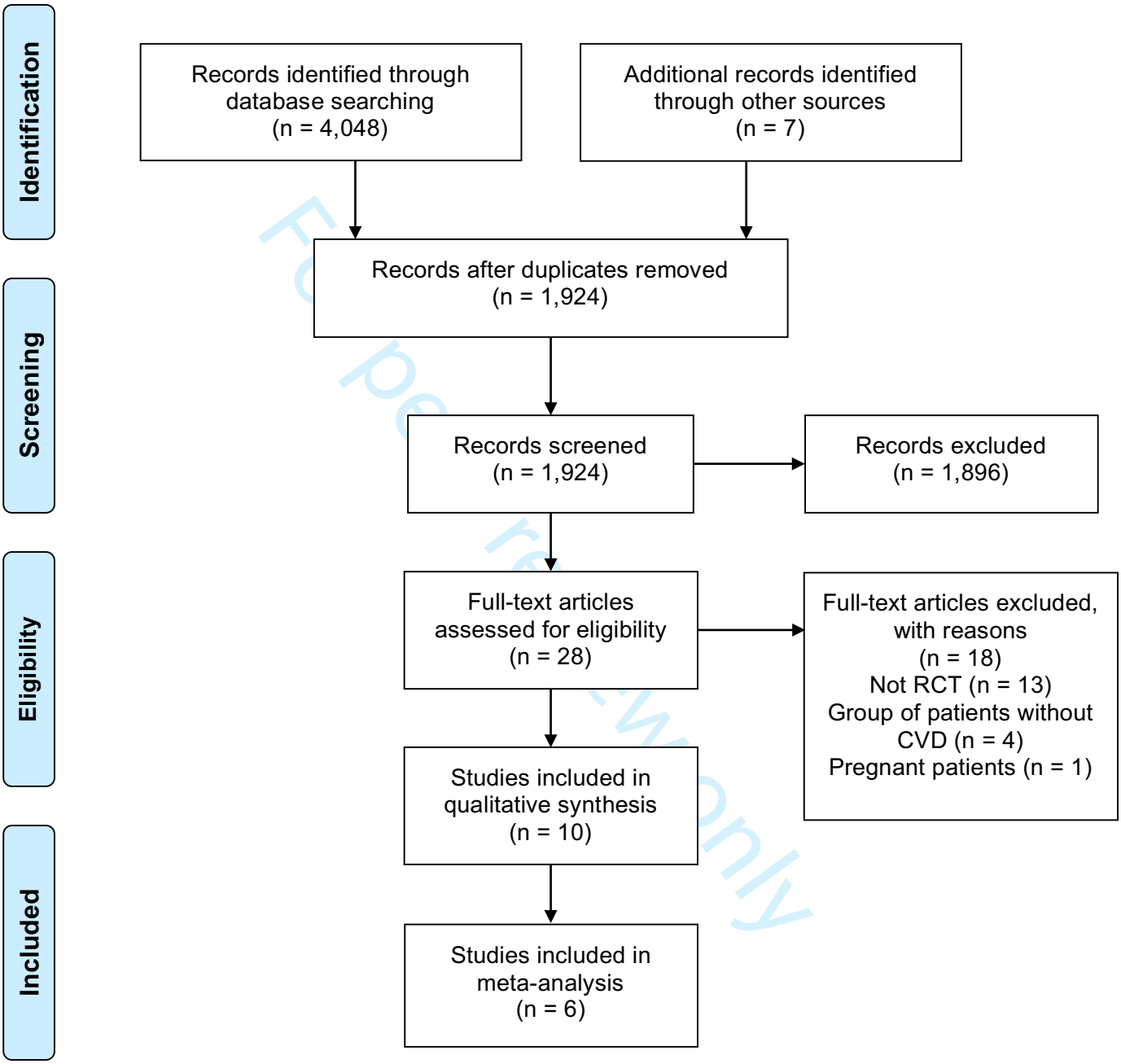
36. Ogunlewe MO, James O, Ajuluchukwu JN, Ladeinde AL, Adeyemo WL, Gbotolorun OM. Evaluation of haemodynamic changes in hypertensive patients during tooth extraction under local anaesthesia. The West Indian medical journal. 2011;60(1):91-5.

37. Santos-Paul MA, Neves IL, Neves RS, Ramires JA. Local anesthesia with epinephrine is safe and effective for oral surgery in patients with type 2 diabetes mellitus and coronary disease: a prospective randomized study. Clinics (Sao Paulo, Brazil). 2015;70(3):185-9.

38. Torres-Lagares D, Serrera-Figallo MA, Machuca-Portillo G, Corcuera-Flores JR, Machuca-Portillo C, Castillo-Oyague R, et al. Cardiovascular effect of dental anesthesia with articaine (40 mg with epinefrine 0,5 mg % and 40 mg with epinefrine 1 mg%) versus mepivacaine (30 mg and 20 mg with epinefrine 1 mg%) in medically compromised cardiac patients: a cross-over, randomized, single blinded study. Medicina oral, patologia oral y cirugia bucal. 2012;17(4):e655-60.

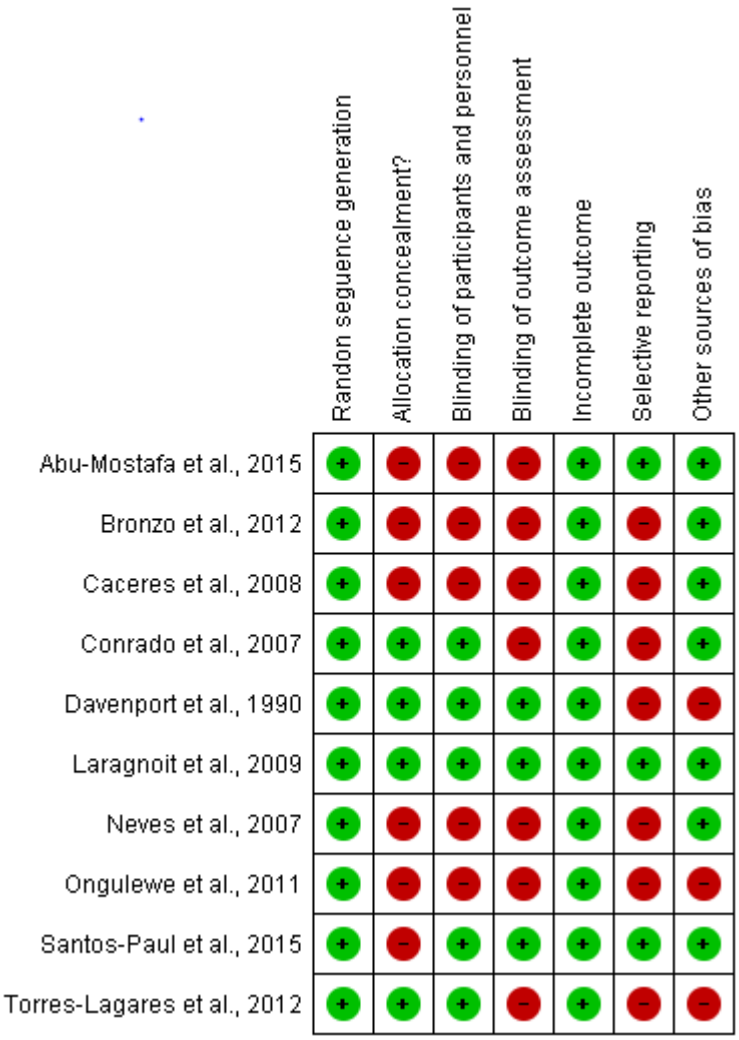
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S1 Figure. Flowchart for literature search and study selection



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S2 Figure. Risk of bias summary: review authors' judgments



S1 Table. Characteristic of included studies

STUDY	Population	Sample size (N)	Country	Female (n)	Mean age (y)	Local anesthetic + vasoconstrictor	Anesthetic without vasoconstrictor	Procedures	Outcomes measured
Abu-mostafa et al., 2015	Essential hypertension	45	Saudia Arabia	14	55.66±NR	Lidocaine 2% (Epinephrine$) Prilocaine 3% (Felypressin&)	Mepivacaine 3%	Dental extraction	Systolic and diastolic blood pressure, and heart rate
Bronzo et al., 2012	Essential hypertension	71	Brazil	44	48±12	Prilocaine 3% (Felypressin&)	Prilocaine 4%	Periodontal treatment	Anxiety, systolic, and diastolic blood pressure
Caceres et al., 2008	Chagas'disease, coronary artery disease	65	Brazil	21	54.73± 7.94	Prilocaine 3% (Felypressin&)	Lidocaine 2%	Dental procedures	Bleeding, arrhythmia, mean blood pressure, and heart rate
Conrado et al., 2007	Coronary artery disease	54	Brazil	20	56.65±8.27	Mepivacaine 2% (Epinephrine$)	Mepivacaine 3%	Dental extraction	Pain, ST segment depression, systolic and diastolic blood pressure, and heart rate
Davenport et al., 1990	Essential hypertension, coronary artery disease	9	US	0	62±NR	Lidocaine 2% (Epinephrine$)	Lidocaine 2%	Periodontal surgery	Bleeding, mean blood pressure, heart rate and plasma epinephrine, and norepinephrine levels
Laragnoit et al., 2009	Valvular disease	59	Brazil	NR	41.25±10.6	Lidocaine 2% (Epinephrine$)	Lidocaine 2%	Dental extraction or restorative procedure	Pain, arrhythmia, anxiety, mean blood pressure, systolic and diastolic blood pressure, and heart rate
Neves et al., 2007	Essential hypertension and coronary artery disease	62	Brazil	11	58.7±8.8	Lidocaine 2% (Epinephrine$)	Lidocaine 2%	Restorative procedure	Arrhythmia, ST segment depression, mean blood pressure, systolic and diastolic blood pressure, and heart rate
Ongulewe et al., 2011	Essential hypertension	33	Nigeria	20	50±11.7	Lidocaine 2% (Epinephrine@)	Lidocaine 2%	Dental extraction	Systolic and diastolic blood pressure, and heart rate
Santos-paul et al., 2015	Coronary disease	70	Brazil	20	63.4±8.3	Lidocaine 2% (Epinephrine$)	Lidocaine 2%	Dental extraction	Anxiety, systolic and diastolic blood pressure, heart rate, and blood glucose levels
Torres-lagares et al., 2012	Essential hypertension	10	Spain	NR	63±12,5	Articaine 4% (Epinephrine$, Epinephrine#) Mepivacaine 2% (Epinephrine$)	Mepivacaine 3%	Periodontal treatment	Mean blood pressure, systolic and diastolic blood pressure, heart rate, and oximetry

NR: not reported

Dose of Epinephrine: $ 1:100,000, # 1:200,000, @ 1:80,000. Dosage of Felypressin: & 0.03 IU/ml

S2 Table. - GRADE Assessment of quality of evidence

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Local Anesthesic with vasoconstrictor	Local Anesthesic without vasoconstrictor	Relative (95% CI)	Absolute (95% CI)		
Mean blood pressure during the procedure (assessed with: mmHg)												
1	randomised trials	serious [a]	not serious	serious [b]	serious [c]	none	10	10	-	SMD **0.05 SD lower** (0.93 lower to 0.82 higher)	⨁◯◯◯ VERY LOW	IMPORTANT
Systolic Blood Pressure during the procedure (assessed with: mmHg)												
3	randomised trials	very serious [d]	not serious	serious [b]	not serious	none	55	57	-	SMD **0.39 SD lower** (0.8 lower to 0.03 higher)	⨁◯◯◯ VERY LOW	NOT IMPORTANT
Diastolic Blood Pressure during the procedure (assessed with: mmHg)												
3	randomised trials	very serious [d]	serious [e]	serious [b]	not serious	none	55	57	-	SMD **0.51 SD lower** (1.07 lower to 0.06 higher)	⨁◯◯◯ VERY LOW	IMPORTANT
Heart Rate during the procedure (assessed with: bpm)												
3	randomised trials	very serious [d]	not serious	serious [f]	not serious	none	55	57	-	SMD **0.24 SD lower** (0.61 lower to 0.13 higher)	⨁◯◯◯ VERY LOW	NOT IMPORTANT
Mean blood Pressure after the procedure (assessed with: mmHg)												
2	randomised trials	serious [g]	not serious	serious [b]	not serious	none	38	41	-	SMD **0.26 SD lower** (0.71 lower to 0.18 higher)	⨁⨁◯◯ LOW	IMPORTANT
Systolic Blood Pressure after the procedure (assessed with: mmHg)												
3	randomised trials	very serious [h]	not serious	serious [b]	not serious	none	53	56	-	SMD **0.95 SD lower** (1.35 lower to 0.55 lower)	⨁◯◯◯ VERY LOW	NOT IMPORTANT
Diastolic Blood Pressure after the procedure (assessed with: mmHg)												
3	randomised trials	very serious [h]	serious [e]	serious [b]	not serious	none	53	56	-	SMD **0.53 SD lower** (1.06 lower to 0.01 higher)	⨁◯◯◯ VERY LOW	IMPORTANT
Heart Rate after the procedure (assessed with: HR)												

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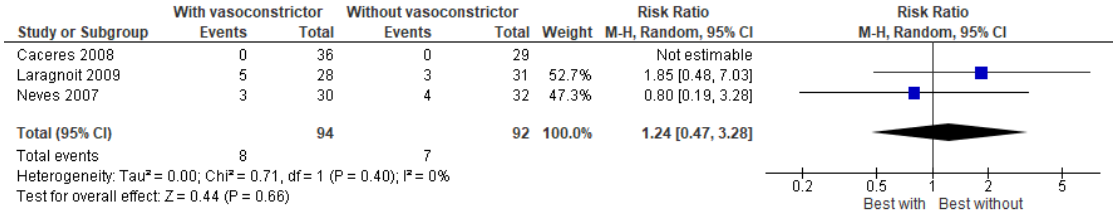
Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Local Anesthesic with vasoconstrictor	Local Anesthesic without vasoconstrictor	Relative (95% CI)	Absolute (95% CI)		
3	randomised trials	very serious [i]	not serious	serious [f]	not serious	none	55	57	-	SMD **0.2 SD lower** (0.57 lower to 0.17 higher)	⨁◯◯◯ VERY LOW	NOT IMPORTANT
Arrhythmia												
3	randomised trials	serious [i]	not serious	serious [j]	serious [c]	none	12/94 (12.8%)	6/92 (6.5%)	**RR 2.16** (0.87 to 5.41)	**76 more per 1.000** (from 8 fewer to 288 more)	⨁◯◯◯ VERY LOW	NÃO IMPORTANTE
ST segment depression												
2	randomised trials	serious [k]	serious [e]	serious [l]	serious [m]	none	7/57 (12.3%)	6/59 (10.2%)	**RR 1.53** (0.17 to 13.72)	**54 more per 1.000** (from 84 fewer to 1.000 more)	⨁◯◯◯ VERY LOW	NÃO IMPORTANTE

CI: Confidence interval; **SMD:** Standardised mean difference; **RR:** Risk ratio

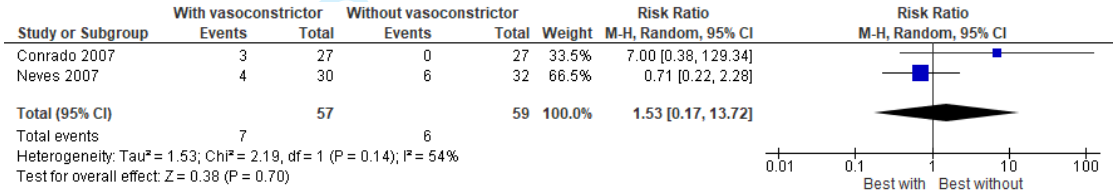
Explanations
a. The study was not blinding for outcome assessment. The protocol of the study was not registered previously.
b. Blood Pressor is a surrogate outcome.
c. Wide confidence interval considering the size of the generated effect
d. Three studies not blinded also alocation not concealment without register of protocols.
e. Moderate heterogeneity without explanation of differences between studies
f. Heart Rate is a surrogate outcome
g. One study was not blinded for outcome assessment.
h. Two studies from three of them were not blinded.
i. Three studies without blinding outcome assessment. Two of them not clear to allocation concealment and two without any previous register of protocol.
j. arrhythmias is a surrogate outcome
k. One study was not blinded for outcome assessment, and two without previous register of protocol.
l. ST segment depression is a surrogate outcome
m. one study with wide confidence interval consideringthe size of the generated effect

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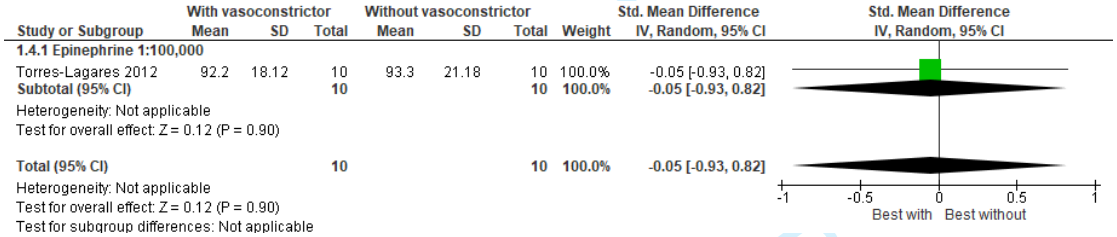
S3 Figure. Presence of arrhythmia associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor.



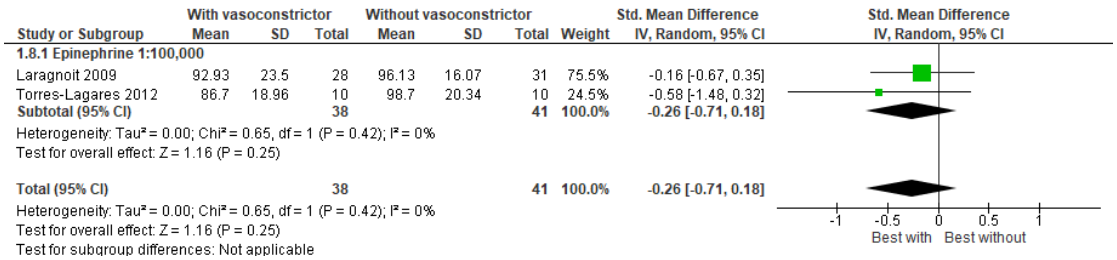
S4 Figure. Presence of ST segment depression associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor.



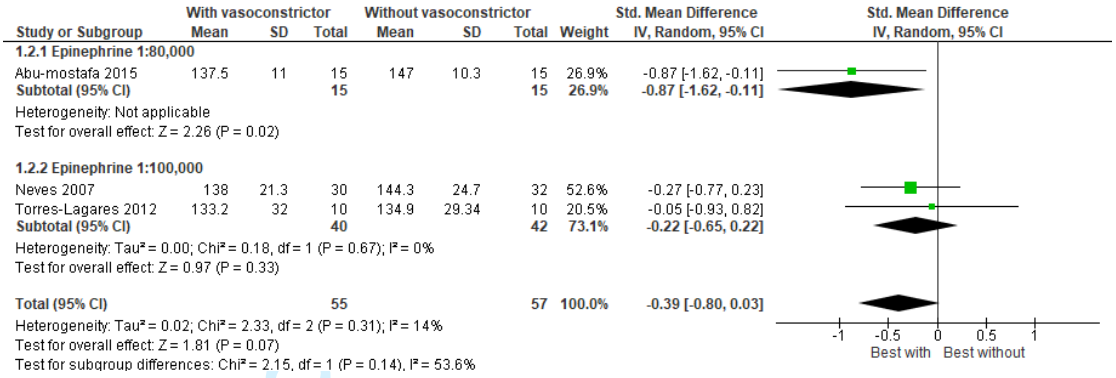
S5 Figure. Mean blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, during the procedure period.



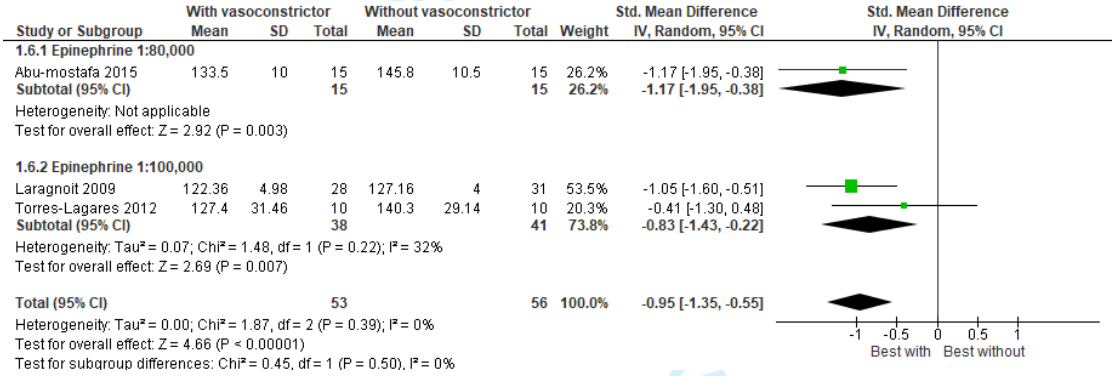
S6 Figure. Mean blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, after the procedure period.



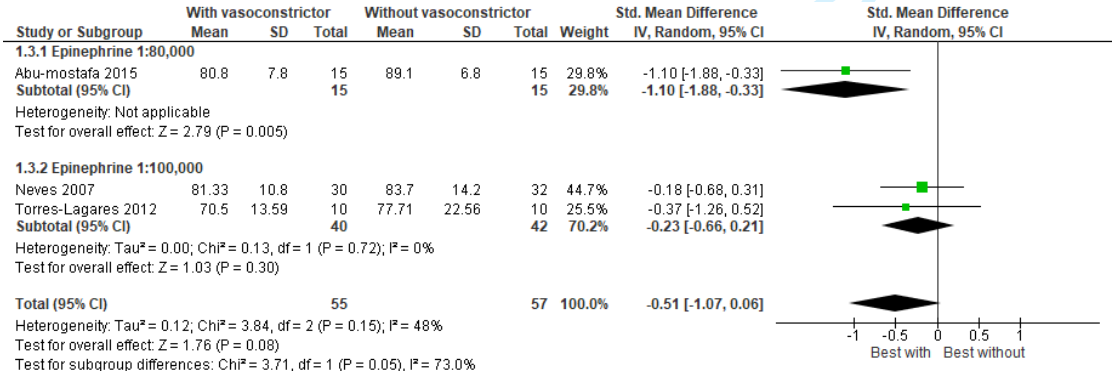
S7 Figure. Systolic blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, during the procedure period.



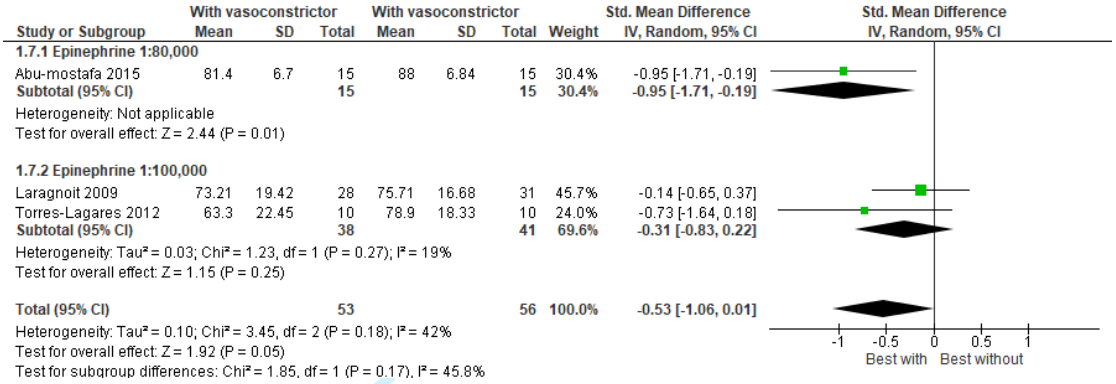
S8 Figure. Systolic blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, after procedure period.



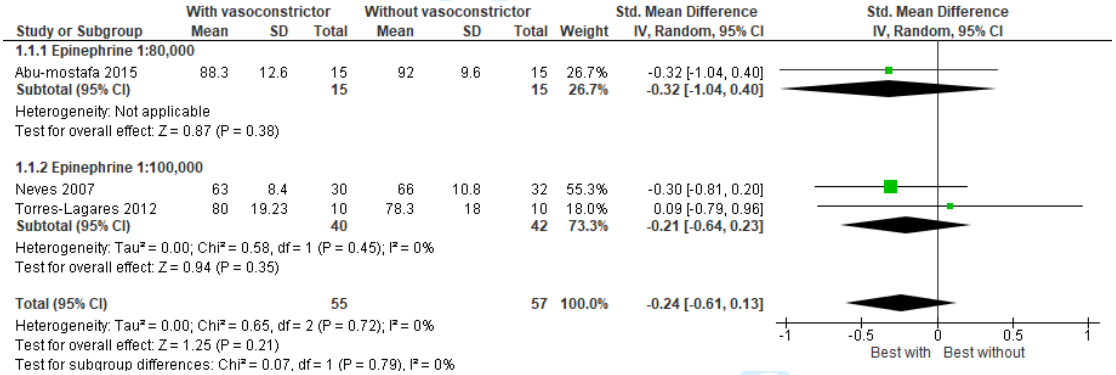
S9 Figure. Diastolic blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, during the procedure period.



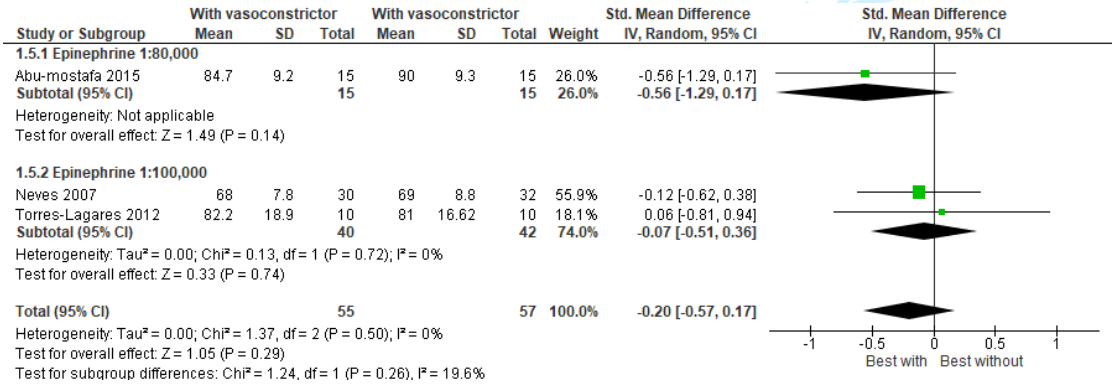
S10 Figure. Diastolic blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, after the procedure period.



S11 Figure. Heart rate associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, during the procedure period.



S12 Figure. Heart rate associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, after the procedure period.



S1 Appendix– Search strategy (via OVID, MEDLINE and EMBASE)

1 exp Dentistry/ (357819)
2 exp Dentistry, Operative/ (32163)
3 exp Dental Care/ (29438)
4 Dental Restoration, Permanent/ (18759)
5 Dental Restoration Repair/ (102)
6 Periodontal Debridement/ (191)
7 Subgingival Curettage/ (977)
8 Dental Scaling/ (3316)
9 Chronic Periodontitis/ (1971)
10 Periodontal Diseases/ (24137)
11 Periodontal Surgery.mp. (1302)
12 Periodontal treatment.mp. (2728)
13 Oral Surgical Procedures/ (5363)
14 exp Surgery, Oral/ (7419)
15 Tooth Extraction/ (17008)
16 Dental Prosthesis/ (3384)
17 “Root Canal Therapy”/ (11735)
18 exp Dental Implants/ (17311)
19 Dental Implants, Single Tooth/ (1901)
20 Dental Implantation/ (3773)
21 1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 (372164)
22 exp Anesthetics, Local/ (96504)
23 exp Anesthesia, Local/ (15673)
24 exp Anesthesia, Dental/ (10417)
25 Lidocaine/ (22512)
26 Prilocaine/ (2018)
27 Bupivacaine/ (10713)
28 Procaine/ (11313)
29 Mepivacaine/ (1899)
30 Carticaine/ (451)
31 Etidocaine/ (288)
32 22 OR 23 OR 24 OR 25 OR 26 OR 27 OR 28 OR 29 OR 30 OR 31 (114729)
33 exp Cardiovascular Diseases/ (2067079)
34 Cardiac.mp. (631932)
35 Coronary Disease/ (128393)
36 Coronary Artery Disease/ (47503)
37 Coronary arteriosclerosis.mp. (733)
38 Coronariopathy.mp. (29)
39 Arrhythmias, cardiac/ (56112)
40 Heart Valve Diseases/ (21116)
41 Heart Diseases/ (63528)
42 Heart Failure/ (63528)
43 Rheumatic Heart Disease/ (12263)
44 Myocardial Ischemia/ (34898)
45 Myocardial Infarction/ (151398)
46 Hypertension (210548)
47 Hypertensive Patients.mp. (25167)
48 33 OR 34 OR 35 OR 36 OR 37 OR 38 OR 39 OR 40 OR 41 OR 42 OR 43 OR 44 OR 45 OR 46 OR 47 (2314784)
49 21 AND 32 AND 48 (752)

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S2 Appendix– Risk of bias of RCts included (modified version of the Cochrane collaboration)

Author	Year	Random sequence generation	Allocation concealment	Blinding of participants and personnel	Blinding of outcome assessment	Incomplete outcome data	Selective reporting	Other sources of bias
Abu-Mostafa, et al. 2015)	2015	Definitely yes	Probably no	Probably no	Probably no	Probably yes	Probably yes	Probably yes
Bronzo, et al. 2012	2012	Probably yes	Probably no	Definitely no	Definitely no	Probably yes	Probably no	Probably yes
Caceres, et al. 2008	2008	Probably yes	Definitely no	Definitely no	Definitely no	Probably yes	Probably no	Probably yes
Conrado, et al. 2007	2007	Probably yes	Probably yes	Probably yes	Definitely no	Probably yes	Probably no	Probably yes
Davenport, et al. 1990	1990	Definitely yes	Probably yes	Probably yes	Probably yes	Probably yes	Probably no	Probably no
Laragnoit, et al. 2009	2009	Definitely yes	Probably yes	Definitely yes	Definitely yes	Probably yes	Probably yes	Probably yes
Neves, et al. 2007	2007	Probably yes	Probably no	Definitely no	Definitely no	Probably yes	Probably no	Probably yes
Ogunlewe, et al. 2011	2011	Probably yes	Probably no	Definitely no	Definitely no	Probably yes	Probably no	Probably no
Santos-Paul, et al. 2015)	2015	Probably yes	Probably no	Probably yes	Probably yes	Probably yes	Probably yes	Probably yes
Torres-Lagares, et al. 2012	2012	Definitely yes	Probably yes	Probably yes	Definitely no	Probably yes	Definitely no	Probably no

S3 Appendix- Characteristics of excluded studies

Excluded studies	**Exclusion reasons**
Abu-Mostafa et al. 2015	Group of patients without CVD
Agani et al. 2015	Not RCT
Bazhanov et al.1983	Not RCT
Brkovic et al. 2008	Group of patients without CVD
Caviedes et al. 1983	Not RCT
Cecava, Mocek, Sel. 1965	Not RCT
Harang. 1949	Not RCT
Montebugnoli, Pelliccioni, Borghi 1990	Not RCT
Nava, Torres-Martinez 1967	Not RCT
Neves et al. 2009	Pregnant patients
Niwa 1999	Group of patients without CVD
Ohgami, Suda 1979	Not RCT
Rollason et al. 1977	Not RCT
Salman, Schwartz 1955	Not RCT
Tucci, Gabrielli, Mendes 1977	Not RCT
Vieillefosse 1973	Not RCT
Yoshimura et al. 1987	Not RCT
Zivotic-Vanovic, Marjanovic 2006	Group of patients without CVD

CAD- Cardiovascular disease. RCT- Randomised controlled trial

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	S1 appendix
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	8
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	9
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	9

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	10
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	–
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	S1 Figure
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	S1 Table
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	11
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	13
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	13
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	S2 Table
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	–
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	17
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	18
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	20
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	20

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Local anesthetics combined with vasoconstrictors in patients with cardiovascular disease undergoing dental procedures: Systematic review and meta-analysis

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Primary Subject Heading:	Anaesthesia
Secondary Subject Heading:	Anaesthesia, Evidence based practice
Keywords:	Adult anaesthesia < ANAESTHETICS, Anaesthesia in cardiology < ANAESTHETICS, ORAL & MAXILLOFACIAL SURGERY, ORAL MEDICINE

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Title: Local anesthetics combined with vasoconstrictors in patients with cardiovascular disease undergoing dental procedures: Systematic review and meta-analysis

Running title: Local anesthetics combined with vasoconstrictors in patients with cardiovascular disease undergoing dental procedures

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ABSTRACT

Objectives: There is a lack of evidence about the use of local anesthetics (LA) in patients with cardiovascular diseases (CVD) in dental procedures. Thus, this study evaluated the safety of using LA with vasoconstrictor to determine the risk of cardiovascular events in patients with CVD.

Design: Systematic Review and meta-analysis.

Methods: We have searched in Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE (via Ovid), EMBASE (via Ovid), Healthstar (via Ovid), CINAHL, Web of Science, and ClinicalTrials.gov for randomized controlled trials (RCTs) up to January of 2020. We have included RCTs involving adults with CVD within two groups: intervention group with LA with vasoconstrictor and control group with LA without vasoconstrictor. The primary outcomes assessed were death, mortality by a specific cause, stroke, acute myocardial infarction, hospitalization, pain, bleeding and arrhythmias. The secondary outcomes were ST segment depression, anxiety, adverse effects and changes in hemodynamic parameters. The data were pooled using random effects meta-analyses and the confidence in the estimates was verified using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE).

Results: Ten RCTs (n=478 participants) were included. Most of them had a high risk of bias. There were more cases of pain and bleeding in groups without vasoconstrictor. Meta-analysis demonstrated a decrease in the systolic blood pressure with the use of local anesthetic with vasoconstrictor (SMD: -0.95, 95% CI: -1.35 to -0.55) after procedure. Overall, for the other outcomes assessed there was no statistical difference. The quality of evidence was considered low according to the GRADE profile.

Conclusions: The results suggest that the use of LA with vasoconstrictors (epinephrine in low doses) is safe in patients with some types of CVD. However, the low quality of evidence demonstrated that literature needs further studies in order to confirm these results.

Protocol registration: PROSPERO (CRD42016045421)

Keywords: Local Anesthetics, Dental Anesthesia, Dentistry, Cardiovascular Disease.

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Strengths and limitations of this study

- This review was conducted with methodological rigor, including assessment of the risk of bias and the quality of evidence; such information is not available in previous systematic reviews related to the subject.
- This study has explicit eligibility criteria, priority of primary and secondary outcomes, comprehensive and extensive search in the databases, with independent and peer reviews for each study and use of GRADE to assess the quality of the evidence and the strength of recommendation for the outcomes included in the meta-analysis.
- The primary studies included were a limiting factor for the findings of this review due to methodological bias observed in the included RCTs, as well as the differences observed in the types of CVDs, types and dosages of local anesthetics and vasoconstrictors.

INTRODUCTION

Cardiovascular diseases (CVD) are one of the causes of most premature deaths worldwide, representing 31% of all deaths. It is estimated that in 2012 CVD were the main cause of death in the world with approximately 17.5 million deaths. In 2002, three-quarters of CVD deaths occurred in low- or middle-income countries. CVD comprises essential hypertension, rheumatic heart diseases, ischemic heart diseases, cerebrovascular diseases, heart inflammatory diseases, and so forth (1).

In the dental office, the presence of patients with complex diseases such as CVD is increasingly common. These patients have access to better treatments and medications, which makes it possible to perform treatment in private dental clinics that previously was only executed by specialized hospitals (2, 3). However, careful evaluations should be conducted before the begin of dental treatment in patients with CVD in order to control stress and mainly to block nerve

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transmission of pain, thereby minimizing transient episodes that may trigger negative cardiovascular outcomes (3).

The use of local anesthetics (LA) and analgesics can control pain, while sedatives control the anxiety during dental procedures (3, 4). However, the use of LA has been the baseline of pain control for dental procedures for several decades (5,6).

Local anesthetics are combined with vasoconstrictors to increase duration of anesthetic effect, reduce systemic toxicity, and optimize soft tissue hemostasis (6). However, despite the beneficial properties of vasoconstrictors when combined with local anesthetics, dental professionals are concerned about the adverse effects associated with the vasoconstrictor, mainly with epinephrine, which is the most used vasoconstrictor in Dentistry (2, 7).

These systemic adverse effects are linked to inadequate procedures such as intravascular injection, high-dose injections, drug interactions, or instant absorption of the anesthetic agent when it is injected into an extremely vascular area that could induce adverse cardiovascular effects, especially in patients with CVD (3, 8-10).

Another factor for adverse effects is the concentration used in LAs. Currently, a greater dilution of the vasoconstrictor is recommended, mostly used at a concentration of 1: 100,000 for surgical procedures and, for other procedures, with a dilution of up to 1: 200,000 (2, 6). Nevertheless, for long and/or surgical procedures, LAs without vasoconstrictor do not offer a satisfactory anesthetic duration, which can cause pain resulting in systemic endogenous release of catecholamines (particularly noradrenaline) and causing to negative cardiovascular outcomes (11, 12, 13).

A previous study reported that the stress-induced release of catecholamines could be more than 10 times greater than the basal level (14). In stressful situations, such as pain and anxiety, the released endogenous catecholamines may also reach concentrations higher than the low epinephrine concentrations used in dental LA (14, 15). Thus, the literature recommends using the lowest effective concentration of epinephrine as possible for CVD patients, both for dermatological and dental procedures (9, 13). A few decades ago, the maximum dose of epinephrine in patients with cardiovascular disease was set at up to 200 µg (16). However, over the years several studies and guidelines on this

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subject have been published and suggested that 36 to 54 µg of epinephrine appear to be tolerated in most patients with hypertension or other cardiovascular diseases (2,17).

Previously two systematic reviews analyzed the most frequent complications in patients with CVD after dental local anesthesia with vasoconstrictor, disclosed in electrocardiogram arrhythmias (12, 13). However, both reviews differed from the present study. Bader et al (2002) did not describe the inclusion criteria and risk of bias (12). In addition, the authors included non-randomized clinical trials and a group of patients without CVD. Godzieba et al (2014) also included non-randomized clinical trials and patients without CVD (13).

LA have a long history of safety and efficacy in dental patients if used in recommended doses and techniques (3,6). However, dental professionals still have doubts about their use in patients with CVD, especially when combined with vasoconstrictor (13). Thus, this systematic review evaluated the safety in the use of LA with vasoconstrictor and determined the risk of cardiovascular events in patients with CVD.

METHODS AND ANALYSES

The systematic review was performed according to the recommendations specified in the Cochrane Handbook for Interventional Reviews and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses statement (PRISMA) (18).

Protocol and Registration

The protocol of this systematic review was registered on the PROSPERO – *International Prospective Register of Systematic Reviews* (registration number - CRD42016045421).

Eligibility criteria

Inclusion criteria

Patients: adult patients with CVD (essential hypertension, rheumatic heart diseases, ischemic heart diseases, cerebrovascular diseases, and heart inflammatory diseases) (1).

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Interventions: one arm wherein patients received LA with vasoconstrictors compared to another arm wherein patients received LAs without vasoconstrictors.

Procedures: patients who undergo tooth extraction, dental restorations, treatment and periodontal surgery, among others.

Type of study: considering that observational studies increase the risk of bias to answer this question, the study design that can generate high quality evidence, according to the evidence hierarchy, would be the randomized controlled studies (RCTs). We include two types of RCT designs. In the first type, patients are randomized to receive either LAs with vasoconstrictors during the first dental procedure or LAs without vasoconstrictors during the second dental procedure or vice versa. In the second type, patients are randomized to receive only one type of LA, with or without vasoconstrictors, during the dental procedure.

Language: any language without restriction.

Outcomes measures

The safety of the use of local anesthetics with vasoconstrictors in patients with CVD was based on the number of reports considering the following primary and secondary outcomes:

Primary outcomes:

- death;
- mortality by a specific cause;
- stroke;
- acute myocardial infarction;
- hospitalization;
- pain;
- bleeding;
- arrhythmias;

Secondary outcomes:

- ST segment depression;
- anxiety;

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- adverse effects;
- changes in hemodynamic parameters.

Exclusion criteria

We exclude studies involving patients with untreated or out-of-control essential hypertension or any type of CAD, who were pregnant or breastfeeding, who were allergic to the LAs used in the studies, with out-of-control diabetes mellitus, or who have had recent myocardial infarction, cancer, and malignant hypertension.

Search methods for primary studies

Electronic searches

The following electronic databases were searched: the Cochrane Central Register of Controlled Trials (CENTRAL) part of The Cochrane Library; MEDLINE (Ovid); EMBASE (Ovid); Healthstar (Ovid); CINAHL (Cumulative Index to Nursing and Allied Health Literature); and Web of Science, from their inception to January of 2020, without restrictions on the language and date of publication.

Searching other resources

We performed search in registration of clinical trials: https://clinicaltrials.gov, WHO clinical trials registry, http://www.ensaiosclinicos.gov.br; trials registry and bank of Brazil thesis (CAPES); conference proceedings of the Brazilian Congress of Cardiology, in the Brazilian Congress of Anesthesiology, and in the International Congress of Dentistry (CIOSP). Two reviewers analyzed the reference list or quotations found in secondary studies to verify and identify possible eligible studies.

Search strategy

The search strategy was individually conducted by type of dental intervention; type of anesthetic; and type of CVD. We have adapted the search strategy according to each database. The search strategy in Medline (via Ovid) is in Appendix 1.

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Eligibility determination

Six reviewers (CCG vs CCB vs JOA and RHLM vs NKA vs JCR) working in trio independently evaluated whether summaries are in accordance with eligibility criteria. Discordances were resolved by consensus, and contentious issues were discussed with a third reviewer (LCL).

To exclude duplicate articles, reviewers analyzed all eligible articles and identify those with one or more authors in common. In case of duplicate publications, we used the article with more complete data.

Data extraction

Six reviewers (CCG vs CCB, JOA vs RHLM and NKA vs JCR), working in pairs and independently extracted data and recorded information regarding patients, methods, interventions, outcomes, and missing outcome data using standardized and pretested data extraction forms with instructions (Figure 1). Before initiating data extraction, we conducted calibration exercises to ensure consistency among the reviewers, both in screening and data extraction. We contacted the study authors as necessary to resolve any uncertainties. Reviewer's resolved disagreements by discussion, and one arbitrator (LCL) judged unresolved disagreements.

Risk of bias in individual studies

Using a modified version of the Cochrane collaboration risk of bias tool (19, 20), The same pairs of reviewers independently rated the risk of bias for each RCT according to the following criteria: random sequence; allocation concealment; blinding of the patient, healthcare professionals, outcome assessors, data collectors, and data analysts; incomplete outcome data; selective outcome reporting; and major baseline imbalance.

Reviewers gave response options of "definitely yes," "probably yes," "probably no," and "definitely no" for each of the domains, with the options "definitely yes" and "probably yes" ultimately being assigned a low risk of bias and "definitely no" and "probably no" as having a high risk of bias (21). Reviewer's resolved disagreements by discussion, and one arbitrator (LCL) judge unresolved disagreements. Risk of bias of RCTs included are in Figure 2 and Appendix 2.

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Explaining the heterogeneity of evidence

Possible explanations for heterogeneity included: a) age- the older the age, the higher the risk of cardiovascular transient episodes; b) gender- women outnumber men in deaths due to CVD; c) vasoconstrictor type- vasoconstrictors are linked to receptors α and β. However, some of these are more often linked to cardiac receptor β (except for felypressin, which links to the vasopressin receptor v1, present in the smooth muscles of blood vessel walls, raise cardiac frequency, and thus, higher risks of transient episodes are expected (22); d) vasoconstrictor concentration - which may vary from a 1:2,500 to a 1:200,000 greater risk is expected with higher vasoconstrictor concentration; e) dental procedure duration- the longer the duration to perform the procedure (surgical or periodontal procedures take longer than restorative procedures), the higher the concentration of anesthetic agent necessary, and the stronger the toxicity to the cardiovascular system, thereby increasing the risks of transient episodes in long-duration procedures; f) dental procedure type: usually surgical procedures (periodontal, extraction, and implantation) trigger great stress in the patient, thus increasing the risk of transient episodes. For the all outcomes, subgroup analyses were not possible due to the low number of studies included.

We ranked heterogeneity associated with pooled effect estimates with the use of the χ^2 test and the I^2 statistic (23). The following heterogeneities were considered: 0–25% (low heterogeneity), 50% (moderate heterogeneity), and 75% (high heterogeneity) (20).

Data synthesis

Analyses were conducted for each vasoconstrictor and for each outcome of interest. Confidence in the findings was determined by estimates for each body of evidence. The studies which reported continuous outcomes presented the date in Standard Mean Difference (SMD) and 95% Confidence Interval (95% CI) and dichotomous outcomes were reported as risk ratio (RR) combined with the 95% CI.

Other details about the methods adopted for the synthesis of the data can be found in the publication of the protocol of this review (24). The random effect in meta-analysis was performed using the Review Manager 5 Software (version 5.3).

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Summarizing evidence

The quality of evidence was independently evaluated (confidence in effect estimates) for each for each outcome reported using GRADE (25). Results are presented in evidence profiles, as recommended by the GRADE Working Group (26, 27).

In this approach, randomized trials begin with evidence of high quality may have their qualities diminished according to the judgment of one or more of the five categories of limitations presented below: risk of bias (assessed for each study as described previously), inconsistency, indirect evidence, imprecision and publication bias (25, 28). We used GRADE methodology to rate confidence in estimates of effect for each outcome as high, moderate, low or very low. Evidence profiles are provided brief presentations of evidence quality and effect magnitude. With the help of the software program GRADEpro (http://ims.cochrane.org/gradepro).

Patient and public involvement

No patient was involved in this study.

RESULTS

Literature search results

From 4,048 publications retrieved from the databases, 1,924 articles remained for the titles and abstracts selection after removing duplicates. From the 28 pre-selected papers, 18 were excluded (Appendix 3) and 10 met eligibility criteria (29-38). Six studies (29, 31, 32, 34, 35, 38) were able to be included in the meta-analysis (Figure 1).

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Description of the studies (Table 1)

Methods

Ten RCTs were included, encompassing a total of 478 patients with CVD, from which 226 patients had coronary artery disease, 226 patients had essential hypertension, 59 patients had valvular diseases and 33 patients had Chagas disease. Table 1 describes the characteristics of the studies in which it was observed that the majority of the studies used lidocaine as LA and epinephrine as a vasoconstrictor. The studies were mainly published in Brazil between 2007 and 2014. The different reasons for exclusion of articles are found in Appendix 3.

Intervention

Eight studies (30, 32-38) used the same local anesthetic both in the control and intervention groups; two studies (29, 38) included more than one group in the intervention group; five studies (33-37) used lidocaine both in the intervention and control groups; one study (30) used prilocaine both in the intervention and control groups; two studies (32, 38) used mepivacaine both in the intervention and control groups; and three studies (29, 31, 38) used different local anesthetic solutions between the intervention and control groups.

Eight studies (29, 32-38) used the vasoconstrictor epinephrine; followed by felypressin, which was present in three studies (29-31) and only one study (29) used both epinephrine and felypressin (Table 1).

Participants

Five studies (29, 30, 33, 36, 38) included patients with essential hypertension. Two studies (32, 37) included patients with coronary artery disease. Two studies (33, 35) included patients with essential hypertension and coronary artery disease. One study (34) included patients with valvular diseases, and one study (31) included patients with Chagas disease and coronary artery disease.

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Table 1. Characteristic of included studies

STUDY	Population	Sample size (N)	Country	Female (n)	Mean age (y)	Local anesthetic + vasoconstrictor	Anesthetic without vasoconstrictor	Procedures	Outcomes measured
Abu-mostafa et al., 2015	Essential hypertension	45	Saudia Arabia	14	55.66±NR	Lidocaine 2% (Epinephrine$) Prilocaine 3% (Felypressin&)	Mepivacaine 3%	Dental extraction	Systolic and diastolic blood pressure, and heart rate
Bronzo et al., 2012	Essential hypertension	71	Brazil	44	48±12	Prilocaine 3% (Felypressin&)	Prilocaine 4%	Periodontal treatment	Anxiety, systolic, and diastolic blood pressure
Caceres et al., 2008	Chagas'disease, coronary artery disease	65	Brazil	21	54.73± 7.94	Prilocaine 3% (Felypressin&)	Lidocaine 2%	Dental procedures	Bleeding, arrhythmia, mean blood pressure, and heart rate
Conrado et al., 2007	Coronary artery disease	54	Brazil	20	56.65±8.27	Mepivacaine 2% (Epinephrine$)	Mepivacaine 3%	Dental extraction	Pain, ST segment depression, systolic and diastolic blood pressure, and heart rate
Davenport et al., 1990	Essential hypertension, coronary artery disease	9	US	0	62±NR	Lidocaine 2% (Epinephrine$)	Lidocaine 2%	Periodontal surgery	Bleeding, mean blood pressure, heart rate and plasma epinephrine, and norepinephrine levels
Laragnoit et al., 2009	Valvular disease	59	Brazil	NR	41.25±10.6	Lidocaine 2% (Epinephrine$)	Lidocaine 2%	Dental extraction or restorative procedure	Pain, arrhythmia, anxiety, mean blood pressure, systolic and diastolic blood pressure, and heart rate
Neves et al., 2007	Essential hypertension and coronary artery disease	62	Brazil	11	58.7±8.8	Lidocaine 2% (Epinephrine$)	Lidocaine 2%	Restorative procedure	Arrhythmia, ST segment depression, mean blood pressure, systolic and diastolic blood pressure, and heart rate
Ongulewe et al., 2011	Essential hypertension	33	Nigeria	20	50±11.7	Lidocaine 2% (Epinephrine@)	Lidocaine 2%	Dental extraction	Systolic and diastolic blood pressure, and heart rate
Santos-paul et al., 2015	Coronary disease	70	Brazil	20	63.4±8.3	Lidocaine 2% (Epinephrine$)	Lidocaine 2%	Dental extraction	Anxiety, systolic and diastolic blood pressure, heart rate, and blood glucose levels
Torres-lagares et al., 2012	Essential hypertension	10	Spain	NR	63±12,5	Articaine 4% (Epinephrine$, Epinephrine#) Mepivacaine 2% (Epinephrine$)	Mepivacaine 3%	Periodontal treatment	Mean blood pressure, systolic and diastolic blood pressure, heart rate, and oximetry

NR: not reported
Dose of Epinephrine: $ 1:100,000, # 1:200,000, @ 1:80,000. Dosage of Felypressin: & 0.03 IU/ml

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Risk of bias in the included studies (Appendix 2 and Figure 2)

Randomization and allocation

From the 10 RCTs included, three studies (30, 33, 38) used the crossover design, what reduces the quality of the evidence, and performed wash-out periods between 10 minutes to 3 weeks. Randomization by computer was performed by only one study (38). Two studies (33, 34) used the random number table. Two studies (29, 32) did a draw of the ID cards. Five studies (30, 31, 35-37) did not mention the randomization method.

The allocation concealment method was considered of low risk of bias in one study (38), when randomization was performed by computer.

Blinding

Two study (34, 37) reported performing double-blinding, but only one study (34) partially described. One study (37) only mentioned that the blinding was performed, without an explanation of the method. Five studies (29-31, 35, 36) were not conducted blinded.

Only two studies (32, 34) did not mention the blinding but used allocation concealment with delivered in sealed envelopes, and the cartridges were covered with a white strip.

Loss to follow-up

The studies reported no loss to follow-up.

Selective reporting of outcomes

Three studies (29, 34, 37) had the registry of clinical trial reporting pre-specified outcomes. Two studies (34, 37) made the registry at ClinicalTrials.gov and one study (29) at the Institute Research Center.

Other potential source of bias

Four studies (29-31, 34) mentioned that there was no external funding. One study (37) received Brazilian funding through the São Paulo State Research Support Foundation (FAPESP), which is non-profit.

Six studies (29-32, 34, 35) declared no potential conflicts of interest. One study

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(34) mentioned both no external funding sources and no potential conflict of interest only in the registry of clinical trials

Reported primary outcomes

None of the studies reported death, mortality by specific cause, stroke, acute myocardial infarction or hospitalization. The only primary outcomes reported by the authors included in this review were pain, bleeding and arrhythmias (Table 1). Due to the small number of studies and the lack of numerical data, it was not possible to perform the data meta-analysis of pain and bleeding. The reports of these outcomes were descriptive and qualitative.

Pain

Two studies (32, 34) evaluated the pain outcome. Conrado et al. (2007) assessed the pain outcome at the extraction site in 54 patients with coronary disease and observed pain report in only one patient from the intervention group (mepivacaine 2% with epinephrine 1:100,000), and in 9 patients from the control group (mepivacaine 3% without vasoconstrictor). The authors did not describe how they have assessed pain and explained that the only one case of pain in the intervention group was caused by a tooth that was ankylosed (32).

Laragnoit, et al. (2009) examined 59 participants with several valvular diseases and described the results only in a qualitative way. It was observed that there were more complaints of pain in the control group (lidocaine 2% without vasoconstrictor) compared to the intervention group (lidocaine 2% with epinephrine 1:100,000) (34). The authors did not describe how they assessed pain.

Bleeding

Two studies (31, 33) evaluated the bleeding outcome. Caceres et al. (2008) did not reported bleeding during the procedures. In both groups, authors did not describe how they assessed this outcome (31). Davenport, et al. (1990) examined 9 participants in the control and intervention groups and observed that all participants in the intervention group (lidocaine 2% with epinephrine 1:100,000) obtained adequate hemostasis versus 7 participants with inadequate hemostasis in the control group (lidocaine 2% without vasoconstrictor). The authors did not describe how they assessed bleeding (33).

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Arrhythmia

Three studies (31, 34, 35) assessed this outcome. Caceres et al. (2008) analyzed the presence of ventricular extrasystoles that remained stable in different measured moments, with a similar behavior in both groups (31).

Laragnoit, et al. (2009) reported eight patients with arrhythmia, being five patients from the intervention group (lidocaine 2% with epinephrine 1:100,000) and three patients in the control group (lidocaine 2%), with no observed statistical difference between both groups (34).

Neves et al. (2007) reported seven patients with arrhythmia, being three (11.1%) from the intervention group (lidocaine 2% with epinephrine 1:100,000) and four (13.8%) from the control group (lidocaine 2% without vasoconstrictor), with no observed significant statistical difference between the two groups (35).

The meta-analysis demonstrated no difference in the arrhythmias between the groups (RR: 1.24, 95% CI: 0.47 to 3.28 low quality evidence) (Figure 3, Table 2).

Reported secondary outcomes

Adverse effects were not reported in the studies.

ST segment depression

Two studies (32, 35) evaluated the presence of ischemic episodes (measured as ST segment depression).

Conrado et al. (2007) observed the occurrence of the ST segment depression greater than 1mm in three patients in the intervention group (mepivacaine 2% with epinephrine 1:100,000) during anesthetic injection. No event was observed in the control group (mepivacaine 3% without vasoconstrictor) and no statistical difference was observed between the groups (32).

Neves et al. (2007) observed ST segment depression greater than 1mm in four patients (14.8%) in the intervention group (lidocaine 2% with epinephrine 1:100,000) and in six patients (20.7%) in the control group (lidocaine 2% without vasoconstrictor), with no significant statistical difference between the groups (35).

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Table 2. - GRADE Assessment of quality of evidence

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Local Anesthesic with vasoconstrictor	Local Anesthesic without vasoconstrictor	Relative (95% CI)	Absolute (95% CI)		
Mean blood pressure during the procedure (assessed with: mmHg)												
1	randomised trials	serious [a]	not serious	serious [b]	serious [c]	none	10	10	-	SMD **0.05 SD lower** (0.93 lower to 0.82 higher)	⨁◯◯◯ VERY LOW	IMPORTANT
Systolic Blood Pressure during the procedure (assessed with: mmHg)												
3	randomised trials	very serious [d]	not serious	serious [b]	not serious	none	55	57	-	SMD **0.39 SD lower** (0.8 lower to 0.03 higher)	⨁◯◯◯ VERY LOW	NOT IMPORTANT
Diastolic Blood Pressure during the procedure (assessed with: mmHg)												
3	randomised trials	very serious [d]	serious [e]	serious [b]	not serious	none	55	57	-	SMD **0.51 SD lower** (1.07 lower to 0.06 higher)	⨁◯◯◯ VERY LOW	IMPORTANT
Heart Rate during the procedure (assessed with: bpm)												
3	randomised trials	very serious [d]	not serious	serious [f]	not serious	none	55	57	-	SMD **0.24 SD lower** (0.61 lower to 0.13 higher)	⨁◯◯◯ VERY LOW	NOT IMPORTANT
Mean blood Pressure after the procedure (assessed with: mmHg)												
2	randomised trials	serious [g]	not serious	serious [b]	not serious	none	38	41	-	SMD **0.26 SD lower** (0.71 lower to 0.18 higher)	⨁⨁◯◯ LOW	IMPORTANT
Systolic Blood Pressure after the procedure (assessed with: mmHg)												
3	randomised trials	very serious [h]	not serious	serious [b]	not serious	none	53	56	-	SMD **0.95 SD lower** (1.35 lower to 0.55 lower)	⨁◯◯◯ VERY LOW	NOT IMPORTANT
Diastolic Blood Pressure after the procedure (assessed with: mmHg)												
3	randomised trials	very serious [h]	serious [e]	serious [b]	not serious	none	53	56	-	SMD **0.53 SD lower** (1.06 lower to 0.01 higher)	⨁◯◯◯ VERY LOW	IMPORTANT
Heart Rate after the procedure (assessed with: HR)												
3	randomised trials	very serious [i]	not serious	serious [f]	not serious	none	55	57	-	SMD **0.2 SD lower** (0.57 lower to 0.17 higher)	⨁◯◯◯ VERY LOW	NOT IMPORTANT

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Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Local Anesthesic with vasoconstrictor	Local Anesthesic without vasoconstrictor	Relative (95% CI)	Absolute (95% CI)		
Arrhythmia												
3	randomised trials	serious [i]	not serious	serious [j]	serious [c]	none	12/94 (12.8%)	6/92 (6.5%)	**RR 2.16** (0.87 to 5.41)	**76 more per 1.000** (from 8 fewer to 288 more)	⨁◯◯◯ VERY LOW	NÃO IMPORTANTE
ST segment depression												
2	randomised trials	serious [k]	serious [e]	serious [l]	serious [m]	none	7/57 (12.3%)	6/59 (10.2%)	**RR 1.53** (0.17 to 13.72)	**54 more per 1.000** (from 84 fewer to 1.000 more)	⨁◯◯◯ VERY LOW	NÃO IMPORTANTE

CI: Confidence interval; **SMD:** Standardised mean difference; **RR:** Risk ratio

Explanations

a. The study was not blinding for outcome assessment. The protocol of the study was not registered previously.
b. Blood Pressor is a surrogate outcome.
c. Wide confidence interval considering the size of the generated effect
d. Three studies not blinded also alocation not concealment without register of protocols.
e. Moderate heterogeneity without explanation of differences between studies
f. Heart Rate is a surrogate outcome
g. One study was not blinded for outcome assessment.
h. Two studies from three of them were not blinded.
i. Three studies without blinding outcome assessment. Two of them not clear to allocation concealment and two without any previous register of protocol.
j. arrhythmias is a surrogate outcome
k. One study was not blinded for outcome assessment, and two without previous register of protocol.
l. ST segment depression is a surrogate outcome
m. one study with wide confidence interval consideringthe size of the generated effect

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The meta-analysis demonstrated no difference in the ST segment depression between the groups (RR: 1.53, 95% CI: 0.17 to 13.72 low quality evidence) (Figure 4, Table 2).

Anxiety

Two studies (34, 37) evaluated the anxiety outcome. Laragnoit et al. (2009) assessed the anxiety of patients using the STAI scale (Spielberger State Anxiety Inventory) before the procedure and reported no difference in the mean values of anxiety levels between the intervention and control groups (34).

Santos-Paul et al. (2015) assessed the anxiety of patients using Buchanan and Niven facial scale in different periods (one hour before the procedure, five minutes after the procedure and one hour after the procedure). They reported no significant difference between the control and intervention groups at any of the evaluated time periods (37).

Mean blood pressure

Five studies (31, 33, 34, 35, 38) evaluated the mean blood pressure values. In four studies (31, 33, 34, 35) there were no significant statistical difference between the assessed groups. Torres-Lagares et al. (2012) reported increase in the average of mean blood pressure in the intervention group (mepivacaine 2% with epinephrine 1:100,000) in comparison to the other three intervention groups (articaine 4% with epinephrine 1:200,000, articaine 4% with epinephrine 1:100,000, mepivacaine 3% without vasoconstrictor) (38). Three studies (31, 33, 35) mention their results only in graphs and it was not possible to perform meta-analysis.

Meta-analysis was conducted with the Laragnoit et al. (2009) and Torres-Lagares et al. (2012) studies (34, 38). No difference was observed in the mean blood pressure between the groups during the procedure period (SMD: -0.05, 95% CI: -0.93 to 0.82, very low-quality evidence), and after the procedure period (SMD: -0.26, 95% CI: -0.71 to 0.18, low quality evidence). Time periods have been adapted to carry out the meta-analysis (Figures 5 and 6, Table 2).

Systolic blood pressure

Eight studies (29, 30, 32, 34-38) evaluated this outcome. In seven studies (30, 32, 34-38) there were no significant statistical difference between the assessed groups.

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Abu-Mostafa et al. (2015) observed a significant increase in the systolic blood pressure after dental extraction in the control group (mepivacaine 3% without vasoconstrictor) compared to the intervention group (lidocaine 2% with epinephrine 1:80,000) after the procedure (29).

Four studies (30, 32, 36, 37) could not be included in the meta-analysis because their data were presented in graphs. Even after several attempts to contact the authors, they did not respond. Additionally, unsuccessful attempts were made to extract data using the Digitizelt software.

Meta-analysis showed no difference in systolic blood pressure while using local anesthetic with vasoconstrictor compared to using only local anesthetic without vasoconstrictor during the procedure period (SMD: -0.39, 95% CI: -0.80 to 0.03, very low-quality evidence) (Figure 7, Table 2). After the procedure period, meta-analysis demonstrated a decrease in the systolic blood pressure with the use of local anesthetic with vasoconstrictor (SMD: -0.95, 95% CI: -1.35 to -0.55.) (Figure 8, Table 2).

Diastolic blood pressure

Eight studies (29, 30, 32, 34-38) evaluated the average diastolic blood pressure. In six studies (29, 32, 34-37) there was no significant statistical difference between the assessed groups.

In Bronzo et al. (2012), there was an increase in diastolic blood pressure during the dental procedures in both groups. However, in the intervention group (prilocaine 2% with felypressin 0,03 IU) diastolic blood pressure was greater than in the control group (prilocaine 4% without vasoconstrictor) (30). Torres-Lagares et al. (2012) reported diastolic blood pressure values significantly higher in the intervention group (mepivacaine 2% with epinephrine 1:100.000) in comparison to the other two groups intervention groups (articaine 4% with epinephrine 1:200,000; articaine 4% with epinephrine 1:100,000) and control group (mepivacaine 3% without vasoconstrictor) (38).

Four studies (30, 32, 36, 37) could not be included in the meta-analysis because their data were presented in graphs. Even after several attempts to contact the authors, they did not respond. Additionally, unsuccessful attempts were made to extract data using the Digitizelt software.

Meta-analysis showed no statistical difference between the diastolic blood pressure during the procedure (SMD: -0.51, 95% CI: -1.07 to 0.06) and after the procedure periods (SMD: -0.53, 95% CI: -1.06 to 0.01). However, it was noted that there

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was a tendency, with no statistical significance, in favor of the use of LA with vasoconstrictor (Figures 9 and 10). The quality of evidence was considered very low for both during and after the procedure periods (Table 2).

Heart rate

Nine studies (29, 31-38) assessed this outcome and in any of them significant difference between the groups was found. Six studies (31-34, 36, 37) could not be included in the meta-analysis because their data were presented in graphs. Even after several attempts to contact the authors, they did not respond. Additionally, unsuccessful attempts were made to extract data using the DigitizeIt software.

Only three studies (29, 35, 38) could be included in meta-analysis from which numerical data was available. Meta-analysis showed no statistical difference for the periods during (SMD: -0.24, 95% CI: -0.61 to 0.13) after the procedure (SMD: -0.20, 95% CI: -0.57 to 0.17) for heart rate between the groups (Figures 11 and 12), being the evidence considered of very low quality (Table 2).

DISCUSSION

Discussion of main findings with the literature

The use of local anesthetic with vasoconstrictor decreased pain and bleeding, but neither outcome could be included in the statistical analysis due to the lack of data. The findings of meta-analysis showed decreasing on the systolic blood pressure with the use of local anesthetic with vasoconstrictor compared to control group after the procedure. No significant difference was observed comparing LA with and without the use of vasoconstrictor for the outcomes anxiety, arrhythmia, ST segment depression, mean blood pressure, diastolic blood pressure and rate heart during and after the dental procedure.

In general, the quality of the evidence was considered low for the outcomes evaluated mainly due to the risk of bias, such as lack of blinding, concealment of the allocation not clearly described and problems due to the high heterogeneity between clinical trials and the inaccuracy of the findings.

Considering previous systematic reviews found in literature related to this subject (12, 13), methodological differences and in the proposed objectives, as well as the

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absence of meta-analysis did not allow the direct comparison of their results with the present study.

Bader et al. (2002) evaluated the cardiovascular adverse effects risk on the use of LA with epinephrine and the use of epinephrine-impregnated retraction cord. The authors included healthy patients and patients with CVD using LA with or without vasoconstrictor, and they also included non-randomized clinical trials, different from the objectives and methods of our review. The authors reported that the adverse effects in patients with essential hypertension are uncommon and that there is a small risk for hemodynamic changes, such as blood pressure and heart rate.

Another systematic review (13) evaluated the use of LA with vasoconstrictor in patients with CVD. However, studies with healthy patients and non-randomized studies were included and outcomes were not prioritized. The review verified 87 complications that occurred after the administration of the local anesthetic. Although cardiac arrhythmia was the most frequent outcome, it was not clinically relevant. The authors concluded that use with the maximum limit of four dental cartridges of lidocaine with epinephrine 1: 100,000 appears to be relatively safe in patients with CVD.

Strengths and limitations of this study

This study was conducted with methodological rigor, including risk assessment of bias and of quality of evidence; such information is not available in reviews related to the subject (12, 13). Furthermore, the study has explicit eligibility criteria, priority of outcomes (primary and secondary), comprehensive and extensive search in the databases, with independent and peer reviews for each study and use of GRADE to assess the quality of the evidence and the strength of recommendation for the outcomes included in the meta-analysis.

Although the search strategy was carried out in different databases (Cochrane Central Register of Controlled Trials, MEDLINE, EMBASE, Healthstar, CINAHL, Web of Science, and ClinicalTrials.gov), coincidentally the included studies were mainly published in Brazil between 2007 and 2014. CVD (mainly stroke and coronary heart disease) has been Brazil´s leading cause of death for half a century (39). However, CVDs have several etiological factors related to the population profile (ethnicity, genetic factors, habits, among others). Considering these factors, and that most of the studies included are from Brazil, the present study also indicates the need for further studies related to the topic of this systematic review with different patient profiles.

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Considering the possible outcomes, conducting clinical studies with CVD patients is highly complex. In addition, the difficulty of including patients with specific systemic profiles may explain the limited number of types of CVD included in this study. The primary studies included were a limiting factor for the findings of this review due to methodological bias observed in the included RCTs, as well as the differences observed in the types of CVDs, types and dosages of LA and vasoconstrictors. Thus, it was noted a need for more primary studies about the subject, with better methodological accuracy and prioritization of outcomes in order to increase confidence in findings.

Implications for clinical practice and research

The LA associated with vasoconstrictor is essential in dental treatment because vasoconstrictor agents counteract the vasodilating action of local anesthetics, increasing anesthetic time and reducing bleeding (6, 8). The findings of this systematic review are limited to some types of CVD and the use of epinephrine and felypressin as vasoconstrictors. Additionally, the low quality of evidence demonstrates that literature needs further studies in order to confirm these results.

Thus, new RCTs with higher methodological rigor are needed to obtain more reliable results, in order to verify the safety of LAs with vasoconstrictor in patients with CVDs. It is also suggested that the professionals use the recommended doses as well as perform the procedure respecting the anesthetic techniques (2, 6, 9). Moreover, communication with the patient's cardiologist can help clarify the patient's clinical condition, to evaluate the care and limitations of the safest dental treatment for each patient (3,9).

Conclusion

None of the included studies in this review reported death, mortality by specific cause, stroke or acute myocardial infarction. The results suggest that the use of LA with epinephrine in low doses in adult patients with CVD reduced systolic blood pressure after dental procedures when compared to the use of LA without vasoconstrictor. Furthermore, the use of epinephrine (in low doses) does not alter the risk of arrhythmia, ST-segment depression, mean arterial pressure, systolic and diastolic blood pressure and heart rate, suggesting safety in its use for these outcomes in adult patients with some types of CVD. However, the low quality of evidence demonstrates that literature needs further studies in order to confirm these results.

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Contributors

CCG is the principal investigator and led the writing of the manuscript. LCL and RHLM are the project managers and co-investigators and contributed to the writing and revision of the manuscript. CCG, CCB, RHLM, JCR, JOA, NKA are co-investigators who contributed to screening the eligible studies and extracted the data of the studies, also participated the revision of the manuscript. MTS performed the statistical analysis, also participated the revision of the manuscript. All authors read and approved the final manuscript.

Competing interests

None declared.

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Data sharing statement

All data relevant to the study are included in the article or uploaded as supplementary information. However, other information is available upon request.

Research Ethics Approval: Human Participants

This study is a systematic review with meta-analysis and does not involve human participants.

Provenance and peer review

Not commissioned, externally peer reviewed.

Abbreviations

Local anesthetic (LA), cardiovascular disease (CVD), Cochrane Central Register of Controlled Trials (CENTRAL), Medical literature Analysis and Retrieval System Online (MEDLINE), Excerpta Medica dataBASE (EMBASE), Cumulative Index to Nursing and Allied Health Literature (CINAHL), Grading of Recommendations Assessment, Development, and Evaluation (GRADE), randomized clinical trial (RCT), bank of Brazil thesis (CAPES), Congresso Internacional de Odontologia de São Paulo (CIOSP),

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confidence interval (CI), weighted mean difference (WMD), minimally important difference (MID), and standardized mean difference (SMD), standard deviation (SD).

References

1. World Health Organization.WHO. Cardiovascular diseases (CVDs). WHO [Internet]. 2017 2017-05-1714:41:03. Available from: http://www.who.int/mediacentre/factsheets/fs317/en/.

2. Becker DE, Reed KL. Local anesthetics: review of pharmacological considerations. Anesth Prog. 2012;59(2):90-101; quiz 2-3. doi: 10.2344/0003-3006-59.2.90.

3. Jowett NI, Cabot LB. Patients with cardiac disease: considerations for the dental practitioner. Br Dent J. 2000;189(6):297-302.

4. American Dental Association A. Guideline for teaching pain control and sedation to dentist and dental students. 2012:1-18.

5. St George G, Morgan A, Meechan J, Moles DR, Needleman I, Ng YL, et al. Injectable local anaesthetic agents for dental anaesthesia. Cochrane Database Syst Rev. 2018;7(7):Cd006487.

6. Haas DA. An update on local anesthetics in dentistry. J Can Dent Assoc. 2002;68(9):546-51.

7. Jage J. Circulatory effects of vasoconstrictors combined with local anesthetics. Anesth Pain Control Dent. 1993;2(2):81-6.

8. Fencl JL. Guideline implementation: local anesthesia. AORN J. 2015;101(6):682-9; quiz 90-2.

9. Kouba DJ, LoPiccolo MC, Alam M, Bordeaux JS, Cohen B, Hanke CW, et al. Guidelines for the use of local anesthesia in office-based dermatologic surgery. J Am Acad Dermatol. 2016;74(6):1201-19.

10. Wadlund DL. Local Anesthetic Systemic Toxicity. Aorn j. 2017;106(5):367-77.

11. Brown RS. Local anesthetics. Dent Clin North Am. 1994;38(4):619-32.

12. Bader JD, Bonito AJ, Shugars DA. A systematic review of cardiovascular effects of epinephrine on hypertensive dental patients. Oral Surg Oral Med Oral Pathol Oral Radiol Endod. 2002;93(6):647-53.

13. Godzieba A, Smektala T, Jedrzejewski M, Sporniak-Tutak K. Clinical assessment of the safe use local anaesthesia with vasoconstrictor agents in cardiovascular compromised patients: a systematic review. Med Sci Monit. 2014;20:393-8.

For peer review only

14. Brand HS, Gortzak RA, Palmer-Bouva CC, Abraham RE, Abraham-Inpijn L. Cardiovascular and neuroendocrine responses during acute stress induced by different types of dental treatment. Int Dent J. 1995;45(1):45-8.

15. Meyer FU. Haemodynamic changes under emotional stress following a minor surgical procedure under local anaesthesia. Int J Oral Maxillofac Surg. 1987;16(6):688-94.

16. New York Heart Association. Use of epinephrine in connection with procaine in dental procedures. J Am Dent Assoc 1955;50:108.

17. Brown RS, Rhodus NL. Epinephrine and local anesthesia revisited. Oral Surg Oral Med Oral Pathol Oral Radiol Endod. 2005 Oct;100(4):401-8.

18. Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. PLoS Med. 2009;6(7):e1000097.

19. Altman R, Alarcon G, Appelrouth D, Bloch D, Borenstein D, Brandt K, et al. The American College of Rheumatology criteria for the classification and reporting of osteoarthritis of the hand. Arthritis Rheum. 1990;33(11):1601-10.

20. Higgins JP, Green S. Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0 (updated March 2011). The Cochrane Collaboration 2011.

21. Akl EA, Sun X, Busse JW, Johnston BC, Briel M, Mulla S, et al. Specific instructions for estimating unclearly reported blinding status in randomized trials were reliable and valid. J Clin Epidemiol. 2012;65(3):262-7.

22. Manning M, Stoev S, Chini B, Durroux T, Mouillac B, Guillon G. Peptide and non-peptide agonists and antagonists for the vasopressin and oxytocin V1a, V1b, V2 and OT receptors: research tools and potential therapeutic agents. Prog Brain Res [Internet]. 2008; 170:[473-512 pp.].

23. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. Stat Med. 2002;21(11):1539-58.

24. Guimaraes CC, Lopes Motta RH, Bergamaschi CC, Araujo JO, Andrade NK, Figueiro MF, et al. Local anaesthetics combined with vasoconstrictors in patients with cardiovascular disease undergoing dental procedures: systematic review and meta-analysis protocol. BMJ Open. 2017;7(11):e014611.

25. Guyatt GH, Oxman AD, Kunz R, Woodcock J, Brozek J, Helfand M, et al. GRADE guidelines: 7. Rating the quality of evidence--inconsistency. J Clin Epidemiol. 2011;64(12):1294-302.

For peer review only

26. Guyatt GH, Oxman AD, Santesso N, Helfand M, Vist G, Kunz R, et al. GRADE guidelines: 12. Preparing summary of findings tables-binary outcomes. J Clin Epidemiol. 2013;66(2):158-72.

27. Guyatt GH, Thorlund K, Oxman AD, Walter SD, Patrick D, Furukawa TA, et al. GRADE guidelines: 13. Preparing summary of findings tables and evidence profiles-continuous outcomes. J Clin Epidemiol. 2013;66(2):173-83.

28. Guyatt GH, Oxman AD, Montori V, Vist G, Kunz R, Brozek J, et al. GRADE guidelines: 5. Rating the quality of evidence--publication bias. J Clin Epidemiol. 2011;64(12):1277-82.

29. Abu-Mostafa N, Aldawssary A, Assari A, Alnujaidy S, Almutlaq A. A prospective randomized clinical trial compared the effect of various types of local anesthetics cartridges on hypertensive patients during dental extraction. Journal of clinical and experimental dentistry. 2015;7(1):e84-8.

30. Bronzo AL, Cardoso CG, Jr., Ortega KC, Mion D, Jr. Felypressin increases blood pressure during dental procedures in hypertensive patients. Arquivos brasileiros de cardiologia. 2012;99(2):724-31.

31. Caceres MT, Ludovice AC, Brito FS, Darrieux FC, Neves RS, Scanavacca MI, et al. Effect of local anesthetics with and without vasoconstrictor agent in patients with ventricular arrhythmias. Arquivos brasileiros de cardiologia. 2008;91(3):128-33, 42-7.

32. Conrado VC, de Andrade J, de Angelis GA, de Andrade AC, Timerman L, Andrade MM, et al. Cardiovascular effects of local anesthesia with vasoconstrictor during dental extraction in coronary patients. Arquivos brasileiros de cardiologia. 2007;88(5):507-13.

33. Davenport RE, Porcelli RJ, Iacono VJ, Bonura CF, Mallis GI, Baer PN. Effects of anesthetics containing epinephrine on catecholamine levels during periodontal surgery. Journal of periodontology. 1990;61(9):553-8.

34. Laragnoit AB, Neves RS, Neves IL, Vieira JE. Locoregional anesthesia for dental treatment in cardiac patients: a comparative study of 2% plain lidocaine and 2% lidocaine with epinephrine (1:100,000). Clinics (Sao Paulo, Brazil). 2009;64(3):177-82.

35. Neves RS, Neves IL, Giorgi DM, Grupi CJ, Cesar LA, Hueb W, et al. Effects of epinephrine in local dental anesthesia in patients with coronary artery disease. Arquivos brasileiros de cardiologia. 2007;88(5):545-51.

36. Ogunlewe MO, James O, Ajuluchukwu JN, Ladeinde AL, Adeyemo WL, Gbotolorun OM. Evaluation of haemodynamic changes in hypertensive patients during

For peer review only

tooth extraction under local anaesthesia. The West Indian medical journal. 2011;60(1):91-5.

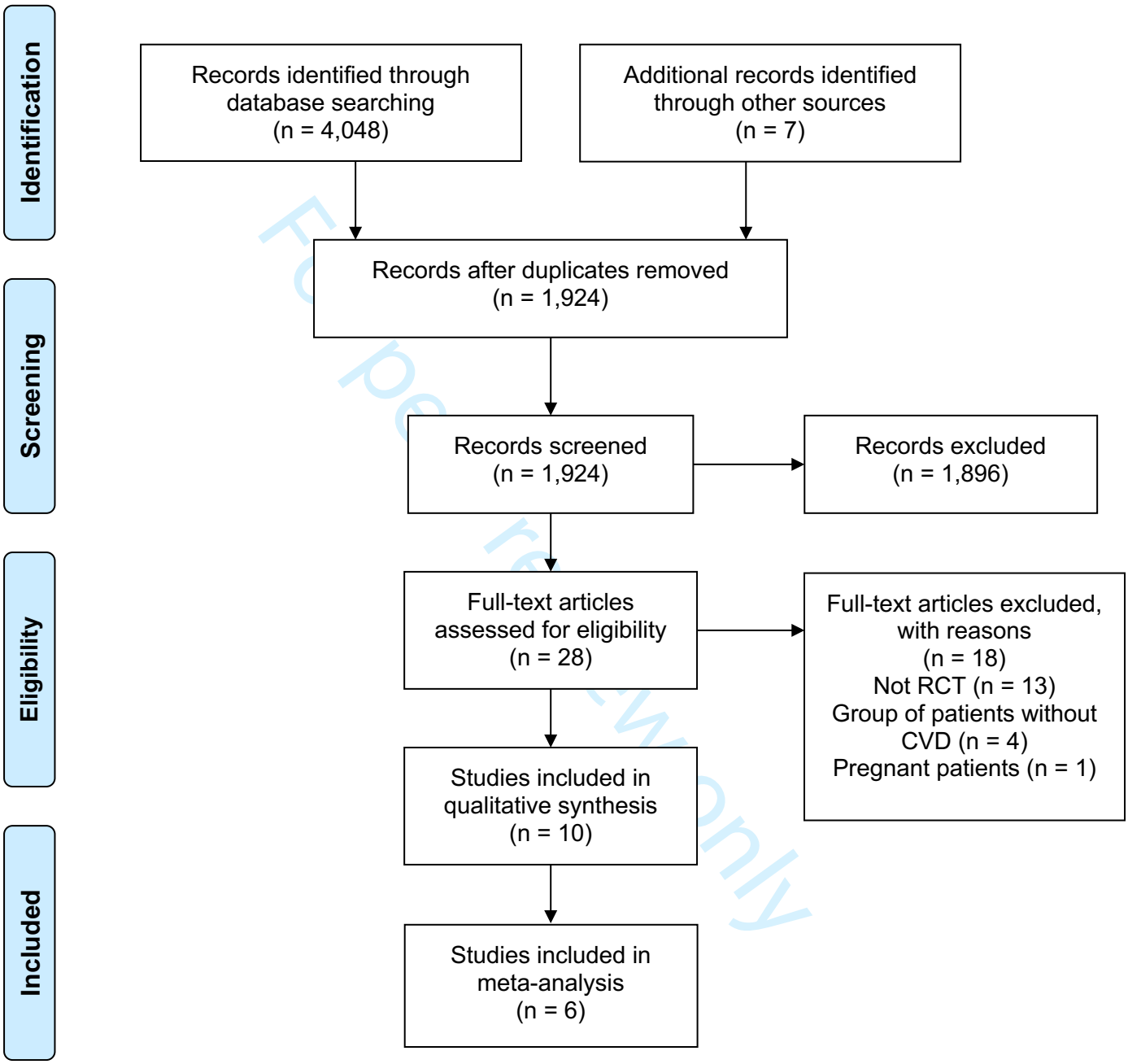
37. Santos-Paul MA, Neves IL, Neves RS, Ramires JA. Local anesthesia with epinephrine is safe and effective for oral surgery in patients with type 2 diabetes mellitus and coronary disease: a prospective randomized study. Clinics (Sao Paulo, Brazil). 2015;70(3):185-9.

38. Torres-Lagares D, Serrera-Figallo MA, Machuca-Portillo G, Corcuera-Flores JR, Machuca-Portillo C, Castillo-Oyague R, et al. Cardiovascular effect of dental anesthesia with articaine (40 mg with epinefrine 0,5 mg % and 40 mg with epinefrine 1 mg%) versus mepivacaine (30 mg and 20 mg with epinefrine 1 mg%) in medically compromised cardiac patients: a cross-over, randomized, single blinded study. Medicina oral, patologia oral y cirugia bucal. 2012;17(4):e655-60.

39. Ribeiro AL, Duncan BB, Brant LC, Lotufo PA, Mill JG, Barreto SM. Cardiovascular Health in Brazil: Trends and Perspectives. Circulation. 2016 Jan 26;133(4):422-3.

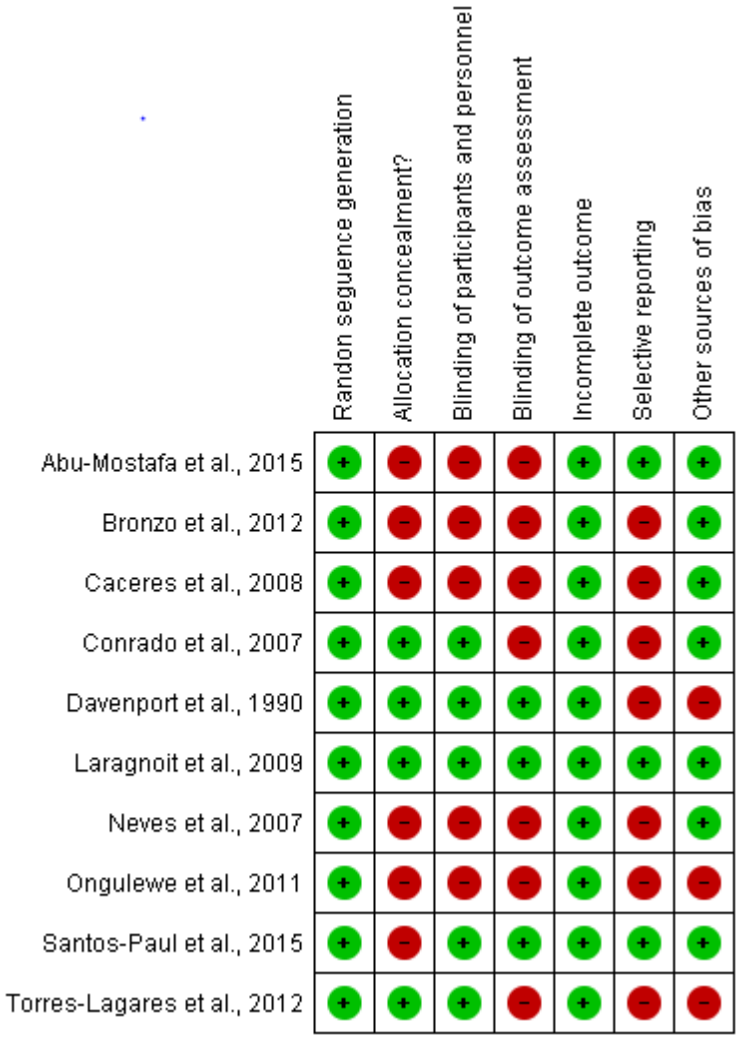
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Figure 1. Flowchart for literature search and study selection



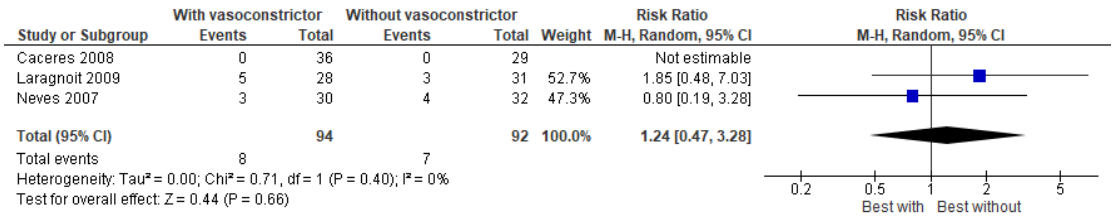
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Figure 2. Risk of bias summary: review authors' judgments



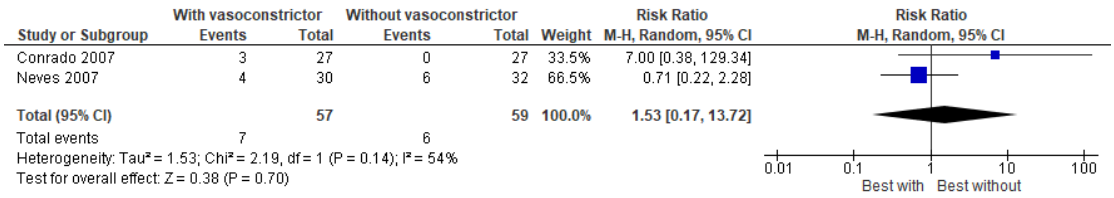
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Figure 3. Presence of arrhythmia associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor.



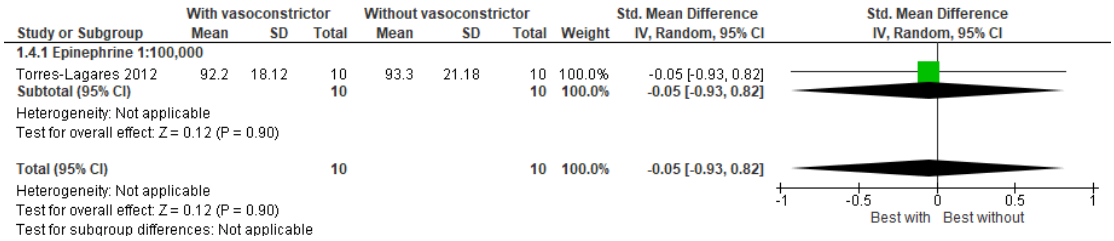
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Figure 4. Presence of ST segment depression associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor.



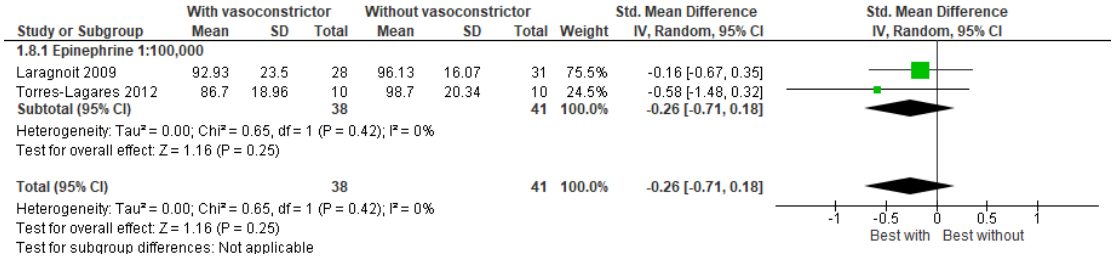
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Figure 5. Mean blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, during the procedure period.



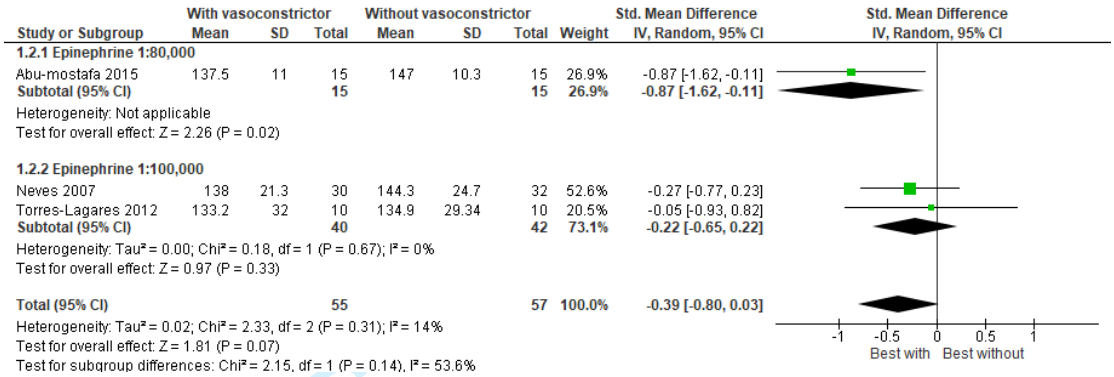
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Figure 6. Mean blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, after the procedure period.



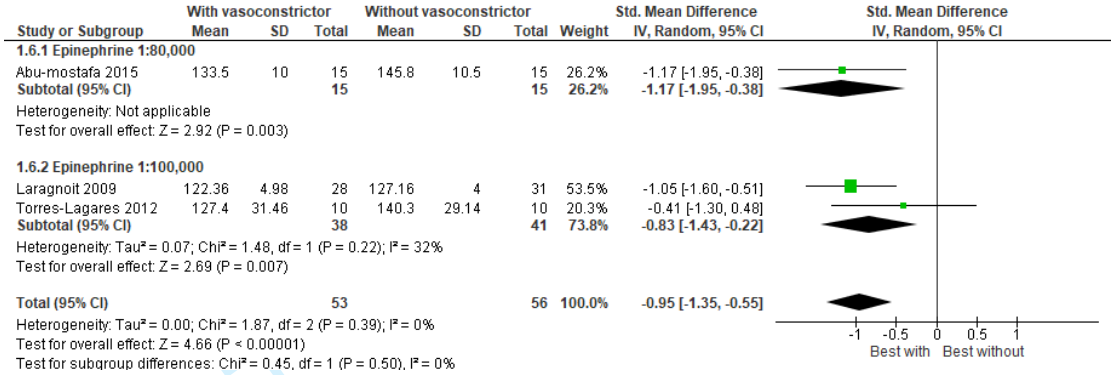
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]Figure 7. Systolic blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, during the procedure period.



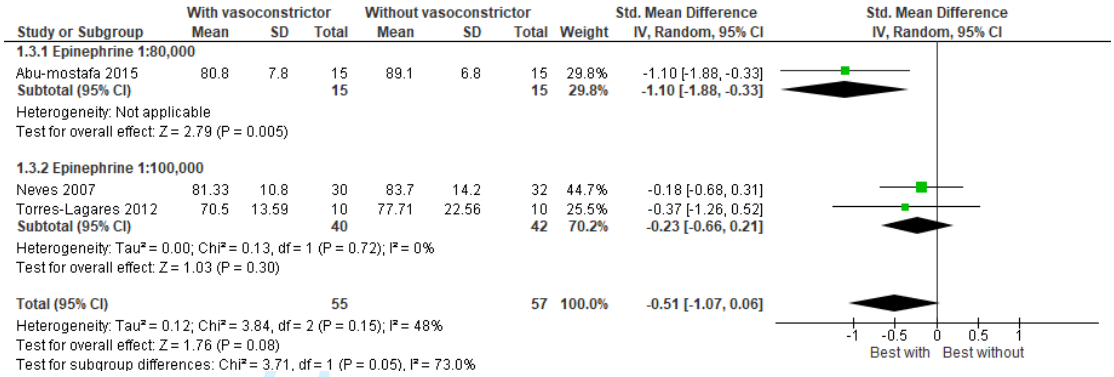
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Figure 8. Systolic blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, after procedure period.



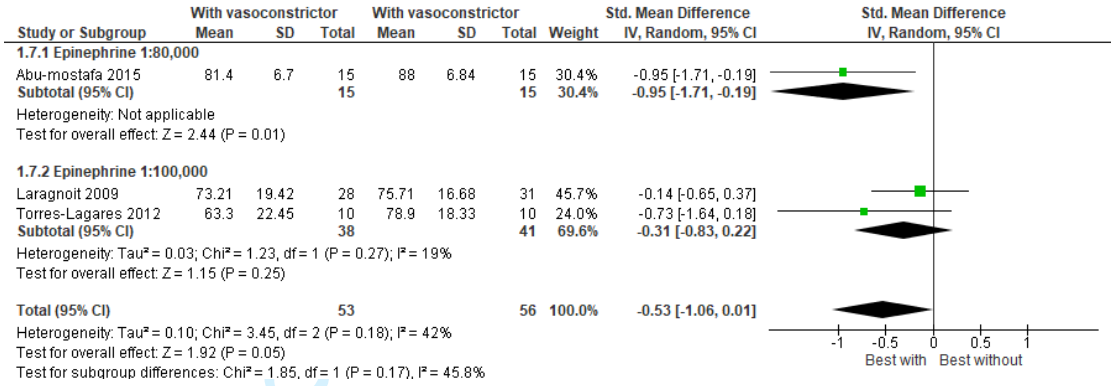
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Figure 9. Diastolic blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, during the procedure period.



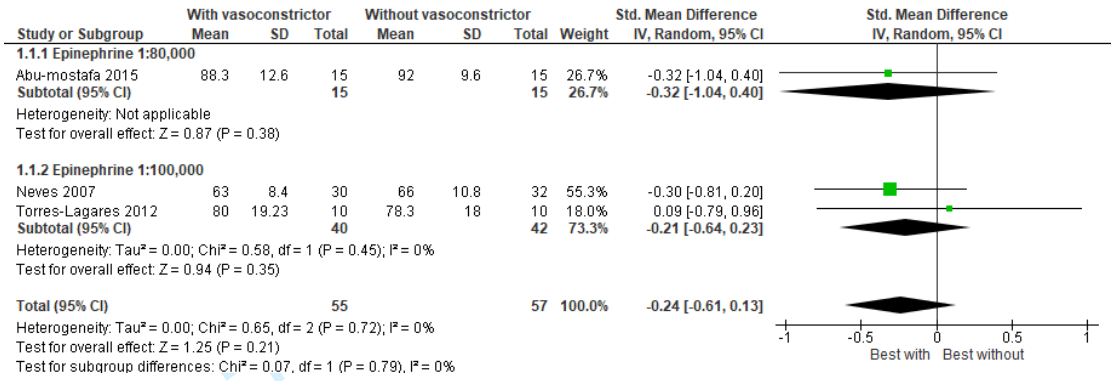
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Figure 10. Diastolic blood pressure values associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, after the procedure period.



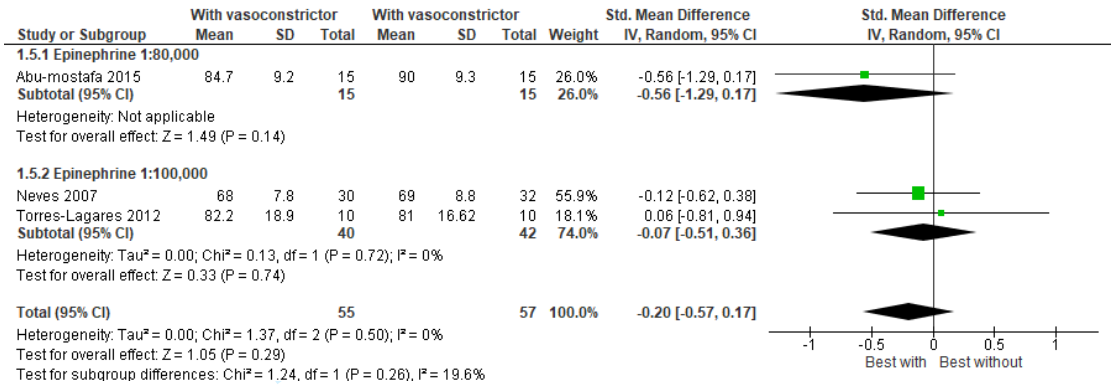
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Figure 11. Heart rate associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, during the procedure period.



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Figure 12. Heart rate associated with the use of anesthetic with vasoconstrictor versus with the use of anesthetic without vasoconstrictor, after the procedure period.



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Appendix 1– Search strategy (via OVID, MEDLINE and EMBASE)

1 exp Dentistry/ (357819)
2 exp Dentistry, Operative/ (32163)
3 exp Dental Care/ (29438)
4 Dental Restoration, Permanent/ (18759)
5 Dental Restoration Repair/ (102)
6 Periodontal Debridement/ (191)
7 Subgingival Curettage/ (977)
8 Dental Scaling/ (3316)
9 Chronic Periodontitis/ (1971)
10 Periodontal Diseases/ (24137)
11 Periodontal Surgery.mp. (1302)
12 Periodontal treatment.mp. (2728)
13 Oral Surgical Procedures/ (5363)
14 exp Surgery, Oral/ (7419)
15 Tooth Extraction/ (17008)
16 Dental Prosthesis/ (3384)
17 "Root Canal Therapy"/ (11735)
18 exp Dental Implants/ (17311)
19 Dental Implants, Single Tooth/ (1901)
20 Dental Implantation/ (3773)
21 1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 (372164)
22 exp Anesthetics, Local/ (96504)
23 exp Anesthesia, Local/ (15673)
24 exp Anesthesia, Dental/ (10417)
25 Lidocaine/ (22512)
26 Prilocaine/ (2018)
27 Bupivacaine/ (10713)
28 Procaine/ (11313)
29 Mepivacaine/ (1899)
30 Carticaine/ (451)
31 Etidocaine/ (288)
32 22 OR 23 OR 24 OR 25 OR 26 OR 27 OR 28 OR 29 OR 30 OR 31 (114729)
33 exp Cardiovascular Diseases/ (2067079)
34 Cardiac.mp. (631932)
35 Coronary Disease/ (128393)
36 Coronary Artery Disease/ (47503)
37 Coronary arteriosclerosis.mp. (733)
38 Coronariopathy.mp. (29)
39 Arrhythmias, cardiac/ (56112)
40 Heart Valve Diseases/ (21116)
41 Heart Diseases/ (63528)
42 Heart Failure/ (63528)
43 Rheumatic Heart Disease/ (12263)
44 Myocardial Ischemia/ (34898)
45 Myocardial Infarction/ (151398)
46 Hypertension (210548)
47 Hypertensive Patients.mp. (25167)
48 33 OR 34 OR 35 OR 36 OR 37 OR 38 OR 39 OR 40 OR 41 OR 42 OR 43 OR 44 OR 45 OR 46 OR 47 (2314784)
49 21 AND 32 AND 48 (752)

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Appendix 2 – Risk of bias of RCts included (modified version of the Cochrane collaboration)

Author	Year	Random sequence generation	Allocation concealment	Blinding of participants and personnel	Blinding of outcome assessment	Incomplete outcome data	Selective reporting	Other sources of bias
Abu-Mostafa, et al. 2015)	2015	Definitely yes	Probably no	Probably no	Probably no	Probably yes	Probably yes	Probably yes
Bronzo, et al. 2012	2012	Probably yes	Probably no	Definitely no	Definitely no	Probably yes	Probably no	Probably yes
Caceres, et al. 2008	2008	Probably yes	Definitely no	Definitely no	Definitely no	Probably yes	Probably no	Probably yes
Conrado, et al. 2007	2007	Probably yes	Probably yes	Probably yes	Definitely no	Probably yes	Probably no	Probably yes
Davenport, et al. 1990	1990	Definitely yes	Probably yes	Probably yes	Probably yes	Probably yes	Probably no	Probably no
Laragnoit, et al. 2009	2009	Definitely yes	Probably yes	Definitely yes	Definitely yes	Probably yes	Probably yes	Probably yes
Neves, et al. 2007	2007	Probably yes	Probably no	Definitely no	Definitely no	Probably yes	Probably no	Probably yes
Ogunlewe, et al. 2011	2011	Probably yes	Probably no	Definitely no	Definitely no	Probably yes	Probably no	Probably no
Santos-Paul, et al. 2015)	2015	Probably yes	Probably no	Probably yes	Probably yes	Probably yes	Probably yes	Probably yes
Torres-Lagares, et al. 2012	2012	Definitely yes	Probably yes	Probably yes	Definitely no	Probably yes	Definitely no	Probably no

Appendix 3- Characteristics of excluded studies

Excluded studies	**Exclusion reasons**
Abu-Mostafa et al. 2015	Group of patients without CVD
Agani et al. 2015	Not RCT
Bazhanov et al.1983	Not RCT
Brkovic et al. 2008	Group of patients without CVD
Caviedes et al. 1983	Not RCT
Cecava, Mocek, Sel. 1965	Not RCT
Harang. 1949	Not RCT
Montebugnoli, Pelliccioni, Borghi 1990	Not RCT
Nava, Torres-Martinez 1967	Not RCT
Neves et al. 2009	Pregnant patients
Niwa 1999	Group of patients without CVD
Ohgami, Suda 1979	Not RCT
Rollason et al. 1977	Not RCT
Salman, Schwartz 1955	Not RCT
Tucci, Gabrielli, Mendes 1977	Not RCT
Vieillefosse 1973	Not RCT
Yoshimura et al. 1987	Not RCT
Zivotic-Vanovic, Marjanovic 2006	Group of patients without CVD

CAD- Cardiovascular disease. RCT- Randomised controlled trial

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	S1 appendix
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	8
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	9
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	9

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	10
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	–
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	S1 Figure
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	S1 Table
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	11
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	13
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	13
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	S2 Table
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	–
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	17
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	18
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	20
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	20

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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